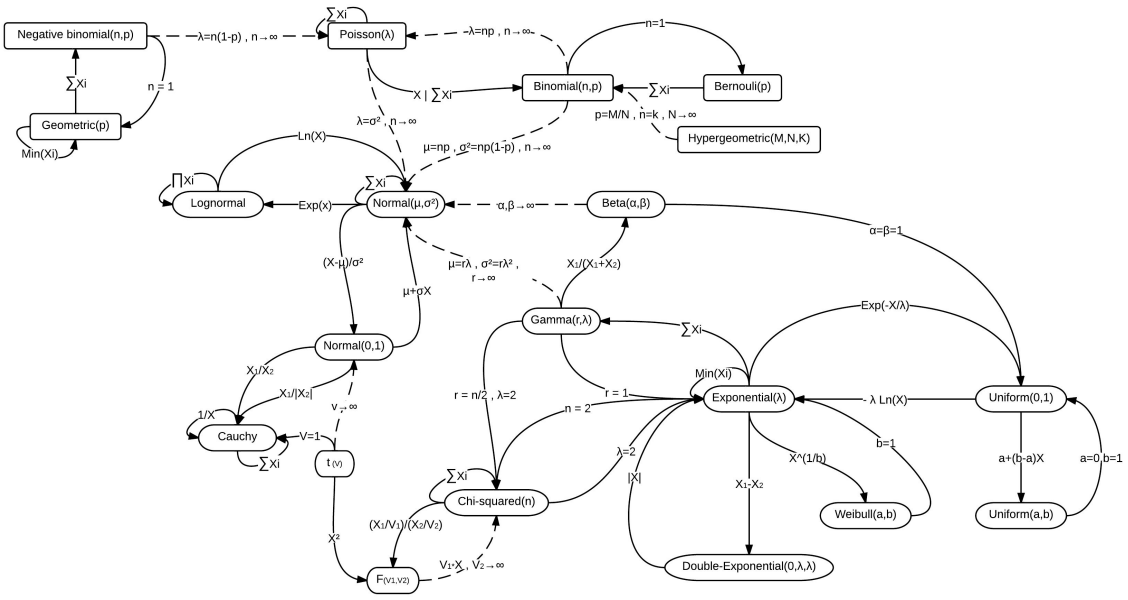


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An Introduction to Mathematical Statistics

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Contents

1 Introduction to Mathematical Statistics 9
1.1 Introduction 9
1.2 What is Mathematical Statistics? 9
1.2.1 What is Statistics? 9
1.3 Populations and Samples 10
1.4 Uses of Statistics 11
1.4.1 Discrete and Continuous Variables 12
1.5 Types of Data 12
1.5.1 Measurement Scales 12
1.6 Sources of Data 14
1.6.1 Internal and External Data Sources 14
1.6.2 Primary and Secondary Data Sources 15
1.7 Conclusion 16

2 Descriptive Statistics 17
2.1 Introduction 17
2.2 Graphical Techniques 17
2.2.1 Tabulation 18
2.2.2 Pie charts 20
2.2.3 Bar Chart 22
2.2.4 Multiple Bar Graph 23
2.2.5 The Component Bar Graph 25
2.2.6 The Percentage Component Bar Graph 27
2.2.7 Pictograms 27
2.2.8 Line Graphs 29
2.2.9 Frequency Distributions 30
2.2.10 Histograms 32
2.2.11 Frequency Polygons 34
2.2.12 Ogive Curves 35
2.3 Summary Statistics 39
2.4 Measures of location 39
2.4.1 The mean [or more precisely the arithmetic mean] . 39
2.4.2 The median 40
2.4.3 The Mode 43
2.4.4 Quantiles/Quartiles/percentiles 44
2.4.5 Skewness 46
2.4.6 Kurtosis 47

2.5 Box-and-Whisker plot 48
2.6 Measures of Dispersion 49
2.6.1 The Range (R) 50
2.6.2 Quartile deviation (IQR) 50
2.6.3 Variance s^2 50
2.7 Conclusion 51

3 Introduction to Probability 53
3.1 Introduction 53
3.2 Terminology 53
3.3 Some Relations from Set Theory 56
3.3.1 De Morgan's Law 58
3.4 Counting Methods: Permutations and Combinations 58
3.4.1 Multiplication Principle 59
3.4.2 Permutation of n objects 60
3.4.3 Permutation n objects taken r at a time 60
3.4.4 Combination of n objects taken r at a time 61
3.4.5 Permutations of n different objects when there are groups of identical items 62
3.5 Rules Governing Probability (Axioms) 63
3.6 Approaches to Probability 64
3.6.1 The Subjective Approach 64
3.6.2 Relative Frequency Approach to Probability 64
3.7 The Classical Approach 65
3.7.1 Addition Rule of Probability 67
3.7.2 Mutually or Disjoint Exclusive Events and Exhaustive Events 68
3.7.3 Multiplicative rule of Probability 69
3.7.4 Independent Events 69
3.7.5 Complementary Events 70
3.7.6 Conditional Probability 70
3.7.7 Probability Tree Diagrams 72
3.7.8 Bayes' Theorem 73
3.7.9 Using counting methods in Probability 75
3.8 Summary 77

4 Random Variables and Distributions 79
4.1 Introduction 79
4.2 Discrete random variables 79
4.2.1 Probability Distribution 79
4.2.2 Cummulative Probability Distribution 80
4.2.3 Expected value of a descrete random variable 80
4.2.4 Variance of a Discrete Random Variable 81
4.3 Moments and Moment Generating Functions of Discrete Random Variables 81
4.4 Continuous Random Variables 82
4.4.1 Cummulative Probability Function 84

4.4.2 The expected value of a continuous random variable 84
4.4.3 The variance of continuous random variable 84
4.5 Empirical Distributions 85
4.6 Some discrete distributions 86
4.6.1 Bernoulli trials 86
4.6.2 Binomial Distribution 86
4.6.3 Poisson Distribution 87
4.6.4 Poisson Distribution's link with the Binomal Distribution 88
4.6.5 The Hypergeometric Distribution 88
4.6.6 Discrete Uniform distribution 89
4.7 Some Common Continuous Empirical distributions 89
4.7.1 The Continuous Uniform probability distribution . . 90
4.7.2 The Normal distribution 90
4.7.3 The Exponential probability distribution 91
4.8 Memoryless Property 91
4.9 Conclusion 92

5 Estimation and Hypothesis Testing 95
5.1 Introduction 95
5.2 Estimation 95
5.2.1 Point Estimation 95
5.2.2 Point Estimate of the Mean 96
5.2.3 Point Estimate of the Population Standard Deviation 96
5.2.4 Point Estimate of a Population Proportion π 96
5.3 Confidence Intervals 98
5.3.1 Case 1: Confidence Interval for μ, when σ^2 is known 98
5.3.2 Case 2: Confidence Interval for μ, when sample size is large, (i.e. $n > 30$) and σ^2 is unknown 99
5.3.3 Case 3: Confidence Interval for μ, when the sample size is small, (i.e. $n < 30$) and σ^2 is unknown 99
5.3.4 Using Student's t Tables 99
5.4 Sampling to a Desired Precision 102
5.5 Hypothesis Testing 104
5.5.1 What is a Statistical Hypothesis 104
5.5.2 How Do You Formulate a Hypothesis test 105
5.5.3 Terms Used In Hypothesis Testing 106
5.5.4 Hypothesis Testing for a Single Population Mean . . 106
5.5.5 Testing $H_0 : \mu = \mu_0$ Versus $H_1 : \mu \neq \mu_0$ When σ^2 is Known 107
5.5.6 Testing $H_0 : \mu = \mu_0$ Versus $H_1 : \mu \neq \mu_0$ When σ^2 is Unknown $n < 30$ 107
5.5.7 Testing $H_0 : \mu = \mu_0$ Versus $H_1 : \mu \neq \mu_0$ When σ^2 is Unknown, n Large 107
5.5.8 Testing $H_0 : \mu = \mu_0$ Versus $H_1 : \mu > \mu_0$ 109
5.5.9 Testing $H_0 : \mu = \mu_0$ Versus $H_1 : \mu < \mu_0$ 110

5.5.10 The p-value . . . 112
5.6 Summary . . . 114

6 Introduction to Correlation and Regression Analysis 115
6.1 Introduction . . . 115
6.2 Scatter Plots . . . 115
6.3 Correlation . . . 116
6.3.1 Correlation Coefficient . . . 116
6.3.2 Correlation Matrix . . . 117
6.4 Simple Linear Regression Analysis . . . 119
6.4.1 Types of Relationship . . . 120
6.5 When To Apply Regression Analysis . . . 124
6.5.1 Some Uses of Regression Analysis . . . 125
6.6 Regression Assumptions . . . 126
6.7 The Least Squares Technique . . . 127
6.7.1 Least Squares Estimation . . . 127
6.8 Properties of Estimates . . . 131
6.9 Making Inferences about β_0 and β_1 . . . 132
6.10 Analysis of Variance of Simple Linear Regression . . . 135
6.10.1 Partitioning the Total Sum of Squares . . . 135
6.11 The Basic ANOVA Table . . . 138
6.12 The Coefficient of (simple) Determination . . . 140
6.12.1 Adjusted R^2 . . . 140
6.13 Conclusion . . . 143
6.14 Exercises . . . 143

List of Figures

2.1 Pie chart of income groups . . . 21
2.2 Bar chart of the Income groups of companies . . . 22
2.3 Bar chart of the Income groups of companies . . . 23
2.4 Multiple Bar graph of the Exports to compare within each year . . . 24
2.5 Multiple Bar graph of the Exports to compare within each product . . . 25
2.6 Simple Bar graph of the Exports to compare the Annual Exports . . . 26
2.7 Component Bar graph of the Exports to compare the Annual Exports . . . 26
2.8 Percentage Component Bar graph of the Exports to compare the Annual Exports . . . 28
2.9 Pictogram of Garden Shovels sold . . . 28

2.10 Line graph of inflation rate figures 29
2.11 Histogram of shipping times 34
2.12 Unknown graph 34
2.13 Frequency Polygon of the shipping times 35
2.14 The more than and less than ogive curve of the of shipping times 37
2.15 Examples of skewed distributions 46
2.16 Boxplot of Accidents before and after Traffic light 49

3.1 Venn Diagram Showing a Universal Set 56
3.2 Venn Diagram Showing a Complement of s set 56
3.3 Venn Diagram Showing a Union of s set A and B 56
3.4 Venn Diagram Showing a Intersection of s set A and B . . . 57
3.5 Number of possible experiments 59
3.6 The Venn Diagram and the a Universal Set 72
3.7 Partition of B by mutually exclusive and exhaustive events A_i 74

4.1 The area representing the probability 83

6.1 Scatter plot of incomes versus entertainment 116
6.2 The relationship between A and L^2 121
6.3 The relationship between A and L 122
6.4 Examination mark versus time spent studying 123
6.5 Scatter plots showing two possible relationships between x and y 125
6.6 The observed values y_i (marked '★'), residuals r_i and fitted values $\hat{y}_i$ (marked '◊') 128
6.7 t-Distribution: when to reject or accept H_0 134
6.8 Variability on the regression line 136
6.9 The general form of the statistical decision rule for an F-test 139

Preface

The aim of this book is to introduce you to Mathematical Statistics and how it is applied to solve real life problems. We will attempt to clarify the links between the techniques and the interpretation of the results obtained. We will be made not to assume that you are familiar with the techniques.

The initial Chapters of this book will introduce you to the ideas of Statistics, the nature of the data you use, and how its measured and focus on summary Statistics and the Graphical or 'Pictorial' components of Statistics. We will then proceed to investigate the central tendency of data as well as its variability. We will then look at the distributions of the data and moments.

We will then discuss probability and permutations and combinations. We will also introduce you to the idea of hypothesis testing.

The best way for you to study this book is to go through all the chapters and sections in a systematic manner. What does this mean? After going through each Chapter you should be able to answer the questions at the end of each Chapter, if not go through the Chapter again.

The diagram on the cover page is taken from Lawrence M. and Jacquelyn T [1].

C. Chimedza and N. Ndebele

[1] American Statistician 62 (1): 45-53

Chapter 1

Introduction to Mathematical Statistics

1.1 Introduction

This chapter will introduce you to Mathematical Statistics. We will discuss statistical techniques, that summarise data and develop the mathematical foundations for these methods. The layout and presentation of the topics should be easy for you to follow.

1.2 What is Mathematical Statistics?

'What is Mathematical Statistics?'. This is perhaps the first question on your mind. Well, it is the application of mathematical concepts to statistics. Mathematical techniques which are used for this include mathematical analysis, linear algebra, stochastic analysis, differential equations, measure-theory and probability theory. In this course we will however, focus on parts of linear Algebra and Mathematical Analysis.

1.2.1 What is Statistics?

1. The word Statistics, in common usage refers to numerical **data**. This means any collection of data or information, constitutes what we refer to as Statistics.

   Example 1.1
   Some examples of Statistics under this definition are:

   (a) Vital Statistics;
   These are numerical data on births, marriages, divorces, communicable diseases e.t.c..

   (b) Business and Economic Statistics;
   These are numerical data on employment, production, prices and Sales e.t.c..

(c) *Social Statistics:*
These are numerical data on housing, crime, education e.t.c.,

2. The other meaning of Statistics is more specialised, that is, Statistics refers to the collection of techniques that are used to collect data, summarise and present data and to make conclusions from the data.

In some other literature Statistics is defined as making sense of data.

In order for you to appreciate data and statistical methods, we will discuss briefly what data is and how it is collected. We will do this by discussing methods used to collect data and what data is.

1.3 Populations and Samples

A population is the collection of all the individual items or points under investigation. A population is usually very large, for instance it could be the number of stars in the sky, the number and types of computers in a firm or the students studying a degree (names, gender, race and test marks). A sample is part of a population under study. It is selected so that it is representative of the population. This means what all the conclusions we make about the sample can be generalised to the whole population. This is why we should take great care collecting data

In the world of computers [as in life] we usually deal with large volumes of information or data [especially these days, where increased use of internet applications and social networks create a lot of data], making it difficult to study each and every observation, in order to draw conclusions about the source of the data. We seek a Statistical method or methods that help us collect and **summarise** the data, so that we can draw conclusions about these data, without scrutinising each and every observation.

The techniques we use to pick a sample from a population are called **sampling**. The methods we use to summarise data fall under the area of Statistics called **Descriptive Statistics**. **Inferential statistics** is that branch of statistics that deals with drawing conclusions based on statistical information that would have been gathered from a sample.

Please note that there are many other areas of Statistics and others are being developed every day. We will however, only introduce you to Descriptive statistics, Inferential statistics (Including Regression Analysis) in this course.

'A Statistician collects data [in an appropriate manner] analyses it using Statistical techniques, interprets the results and makes conclusions and recommendations on the basis of the data analysis'

1.4 Uses of Statistics

Statistics plays a very important role in Business, education, mining, health indeed all aspects of life. Some of the roles of statistics are listed below.

- Statistics in business plays an important role in the calculation of interest on investments, loans and Bonuses on retirement annuities. This means the banking and life assurance sectors depend on the mathematics of finance component of statistics. This leads us to the world of an Actuary which deals with the determination of rates and premiums in insurance that guarantee that a company will not go under.
- Statistics plays the important role of modelling and forecasting future trends of events. Examples are the monitoring of the exchange rates, production (mining, agriculture, manufacturing e.t.c.) and inflation done by central banks.
- Modelling the rainfall patterns for future planning, which leads to the organisation of supplementary food imports if necessary, for food security in the event of anticipated poor rains.
- Presenting large data sets to the public in an understandable way. Statisticians will take data collected from different sectors of the economy and present it to the public in such a way that its easy to understand. A good example is showing how the budget was allocated to the different ministries using a pie chart.
- Collection of information concerning the operation of a business firm to analyse its position and make decisions concerning the future.
- In the government, research costs must be estimated for various projects, manpower, skill, equipment and time requirements using appropriate sampling methods.

These are some of the uses of statistics, there are many other applications which have not been mentioned here. We will give you relevant examples in the sections that follow.

The word **Data** keeps turning up in our discussion of Statistics, data is the 'blood' of Statistics or the raw material of statistics. We can formally

describe data as information contained in all observations of interest (Population).

How is it collected? Why do we collect it? To answer these questions, let us define a few important terms.

1.4.1 Discrete and Continuous Variables

A **variable** is a quantity that can assume any of prescribed set of values. The set is called the domain of the variable. An example of a variable is X. Here X represents the salary of each individual in Zimbabwe. An example of a variable is the salary of each worker in South Africa. Here X may be used to represent this variable. A variable that can only assume one value is called a **constant**, e.g., $h = 4$ (h's value is fixed at 4).

A variable that can assume any value between two given values is called a
continuous variable, e.g., the amount of rainfall received (can be 1cm, 1.12cm or 0.005cm [includes decimals]). A variable that can only take integer values is called a **discrete variable**, e.g., the number of tertiary institutions in South Africa (can be 0, 2, 45 [integers only]). The type of data which results from measurement of continuous variables is called continuous data, and that from discrete variables is called discrete data.

Example 1.2

If the number of people migrating into South Africa is recorded every month, then the variable of concern (number of people) is discrete, i.e., it can only take integer values (values are not continuous).

1.5 Types of Data

Let us discuss the types of data used in data analysis. In order for us to understand these types of data, let us look at how the data are measured. We shall refer to these ways of measuring data as **scales** of measurement.

1.5.1 Measurement Scales

Lets us formally define **measurement** first.

Definition 1.1

Measurement is the assignment of numerals to objects or events according to rules, (different rules lead to different kinds of scale and different kinds of data).

The four common types of measurement scales are

1. Nominal,
2. Ordinal,
3. Interval, and
4. Ratio.

Nominal scale

This is the weakest of the four measurement scales of data. It distinguishes one object or event from another on the basis of a 'name'. An example of this is classifying [naming], items coming off an assembly line as defective or non-defective, or classifying a bank as sound or not sound. We can choose to code these, e.g. if the bank is sound use the number 1 and 2 for a bank that is not sound. Data of this type are frequently referred to as

- count data,
- frequency data or
- categorical data.

Ordinal Scale

In this scale objects or events are distinguished on the basis of the relative amounts of some characteristic they possess. These measurements enable observations to be ranked [given positions smallest to largest]. An example which puts this in perspective is ranking differently sized jerseys from smallest to largest, i.e., assigning rank=1 to the smallest jersey and increasing the rank by one up to the largest jersey. This leads us to a type of data known as **rank data** or **ordinal data**. An example would be small=1, medium=2, large=3 and extra large=4.

Note:

The magnitude of the differences between measurements is not reflected in the rank.

Interval Scale

This scale is applied when objects, or events can be distinguished one from another and ranked, and when the differences between measurements have meaning. An example to clarify how this scale works, follows. Suppose that four objects A, B, C and D are assigned scores

of $20,\ 30,\ 60$, and 70 respectively, if the interval scale is used then we can say that the difference between A and B is equal to the difference between C and D, i.e., equal differences in the amount of trait [characteristic] being measured. However, the ratios of the scores cannot be used i.e the score of 60 for C does not mean that C has twice as much of trait as B which has score 30.

Note:

The numbers $20,\ 30,\ 60$ and 70 are **scores** assigned, not measurements as it were.

Ratio scale

This kind of scale applies all the scales above and has the additional property that the ratios are meaningful. This scale includes the familiar measurements of height, weight e.t.c., that is, quantitative data. Each measurement makes sense, i.e., the difference in magnitude and the Ratio can all be used for analysis.

Now that we have discussed data types and their measurement scales let us now look at a few techniques which we can use to summarise our data.

Exercise 1.1.

1. *The banks in South Africa are assigned positions according to their reported profits, the bank with the highest profit is given position 1, and that with the second highest profit is given position 2, e.t.c., What kind of data do we get?*
2. *If we recorded the actual profit made by each bank, what kind of data do we get?*

1.6 Sources of Data

Now that we know the different types of data let us discuss the different data sources. There are two main sources of data, these are;

- internal and external sources.
- primary and secondary sources.

1.6.1 Internal and External Data Sources

During the normal operations of any business, government department or system, information or data is generated. Such data is known as **internal data** since its generated internally. Examples of data from internal sources are;

- daily balances on accounts in banks or shops.
- daily income generated by a small business.
- Number of visitors to a hotel.

Data that is generated from outside an organisation is known as **external data**, as it is generated from an external source. Examples of external sources of data are;

- statistics published by Statistics South Africa.
- statistics published by the World bank.
- statistics published in journals and magazines.

1.6.2 Primary and Secondary Data Sources

Primary data is data that is captured at its source for a specific function. The source of data is said to be a primary source. The most common primary data source is carrying out a survey to investigate a specific need. Examples of primary data sources are;

- a population census.
- a poverty assessment study.

Secondary data on the other hand, is data that which was collected elsewhere for some other purpose. Secondary data will already be in existence somewhere and the researcher uses this data to solve their business problem, without having to directly collect the data as is the case with primary data. Examples of secondary data are:

- Using past records published by some other organisation to predict something about your own organisation.
- financial records of your organisation from 2 years back.

Statistics plays an important role in all sectors of the economy and the society as it helps decision makers in their decision making. Before a decision can be made about any business or social problem it is often necessary to gather enough data or information to support what ever decision is made. Statistics helps in the collection of information in an scientific and systematic manner. Summarising the information in an easily understandable way and finally making decisions based on this information.

Steps taken in statistics are;

1. Collect data

2. Classify data [usually decided by the data collection method]
3. Summarize data
4. Present data
5. Proceed to inferential statistics if there is enough data to draw a conclusion.

1.7 Conclusion

In this Chapter, Statistics was defined. The uses of statistics were also discussed. Data was defined and the measurement scales that are available were also discussed with appropriate examples given. The major sources of data were discussed and examples of each source were given.

Exercise 1.2

1. *Distinguish between Internal data and primary data.*
2. *What do you understand about nominal data?*
3. *Data is collected from a shop's daily sales. The number of customers and the total daily sales were recorded.*
   (a) *What scales are the data measured on?*
   (b) *Classify the source of data as primary, secondary, internal or external.*
4. *Classify the following data as nominal, ordinal, interval or ratio;*
   (a) *Temperature.*
   (b) *Shoe size.*
   (c) *Denominations of money in South Africa.*
   (d) *Names of banks.*

Chapter 2

Descriptive Statistics

2.1 Introduction

In this chapter we going to discuss **Descriptive statistics** in two parts, that is, **Graphical Statistics** which include tables and graphs and **Summary statistics** involving numerical calculations.

2.2 Graphical Techniques

What are graphical techniques? These are pictorial or graphical representations of the data such that the main features of the data are captured. Let us now look at the various graphical techniques which we will cover in this chapter. These are, namely;

1. tables,
2. pie charts,
3. bar charts,
4. multiple bar charts,
5. stem-and-leaf displays,
6. pictograms,
7. line graphs,
8. histograms,
9. frequency polygons and ogive curves.
10. Box and Wisker plots (later).

These Box and Whisker plots or Box plots will be covered later because they require knowledge of summary statistics which we will discuss later in the chapter.

2.2.1 Tabulation

When you collect data it is important that you present it in an easily understandable way. One of the aims of statistics is to summarise data in the easiest way. One of best ways of summarising data is to use tables. Let us look at an example to illustrate this point.

Example 2.1

Suppose the genders of lecturers in a school are recorded and the results obtained are recorded below.

man	*man*	*woman*	*man*	*woman*	*woman*	*man*
woman	*man*	*woman*	*man*	*man*	*woman*	

Looking at this information, the distribution of the genders of the lecturers is not immediately clear. We need a better way of presenting such information *Imagine they were 1000 lecturers*. *Tabulation is one way of doing this without loosing any of the information detail.*

Table 2.1: Table Lecturers by Gender

Gender	Number of Lecturers
Men	7
Women	6

This is especially useful if the number of observations is large and the distinct categories are few. This leads us to the idea of **Contingency tables**. A contingency table is a convenient way of summarising data if we are dealing with more than one variable. A contingency table consists of row(s) and columns(s) of data. The rows and columns represent the variables.

An example of a 2 by 2 contingency table is given by

Table 2.2: An example of a 2 by 2 Contingency Table

	Column 1	Column 2
Row 1		
Row 2		

We can construct 3 by 3, 4 by 5 e.t.c., contingency tables, depending on the situation. Let us look at an example of how to construct a contingency table.

Example 2.2

Suppose in Example 2.1, more information on the lecturers was recorded as shown in Table 2.3.

Table 2.3: Lecturers by Gender by school

Lecturers	gender	School
1	man	Maths
2	man	Statistics and Actuarial
3	woman	Maths
4	man	Statistics and Actuarial
5	woman	Maths
6	woman	CSAM
7	man	CSAM
8	woman	CSAM
9	man	Maths
10	woman	Statistics and Actuarial
11	man	CAM
12	man	Statistics and Actuarial
13	woman	Statistics and Actuarial

We can put this information in a contingency table as shown in Table 3.4.

Table 2.4: Contingency Table of Gender by sector

		Gender		
		male	female	total
School	CSAM	2	2	4
	Maths	2	2	4
	Statistics and Actuarial	3	2	5
Total		7	6	13

Table 2.4 summarise the information in Table 2.3 more clearly, showing almost the same number of male and female lecturers in each school.

Contingency tables are sometimes referred to as cross tabulations or in some cases crosstabs.

For you to appreciate how useful these tables are try to imagine a situation where we were discussing all the lecturers in Africa in Example 2.1, there would be a lot of information to be studied, however, the contingency table may have the same rows and columns as Table 2.4 with bigger values in the cells.

2.2.2 Pie charts

A **pie chart** as the name suggests, is a circle divided into segments like a pie cut into pieces from the center outwards. Each segment represents one or more values taken by a variable. Such charts are used to display qualitative data. They become impractical if the number of categories become too many though, e.g., representing all the countries in the world.

Let us now look at an example, and see how we can construct and interpret a pie chart.

Example 2.3

The incomes of 10 services companies, were recorded in millions of Rands.

Incomes: 260 280 280 160 220 350 420 190 550 280

Expressing the incomes in groups, i.e. R150000000 and below, R150000000 $\leq$ R250000000, R250000000 $\leq$ R350000000, R350000000$\leq$R450000000, R450000000 and above, leads us to Table 2.5.

Table 2.5: Arranging Data into Groups

Income Group (in Millions of Rands)	Number
Below 150	0
150-250	3
250-350	5
350-450	1
Above 450	1

We now express these income groups as proportions or percentages (Table 2.6). There are only 4 groups, since there are no observations below $R150$

Table 2.6: Calculating Percentages for the Pie Chart

Income Group (in Millions)	Number	Proportion	Percentage
Below 150	0	$\frac{0}{0+3+5+1+1} = 0$	$0 \times 100 = 0$
$150 - 250$	3	$\frac{3}{0+3+5+1+1} = \frac{3}{10}$	$\frac{3}{10} \times 100 = 30$
$250 - 350$	5	$\frac{5}{0+3+5+1+1} = \frac{5}{10}$	$\frac{5}{10} \times 100 = 50$
$350 - 450$	1	$\frac{1}{0+3+5+1+1} = \frac{1}{10}$	$\frac{1}{10} \times 100 = 10$
$Above$ 450	1	$\frac{1}{0+3+5+1+1} = \frac{1}{10}$	$\frac{1}{10} \times 100 = 10$

Million. What we wish to do now is to represent these percentages as angles (^{o}), i.e., instead of everything in Table 2.6 (column 2) above adding up to 10 (or 100 in the case of percentages) we want them to add up to 360 (the total number of degrees in a circle). The pie chart in Fig 2.1 is constructed using the angles we obtained above. It is clear that half of

Table 2.7: Calculating Angles for the Pie Chart

Income Group (in Millions)	Angle
Below 150	$\frac{0}{100} \times 360 = 0$
$150 - 250$	$\frac{30}{100} \times 360 = 108$
$250 - 350$	$\frac{50}{100} \times 360 = 180$
$350 - 450$	$\frac{10}{100} \times 360 = 36$
Above 450	$\frac{10}{100} \times 360 = 36$

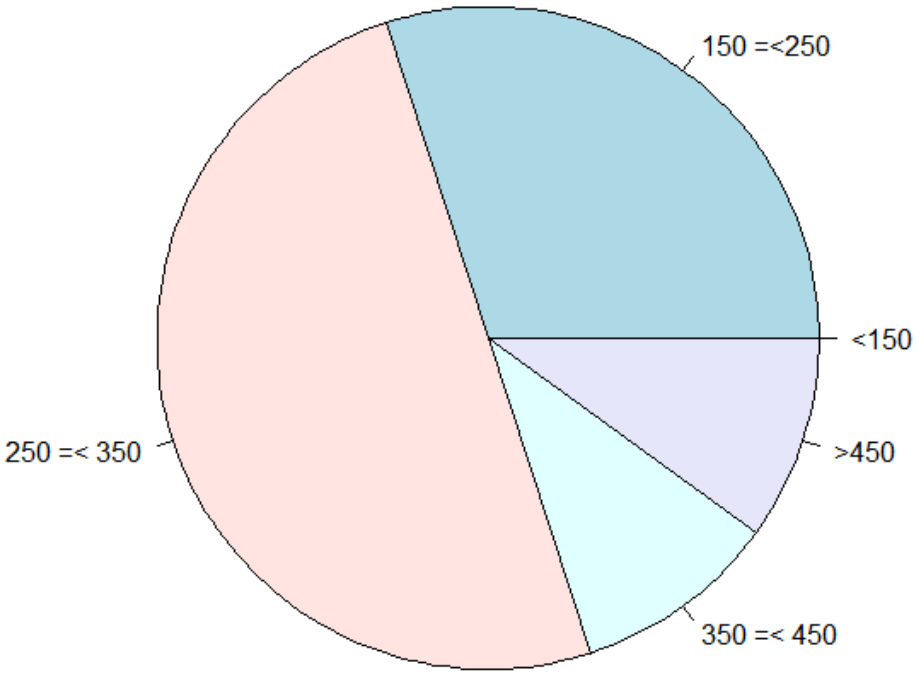


Figure 2.1: Pie chart of income groups

all the companies get incomes between 250 *and* 350 *million Rands. This is because they are accounting for half of the pie.*

The calculation of the angle of the ith *category can be done directly from the observations by using*

$$\frac{x_i}{\sum_{i=1}^{n} x_i} \times 360^o$$

i.e., where the x_i*s are the number of observations in each category.*

We often indicate the different categories by different shades on the pie chart. The pie chart gives us a 'picture' or impression of the **share** of each category in the data. From the pie chart we cannot tell the actual magnitude, but the **share** each income group contributes (unless the actual values are indicated).

2.2.3 Bar Chart

A **bar chart** [also known as a bar graph or a simple bar graph], as the name suggests, is a visual presentation of the data by means of **bars** or blocks put side by side. Usually the bars are separate from each other. Each bar represents a count of the different categories of the data. Although both pie charts and bar graphs are used to illustrate qualitative data or discrete quantitative data, bar charts use the actual counts or frequencies of occurrences of each category of data. We need not use the actual data, we can use the percentage to come up with the Bar graph. Let us use the data in Example 2.3 to illustrate the bar chart.

Example 2.4

We will now construct the bar chart using the data in Example 2.3. We come up with suitable scales for the height and width of the graph, which are such that the graph is clear and representative, i.e. relatively proportionate as in Fig. 2.1.

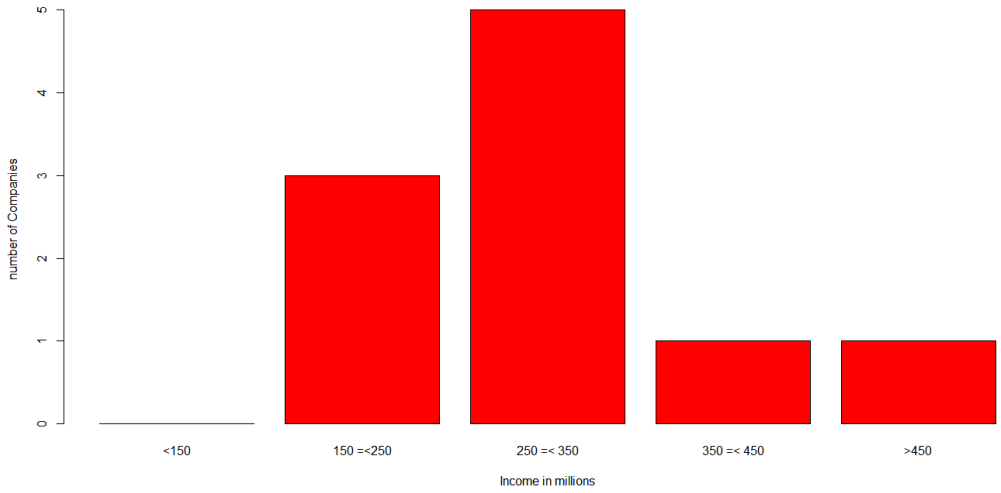


Figure 2.2: Bar chart of the Income groups of companies

The bars can be represented horizontal as well, see Figure 2.3. The bars represent each income group count. You can choose to make the bars thin or wide, its up to you all you need to be certain of is that the bars are of equal width. You can also use percentages instead of the counts.

Looking at the graph its clear that the 250 to 350 million Rand income group has the highest number of companies. The bars should also have the same shade. For the bar chart to be of reasonable size, the highest frequency or number should be close to the highest number on your vertical axis.

We some times need to compare different commodities and give a quick and clear picture of the information in the data. The Multiple bar graph is one useful tool in doing these comparisons.

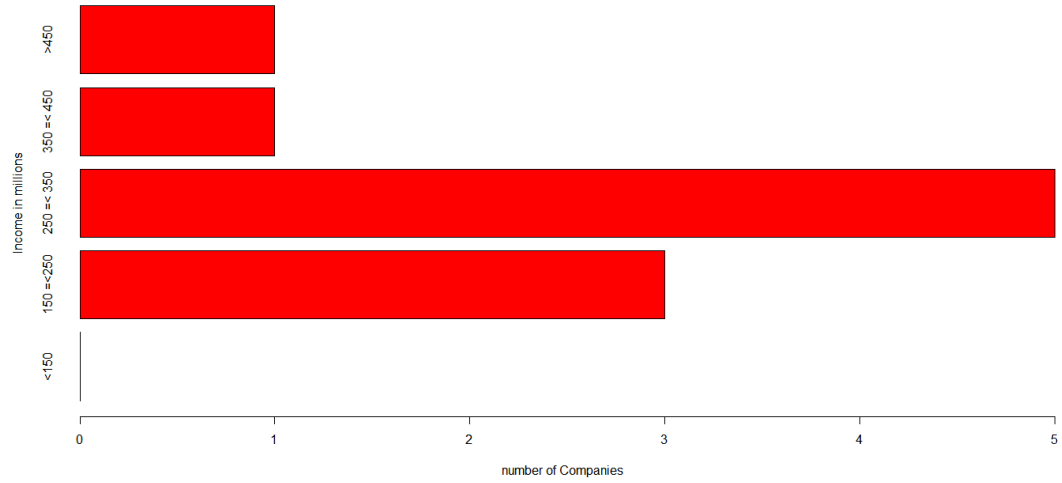


Figure 2.3: Bar chart of the Income groups of companies

2.2.4 Multiple Bar Graph

The **Multiple bar graph** is a number of bar graphs put on the same graph. It gives you the ability to make comparisons between products, quantities or years. In the multiple bar graph each entity to be compared is represented by a different shade or pattern. Let look at an example of how we create tables and construct multiple bar graphs.

Example 2.5

The exports of cotton, sugar and gold from a certain business from the year 2000 to 2003 were recorded as shown below.

Export product	*Year*	*Amount (in Rands)*
Cotton	*2000*	*R2,000,000*
Cotton	*2001*	*R1,861,000*
Cotton	*2002*	*R1,250,000*
Cotton	*2003*	*R985,000*
Gold	*2000*	*R1,300,000*
Gold	*2001*	*R2,100,000*
Gold	*2002*	*R2,800,000*
Gold	*2003*	*R3,100,000*
Sugar	*2000*	*R580,000*
Sugar	*2001*	*R610,000*
Sugar	*2002*	*R590,000*
Sugar	*2003*	*R600,000*

Construct a multiple bar graph to compare the exports of products in each year.

Solution

The first thing we do is to create a table of the exports as shown in Table 2.8. This makes it easy to read our data.

Table 2.8: Table of Exports

Year	Cotton	Gold	Sugar
2000	R2,000,000	R1,300,000	R580,000
2001	R1,861,000	R2,100,000	R610,000
2002	R1,250,000	R2,800,000	R590,000
2003	R985,000	R3,100,000	R600,000

Since the comparison is of products in each year we have the year on the bottom axis. We then construct simple bar graphs of each product for each year on the same graph as shown in fig 2.4.

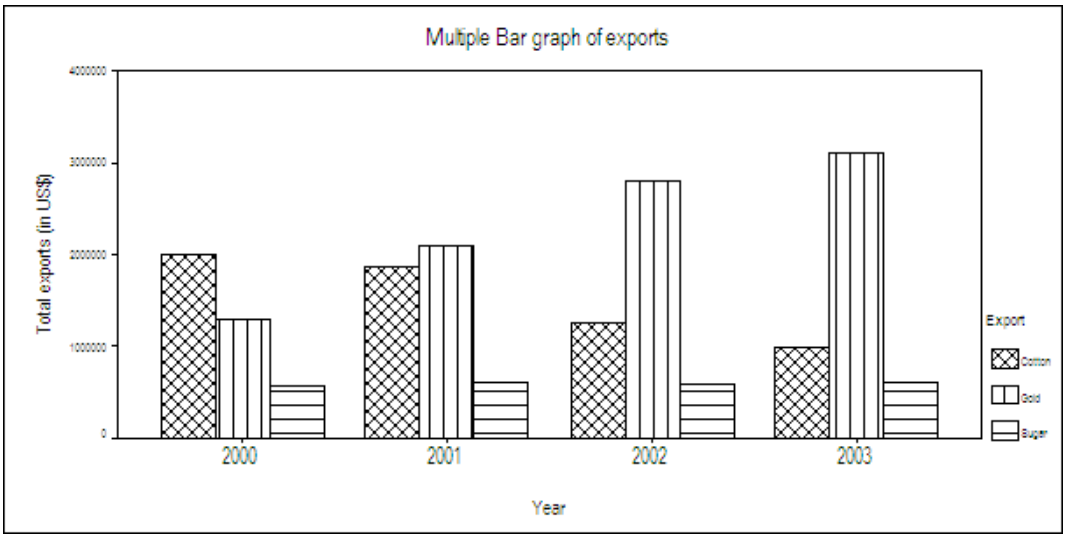


Figure 2.4: Multiple Bar graph of the Exports to compare within each year

In the year 2000 we had more cotton exports than any other product while sugar generated the least income. In the years 2001 to 2003 Gold over took cotton as the top earner and remained at the top.

Example 2.6

Using the data in Example 2.5, construct a multiple bar graph to compare the exports of each product, over the years.

Solution

This time the product are in the bottom axis as the question wants a comparison of each product over the years.

This multiple bar graph helps us explain what is going on within each product. We quickly notice that the cotton exports are on the decline over the years. Gold exports are on the increase and there is little or no change in the sugar exports.

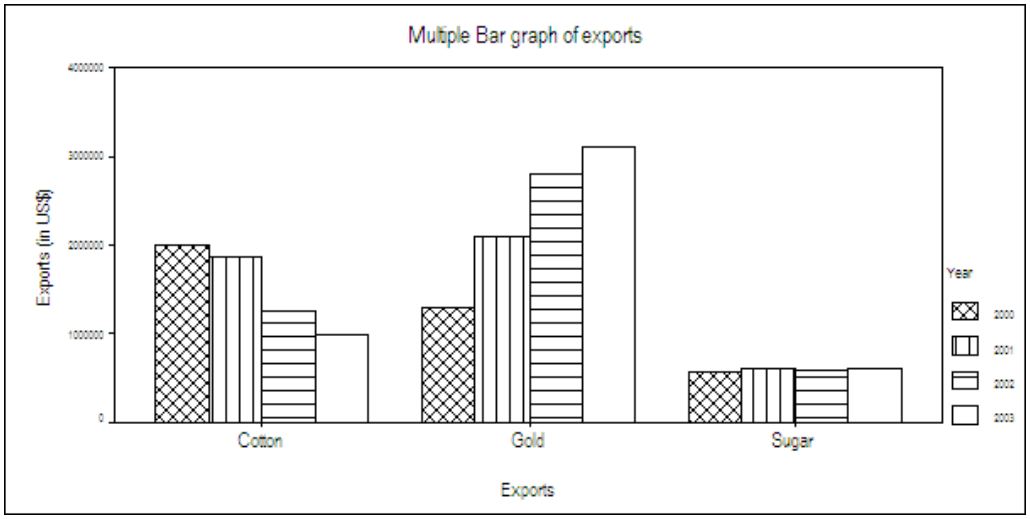


Figure 2.5: Multiple Bar graph of the Exports to compare within each product

Example 2.7

Using the data in Example 2.5, construct a simple bar graph of the exports over the years for each product.

Solution

In order to construct a simple bar graph of the annual exports we need to summarise the annual sales as shown in Table 2.9 below.

Table 2.9: Table of Summaries

Year	*Cotton*	*Gold*	*Sugar*	*Total*
2000	*R2,000,000*	*R1,300,000*	*R580,000*	*R3,880,000*
2001	*R1,861,000*	*R2,100,000*	*R610,000*	*R4,571,000*
2002	*R1,250,000*	*R2,800,000*	*R590,000*	*R4,640,000*
2003	*R985,000*	*R3,100,000*	*R600,000*	*R4,685,000*

Using the totals we construct the simple bar graph as shown in fig 2.6. From fig 2.6, we see that the exports are generally on the increase.

Charts help us answer and explain a the main characteristics of the data, without the need to browse through all the observations. Lets look at another variation in the bar chart family, the component bar graph.

2.2.5 The Component Bar Graph

The **component bar graph** also known as a stacked bar graph, allows you to examine the composition of several variables over time or some other entity. It is important to note at this stage that the variables you

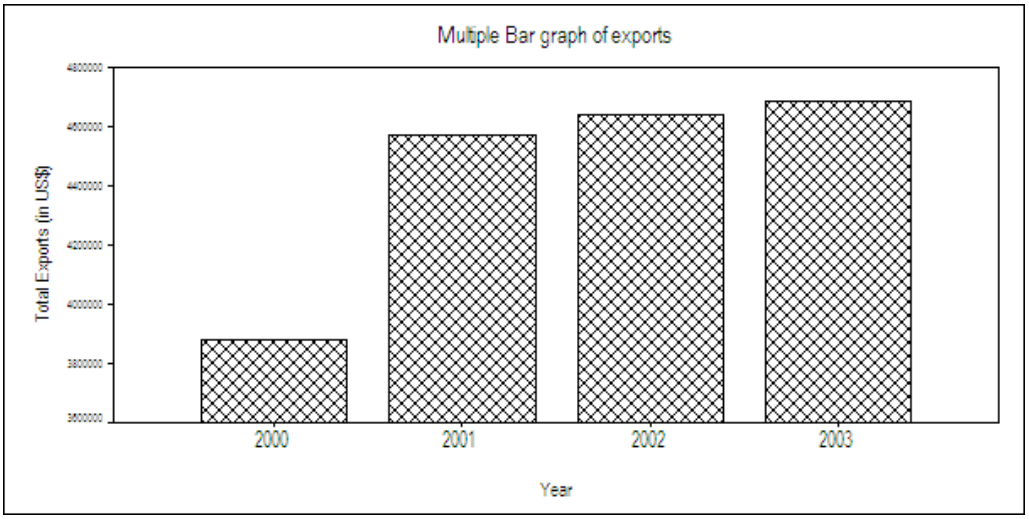


Figure 2.6: Simple Bar graph of the Exports to compare the Annual Exports

compare should have the same units of measurement. In the component bar graph we stack related simple bar graphs. Let us look at an example to "internalise" this graph.

Example 2.8 *In Example 2.5 construct a component bar graph to show how the exports of cotton, sugar and gold vary over time.*

Solution

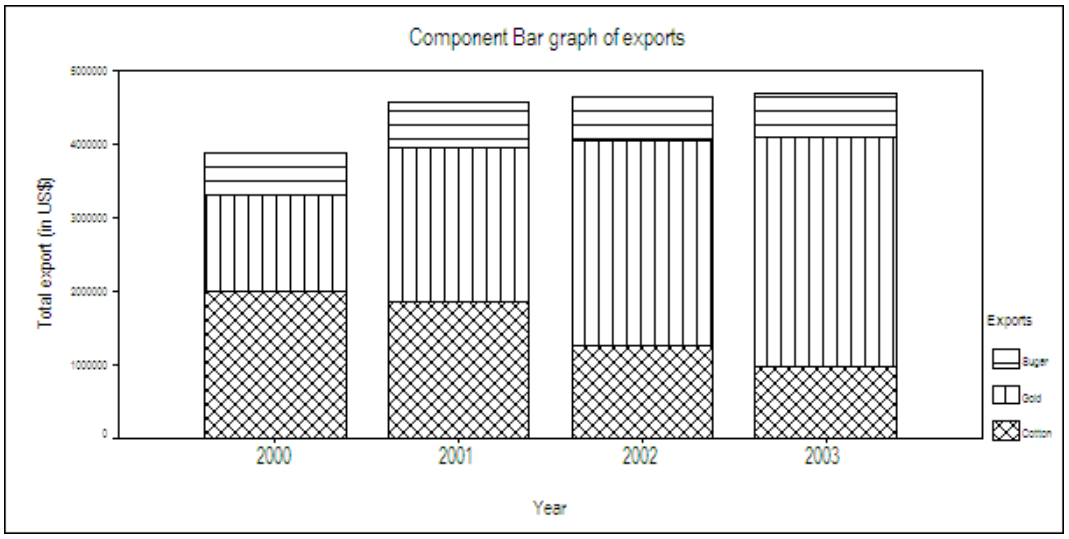


Figure 2.7: Component Bar graph of the Exports to compare the Annual Exports

Fig 2.7 helps compare the total annual exports of products. From this graph we can see that the total exports have been more or less static at around R4.6 Million.

The component bar graph has another advantage in that it can be used to compare the relative size of each variable's contribution (as in a pie chart). This can be done by constructing a percentage component bar graph.

2.2.6 The Percentage Component Bar Graph

The **percentage component bar graph** is a component bar graph constructed using the percentages instead of the observed values. Let us see how to construct one.

Example 2.9

Using the data in Example 2.5, construct a percentage component bar graph of the exports over time.

Solution

The first thing we do is to convert the observed values to percentages for each year as shown in Table 2.10.

Table 2.10: Calculating the percentages of exports

Year	Cotton	%	Gold	%	Sugar	%	Total	%
2000	R2,000,000	51.5%	R1,300,000	33.5 %	R580,000	14.9%	R3,880,000	100%
2001	R1,861,000	40.7%	R2,100,000	45.9%	R610,000	13.3%	R4,571,000	100%
2002	R1,250,000	26.9%	R2,800,000	60.3%	R590,000	12.7 %	R4,640,000	100%
2003	R985,000	21.0%	R3,100,000	66.2%	R600,000	12.8%	R4,685,000	100%

We use the calculated percentages to construct the percentage component bar graph as shown in fig 2.8. From the percentage component bar graph we can see that gold has been increasing its contribution to the country's exports.

2.2.7 Pictograms

You have no doubt met a pictogram. This is a common way of presenting statistics in newspapers, when they present statistics. In a **pictogram**, we represent groups of observations by symbols. For example when comparing populations, you may represent every 500,000 people by a symbol. For populations, the symbol is usually that of a small person, for cars one might chose to use a small car to represent a number of cars. Pictograms are perhaps the simplest graphs to understand. In a pictogram pictures or symbols are used to display statistical data. In a pictogram each picture represents a specified number of objects. The symbol or picture used depends on the kind of data under discussion. Each pictogram comes with a key that tells the reader what each symbols represents.

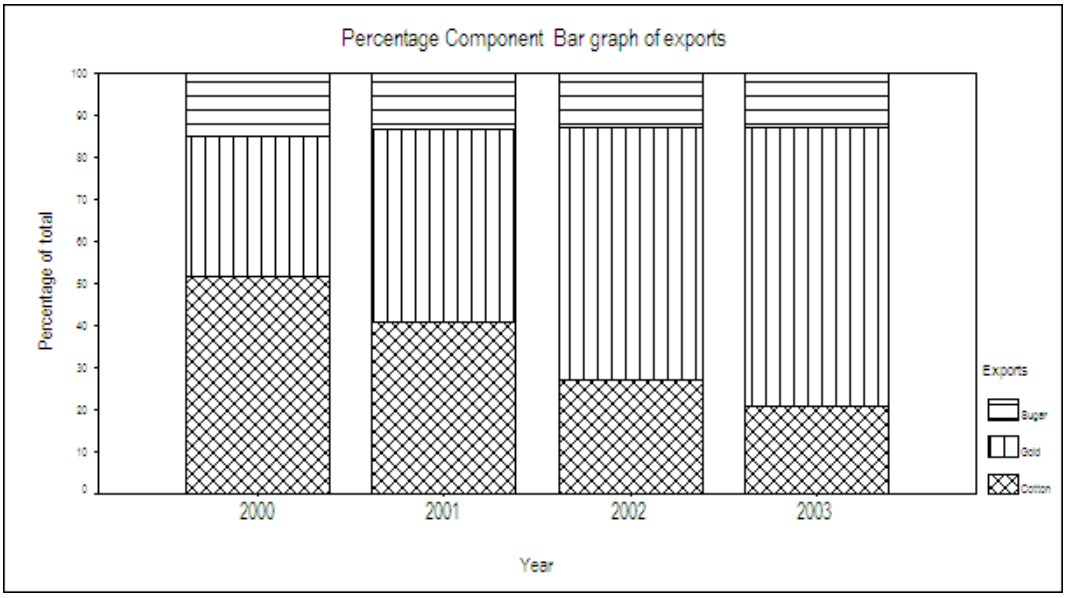


Figure 2.8: Percentage Component Bar graph of the Exports to compare the Annual Exports

Example 2.10

A company which manufactures shovels recorded the following sales in the period 1990 to 1994

Table 2.11: Number of shovels sold between 1990 and 1994

Year	1990	1991	1992	1993	1994
Number of Units sold	700000	500000	300000	400000	600000

Draw a pictogram of these sales.

Solution

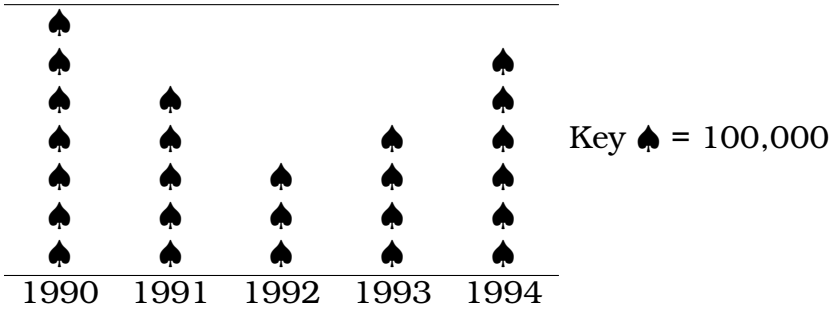


Figure 2.9: Pictogram of Garden Shovels sold

From the pictogram we see that the company had the highest sales in 1990 and a slump in 1992 but recovered in 1994.

The major disadvantage of pictograms is that they lack accuracy. For instance we cannot represent a figure like 345784 sales in our example.

2.2.8 Line Graphs

A line graph is a graph used to show the changes in a variable over time, it can also be used to investigate trends in two variables which are related. It basically consists of a line. You can superimpose two or more lines on one graph in order to compare their behaviour over time.

Example 2.11

An Economist who had been studying the rate of inflation observed the following rates from 1990 to 2005. He put forward a series of suggestions to reduce the rising inflation rate in 1996.

Year	*1990*	*1991*	*1992*	*1993*	*1994*	*1995*	*1996*	*1997*
Inflation rate	*2.3*	*3.4*	*4.3*	*5.6*	*6.7*	*8.7*	*9.6*	*13.0*

year	*1998*	*1999*	*2000*	*2001*	*2002*	*2003*	*2004*	*2005*
Inflation rate	*12.2*	*11.0*	*10.0*	*9.0*	*8.7*	*7.6*	*6.5*	*45*

Construct a line graph of the inflation rate figures. Do you think the economist's intervention worked?

Solution

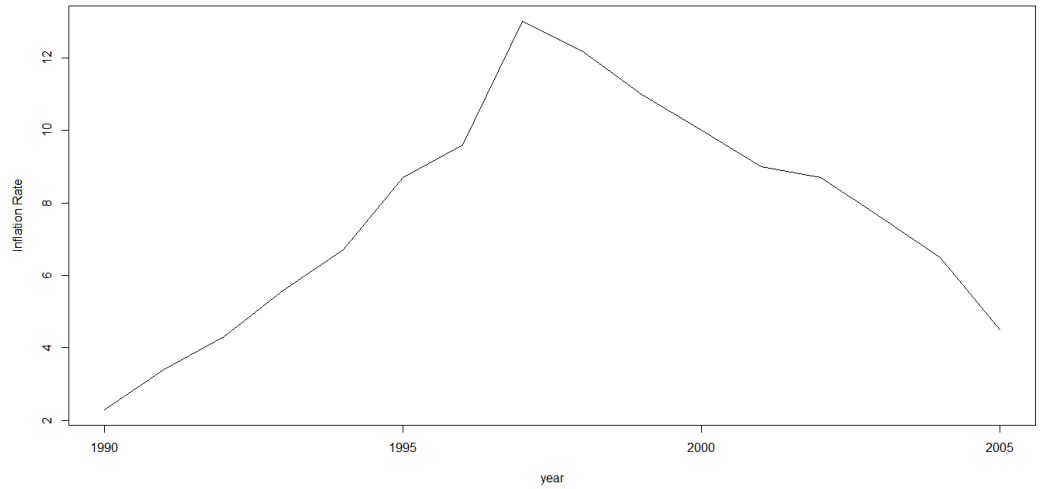


Figure 2.10: Line graph of inflation rate figures

The line graph suggests that inflation steadily rose to a peak in 1997 and started declining. Since the economist intervened in 1996 and nothing happened until 1997, we might argue that the effects of his measures can only be detected the year after they are implemented. We can conclude that perhaps his suggestions worked.

The kind of data that we have been summarising has been qualitative or categorical. This made it ease to group it and summarise it graphically. Let us consider a situation were quantitative data has been collected, say the amount of money you use on public transport on a daily basis. Typical amounts could vary from R0 to R200.00. For example 19 individuals gave the following responses.

Example 2.12

80.0 100.0 0.0 40.0 40.0 80.0 0.0 40.0 20.0 40.0
80.0 60.0 70.0 50.0 30.0 20.0 0.0 20.0 60.0

We can graphically present such data, now imagine 200,000 such observations. To get an understanding of what the data is telling us would not be so easy and straight forward in such cases. When we hav so much data we need to group it, and work with the frequencies of the groups to get a feel of the data.

In the next section we are going to discuss frequency based graphical methods. These are namely; Frequency distribution, Histogram, Polygon and Ogive curves.

2.2.9 Frequency Distributions

A frequency distribution is basically a table that displays the frequency of various outcomes. Thus, a **frequency distribution** is a listing of observation according to their frequencies of occurrence. Let us briefly look at an example.

Example 2.13

An employee of a service station has been asked to study the daily fuel sales around town. He wrote down the following daily sales.

120,000.00 120,000.00 130,000.00 130,000.00 130,000.00 120,000.00 120,000.00 140,000.00 130,000.00 140,000.00 150,000.00 120,000.00 140,000.00 160,000.00 150,000.00

Construct a frequency distribution of the prices.

Solution

The frequency distribution is given below.

Daily takings	*frequency*
120000	*5*
130000	*4*
140000	*3*
150000	*2*
160000	*1*

Sometimes the frequency distribution is not straight forward to construct as the observations are not easy to group as there maybe too many unique values. There is a common approach to constructing frequency distributions, this approach is outlined below.

To construct a frequency distribution we should follow the following steps.

1. Determine the smallest and largest numbers in the data.
2. Calculate the range of the data, which is the difference between the largest number and the smallest number.
3. Calculate $k = 1 + 3.22log(n)$ [Using Sturge's formula, where n is the number of observations]. k represents the number of classes
4. Estimate the approximate class size by dividing the range (largest value - lowest value) by k.
5. Determine the lower end of the first class making sure the smallest value is equal to or more than the lower end.
6. Determine the frequencies for each class by counting the number of observations falling in each class.

Lets look at an example of how to create a frequency distribution.

Example 2.14

The number of loan applications that are received everyday at a bank were recorded over a number days. The results are displayed below.

34	*30*	*54*	*38*	*42*	*26*	*38*	*38*	*52*	*38*	*32*	*32*	*34*	*48*	*50*
28	*34*	*50*	*28*	*34*	*34*	*32*	*40*	*56*	*34*	*34*	*34*	*50*	*30*	*38*
32	*46*	*48*	*42*	*32*	*56*	*40*	*48*	*68*	*22*	*52*	*44*	*46*	*40*	*44*
36	*48*	*38*	*34*	*34*										

Construct a frequency distribution of the data.

Solution

It is not helpful in this case to get the frequency of each value as there are to many unique values. So in such a case we create classes using Sturge's formula.

1. *The largest value is 68 and the smallest on is 22.*
2. *The range is 68-22=46.*
3. $k = 1 + 3.22log(n) = 1 + 3.22 \times log(50) = 1 + 3.22 \times 1.69897 = 6.47068 \approx 7$, *in this case we have 7 classes.*
4. *The approximate class size is* $k = \frac{46}{7} = 6.57142 \approx 7$
5. *Choose the starting point to be 22, this means the classes are 22 to 28, 29 to 35, etc.*

6. *The frequency distribution is given below.*

Class Interval	*frequency*
22 - 28	*4*
29 - 35	*17*
36 - 42	*12*
43 - 49	*8*
50 - 56	*8*
57 - 63	*0*
64 - 70	*1*

from our frequency distribution we can say on average the bank receives between 29 and 42 loan applications.

Remember to avoid choosing classes that will be hard to work with. You also need to remember the following terms, as they will keep appearing in our discussions;

1. *The* *class limits* *which are the starting and ending point of each class*

2. *The* *class boundary* *which is the average of the end of the current class limit and the beginning of the next class limit.*

3. *The* *class midpoint* *which is the average of the current class upper and lower limits*

Let us now proceed to discuss the frequency histogram.

2.2.10 Histograms

A **histogram** is a picture of a frequency distribution. It is used to represent continuous quantitative data. It usually consists of adjacent, rectangles that are not separated (touching). The area of each rectangle is drawn in proportion to the frequency corresponding to that frequency class. When the class intervals are equal, the area of each rectangle is a constant multiple of height and so the histogram can be drawn as for a bar chart, except that the rectangles are not separated. (Notice that the class intervals do not need to be equal.) When all class intervals are not equal, we can determine the height of each rectangle from the Equation 2.1.

$$area = width \times height \tag{2.1}$$

Example 2.15

Consider the results of a survey carried on 45 *shipping and forwarding companies. They were asked how long it took do ship heavy equipment*

Table 2.12: Shipping time of heavy duty equipment

Shipping time	Frequencies
10-19	7
20-29	20
30-39	9
40-49	3
50-59	5
60-69	1

from the far east. The results of their responses in days are shown in Table 2.12.

Note

In constructing a histogram the class intervals should be continuous this means you should not have discontinuous classes such as the one above, i.e., 10-19, 20-29, 30-39, e.t.c., there are gaps between these classes (for example where do you place 19.5 ?). When ever there are gaps between the classes, we have imaginary limits.

When ever the class boundaries and the class limits are not the same, then we know we have imaginary limits. **We never use imaginary limits in constructing Histograms**. We deal with imaginary limits by constructing real limits from the imaginary limits as shown below (i.e., averaging).

Shipping time	Frequencies	lower imaginary limit	Upper imaginary limit	Lower real limit	Upper real limit
10-19	7	10	19	$\frac{10+9}{2} = 9.5$	$\frac{19+20}{2} = 19.5$
20-29	20	20	29	$\frac{20+19}{2} = 19.5$	$\frac{29+30}{2} = 29.5$
30-39	9	30	39	$\frac{30+29}{2} = 29.5$	$\frac{39+40}{2} = 39.5$
40-49	3	40	49	$\frac{40+39}{2} = 39.5$	$\frac{49+50}{2} = 49.5$
50-59	5	50	59	$\frac{50+49}{2} = 49.5$	$\frac{59+60}{2} = 59.5$
60-69	1	60	69	$\frac{60+59}{2} = 59.5$	$\frac{69+70}{2} = 69.5$

We use the real limits to construct the Histogram so that we get Figure 2.11.

The number of objects falling in each class, i.e. the class frequencies, are sometimes found by using a **tally** or **score sheet**. These are symbols or strokes representing the counts. For example, we can represent each count by stroke, a frequency of 3 would be |||, 2 would be ||, we can represent every group of five say, as $\widehat{||||}$ so that 11 is expressed as $\widehat{||||}$ $\widehat{||||}$ |.

Exercise 2.1

1. *What is the difference between a histogram and a bar graph?*

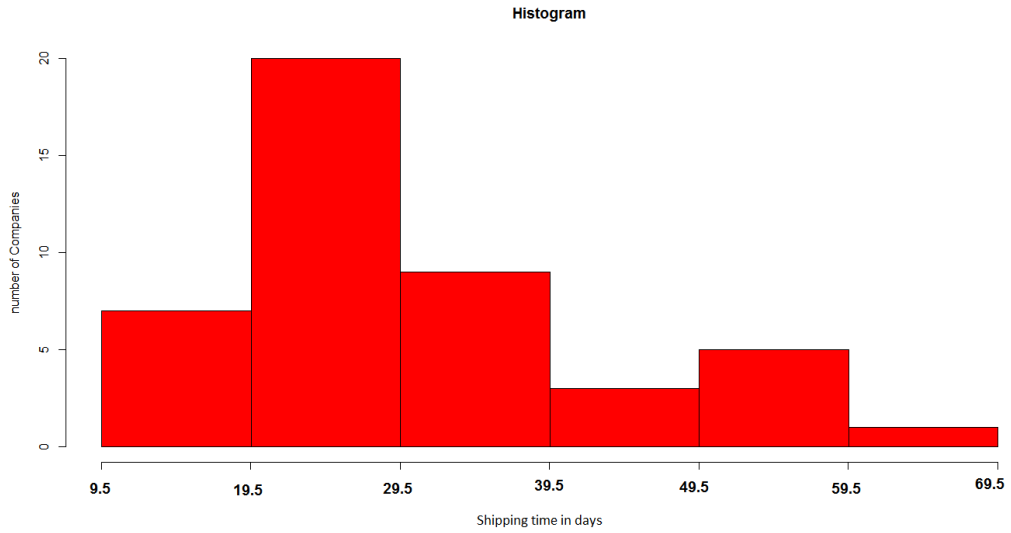


Figure 2.11: Histogram of shipping times

2. *Obtain a histogram for the data in Example 2.13.*

3. *Obtain a histogram for the data in Example 2.14.*

4. *Is the graph in Figure 2.12 a bar graph or a histogram?*

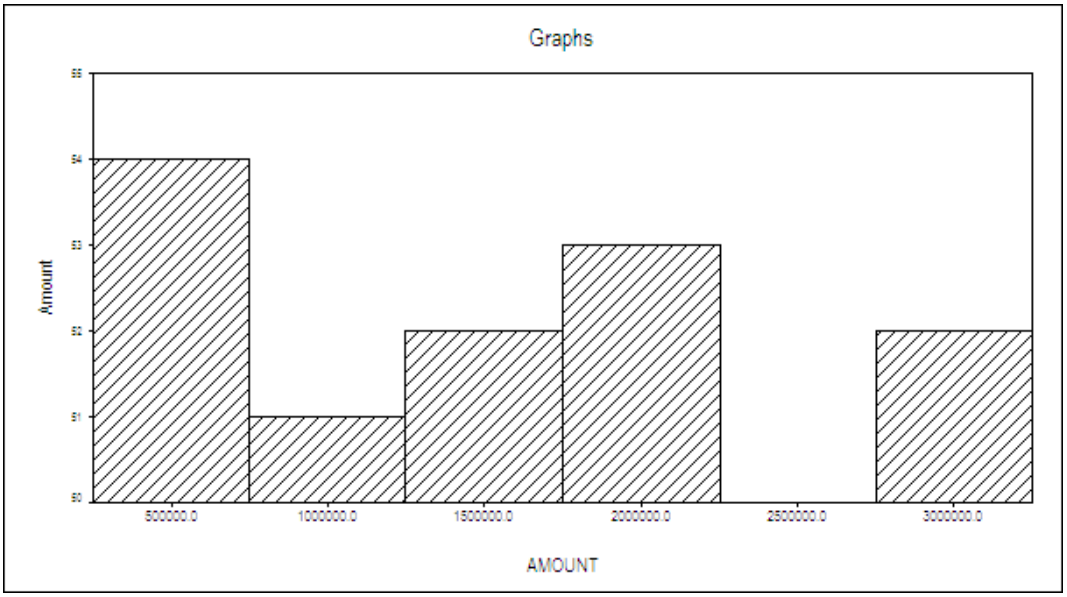


Figure 2.12: Unknown graph

2.2.11 Frequency Polygons

Frequency polygons are one alternative to histograms. The only difference here is that a **frequency polygon** is a line plot of the frequencies

against the corresponding class mid-points. The points are joined by straight lines. This graph helps us to comment on the distribution of our data.

Example 2.16

Construct a frequency polygon of the data in Example 2.15.

Solution

Using the mid points we get

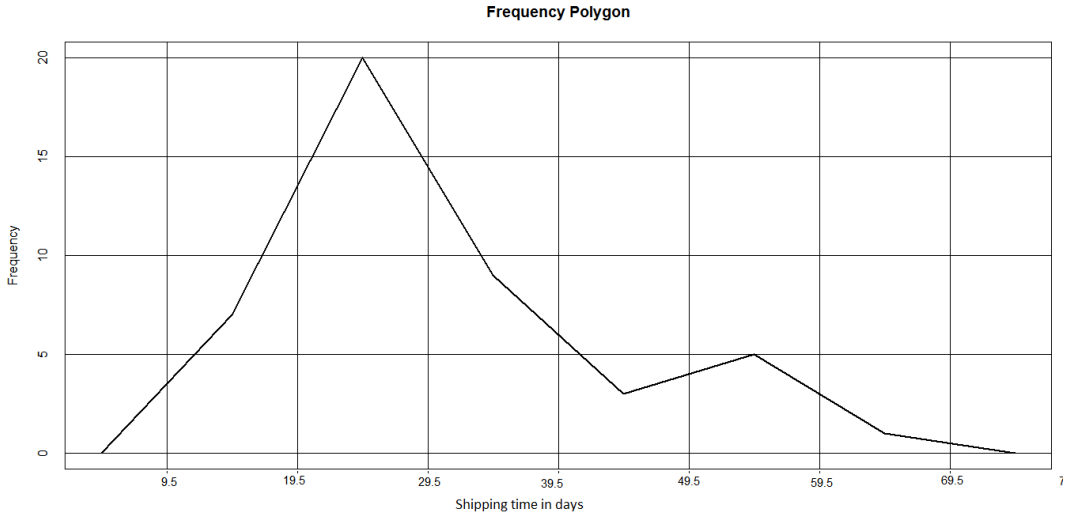


Figure 2.13: Frequency Polygon of the shipping times

2.2.12 Ogive Curves

An **ogive Curve** is a curve drawn using the sum of all previous frequencies at each point [also known as **cumulative frequency**], i.e., plot the lower limits of the classes against the cumulative frequencies.

If the frequency is divided by the total of all the available frequencies then the frequency becomes a **relative frequency**. If the relative frequencies are multiplied by 100 and used in an ogive, we get a **percentage ogive curve**. The beauty of a percentage ogive curve is that several statistics can be deduced from it, e.g., the 50% point on the $y-axis$ represents the **median** on the $x-axis$, (See Section 2.4.2)

If we accumulate the frequencies starting from the lowest class we get what we call a **less than ogive curve**, while accumulating the frequencies starting from the highest class leads to a **more than ogive curve**.

Example 2.17

In Example 2.16, construct the less than and more than ogive curves on the same plot.

Solution

Calculating the less than cumulative frequencies, w
e use the lower class limits to construct the less than ogive curve.

Shipping time	*Frequencies*	*Lower real limit*	*Upper real limit*	*Less than Cumulative frequencies*
$10-19$	7	9.5	19.5	7
$20-29$	20	19.5	29.5	27
$30-39$	9	29.5	39.5	36
$40-49$	3	39.5	49.5	39
$50-59$	5	49.5	59.5	44
$60-69$	1	59.5	69.5	45

Calculating the more than cumulative frequencies. We use the Upper class limits to construct the less than ogive curve.

Shipping time	*Frequencies*	*Lower real limit*	*Upper real limit*	*More than Cumulative frequencies*
$10-19$	7	9.5	19.5	45
$20-29$	20	19.5	29.5	38
$30-39$	9	29.5	39.5	18
$40-49$	3	39.5	49.5	9
$50-59$	5	49.5	59.5	6
$60-69$	1	59.5	69.5	1

When the two ogive curves in Examples 2.15 are combined they should be symmetric and meet at the median at around 27.7, see Figure 2.14.

There are many other graphical techniques which, we have not mentioned here, these include **Lorenz curves Z curves** and **Philips curves**, etc. Most of these, are met in branches of statistics like **Official statistics**.

Exercise 2.2

1. *A sample of 10 new men working in a mine was collected during an investigation into allegations that the mine only employs men who are more than 25years old.*

   19.0 30.0 29.1 20.6 27.9 26.9 23.3 31.1 32.3 24.8 29.9 21.5 26.6

   (a) *Classify these ages into four classes, namely class A: ages below 22.5, B: ages between 22.5 and 25 inclusive, C: ages between 25 and 27.5 years inclusive, and D: above 27.5 years.*

   (b) *Construct a pie chart from these classes.*

   (c) *Construct a bar chart from these data.*

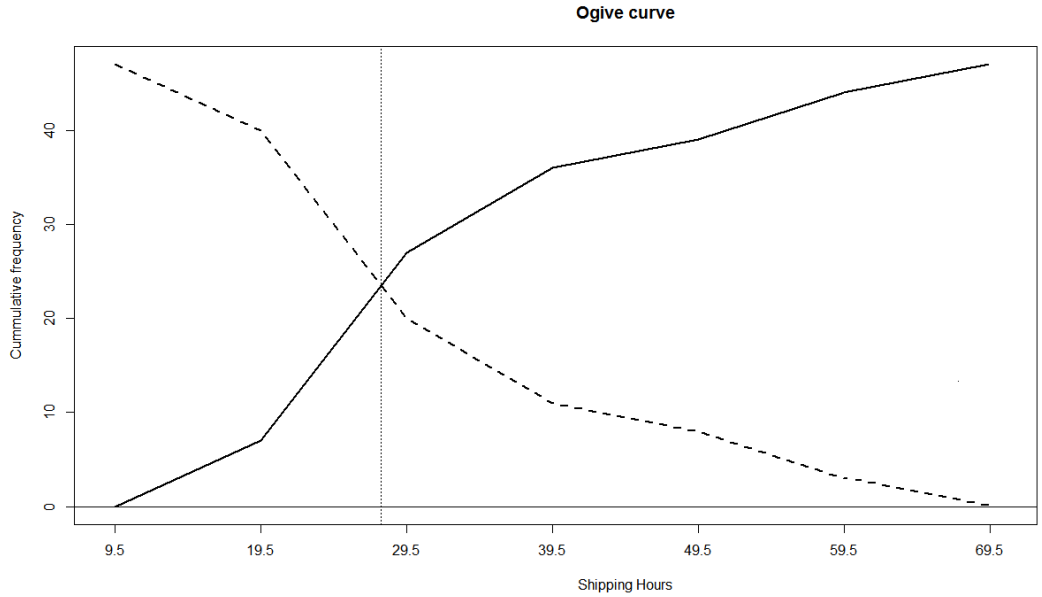


Figure 2.14: The more than and less than ogive curve of the of shipping times

(d) *Is it possible to construct a stem-and-leaf diagram for the original data?*

(e) *Comment on the allegations.*

2. *What is the advantage of using a stem-and-leaf diagram over the histogram?*

3. *Given the following frequency distribution.*

Class	f_i
5-9	*1*
10-14	*9*
15-19	*20*
20-24	*12*
25-29	*5*

(a) *What is the sample size?*

(b) *What is the class size?*

(c) *What are the boundaries of the third class?*

(d) *What is the midpoint of the second class?*

(e) *Obtain the real limits.*

4. *A man has two shops shop 1 and shop 2, he wanted to compare the sales at the two shops, and collected the following monthly sales in Millions.*

shop 1			*shop 2*		
155	*115*	*156*	*156*	*109*	*127*
150	*159*	*163*	*148*	*135*	*119*
172	*143*	*159*	*140*	*127*	*115*
166	*148*	*175*	*122*	*99*	*106*
151	*161*	*138*	*171*	*123*	*135*
148	*129*	*135*	*125*	*107*	*152*
140	*152*	*139*	*111*	*137*	*161*

(a) *Draw the two stem and leaf display for the data sets.*

(b) *Do the stem and leaf displays, suggest that the shops are performing equally?*

5. *The monthly sales (in millions of dollars) of a small business are given below.*

149	*148*	*189*	*167*
380	*170*	*216*	*155*
280	*655*	*250*	*235*
221	*950*	*750*	*912*
510	*565*	*215*	*842*

6. (a) *Construct a frequency distribution.*

(b) *Draw a histogram.*

(c) *Compute the more than and less than cumulative frequencies.*

(d) *Draw the less than and more than cumulative curves on the same graph and hence determine the median.*

7. *A company has been selling two types of cars A and B from 1992 to 1998. The number of sales obtained (in billions of dollars) is given below.*

Year	*A*	*B*
1992	*134*	*119*
1993	*126*	*96*
1994	*198*	*182*
1995	*144*	*98*
1996	*164*	*78*
1997	*200*	*197*
1998	*213*	*187*

Draw line graphs of the sales of the two data sets on the same plot and comment on your findings.

2.3 Summary Statistics

In the previous sections we discussed methods of describing data by considering their distribution, using tabulations and graphs. Whilst such techniques are useful for **exploratory data analysis**, [describing data using summary measures] we also need to be able to summarise our data in a more precise manner. For this purpose we use summary statistics, which are numerical measures characterising the distribution of the data.

The distribution of a quantitative variable is generally described in terms of

1. its position or location, centrality, and
2. its spread or dispersion, variation

and we use summary statistics to describe these features of a distribution. We will summarise all these statistics using a diagram which we refer to as a **Box plot** or a **Box and Whisker plot**.

2.4 Measures of location

For quantitative variables, the position or location of the distribution can be characterised by its center, or central tendency. Qualitative variables do not, of course, have a center. The center or average of a quantitative variable can be summarised by using any of the three measures given below.

2.4.1 The mean [or more precisely the arithmetic mean]

The mean is defined as the sum of all observations, divided by the number of observations.

Suppose we have n observations denoted by x_1, x_2,$\cdots$, x_n, then

$$mean = \overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

for non-frequency data,[data which is not grouped for example 1.1, 2.2, 6.5, 4.0]. If on the other hand we have grouped data, with x_i occurring f_i times with a total of n observations then

$$mean = \overline{x} = \frac{1}{n}\sum_{i=1}^{k} f_i x_i, \qquad n = \sum_{i=1}^{k} f_i$$

for frequency data.

Example 2.18

A scientist interested in finding out the age groups of people interested in cultural movies, went to a movie theatre and collected the following information.

Ages of folks watching the movie

7	9	11	12	12	12	13	13	14	14	14	14	15	15	15	16
17	18	18	19	19	19	20	20	20	21	22	22	22	23	24	24
25	26	28	29	31	31	32	34	38	39	39	16	25			

Let us start by calculating the mean of the above data.

The mean $\overline{x}$*,*

$$\begin{aligned} \overline{x} = & \ \tfrac{1}{45}[7+9+11+12+...+39+39] \\ = & \ \tfrac{1}{45}[927] \\ = & \ 20.6 \end{aligned}$$

Let us look at another example, this time for grouped data.

Example 2.19

In order to understand the background of listeners to a certain radio program the sponsors of the program decided to carry out a small survey. They sent out 2000 questionnaires to listeners, and requested information regarding their income per hour [on average]. The results they obtained are summarised below:

Income per day [R]	*Frequency*
$R0$	*1235*
$R1 - R50$	*459*
$R51 - R100$	*121*
$R101 - R200$	*29*

Calculate the mean income per day of this sample.

Solution

Total is $1235+459+121+29 = 1844$*, so some listeners did not respond. The mean is;*

$$mean = \overline{x} = \frac{1}{n}\sum_{i=1}^{k} f_i x_i = \frac{1235 \times 0 + 25.5 \times 459 + 75.5 \times 121 + 150.5 * 29}{1844} = 13.67.$$

That is R13.67 per day, a low value because most of the listeners have no income. Note that, we used the mid points of the classes

2.4.2 The median

Suppose we sort all our observations into numerical order ranging from lowest to highest. The **median** is the middle value in the sorted list. Half of all the observations will be greater than, the median and the

other half, less than it. If the mean is not situated at the center of the distribution, i.e, if a histogram or a bar chart of values is not symmetric about the median, we say the distribution is **skewed**. **Skewness** is thus, a measure of symmetry (see later discussion in Section 2.4.5;) in a perfect situation we expect values to be just as likely to be less than the mean, as they are likely to be greater than the mean. The median is also known as the 50th **percentile** or the **second quartile**.

Our median $\tilde{x}$ is calculated by first sorting the data in ascending order to get the new data array $x_{(1)}$, $x_{(2)}$, $\cdots$, $x_{(n)}$, (this is the notation used to indicate that x_1, x_2 $\cdots$, x_n have been ordered) and then getting the most central value. If n is odd then our median is the $\frac{n+1}{2}^{th}$ value i.e.,

$$\tilde{x} = x_{(\frac{n+1}{2})}$$

while on the other hand an even n gives the median as

$$\tilde{x} = \frac{1}{2}[x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}]$$

Example 2.20

Calculate the median of the data in Example 2.18.

Solution

In this particular case our data is already sorted and an odd number, so our median is given by;

$$\begin{aligned}\tilde{x} &= x_{(\frac{45+1}{2})} \\ &= x_{(23)} \\ &= 19\end{aligned}$$

The computation of the median for grouped data is a bit more involved that of its ungrouped data counterpart. It is given by;

$$Median = L_m + \frac{c_m(\frac{n}{2} - F_{m-1})}{f_m} \tag{2.2}$$

where

L_m = lower limit of the class containing the median
c_m = difference between the upper end and lower end of the median class
f_m = frequency of the median class
F_{m-1} = cumulative frequency of the class just before the median class
n = $\sum_{i=1}^{k} f_i$ = the sum of frequencies
k = number of classes

When calculating the median for grouped data it is important to remember that we use real limits or the **class boundaries**. If the limits are already real we use them as they are.

Example 2.21

The time it takes to build a three-roomed house is believed to be at most 12 weeks. The man in charge of time and service delivery for a building company took a random sample of three-roomed house constructions and inquired how long it took to build them. He obtained the data below.

Time in weeks	*Freq*
$5 \leq 7$	*5*
$8 \leq 10$	*20*
$11 \leq 13$	*45*
$14 \leq 16$	*10*
$17 \leq 19$	*6*
$20 \leq 22$	*4*

Calculate the median time it takes to construct a three-roomed house.

Solution

The upper class limit (7) of the first class is not the same as the upper class boundary ($\frac{7+8}{2} = 7.5$) of the first class. This means we need do calculate the real limits. (This should also be done for Histograms and ogive curves. The mean gives the same results whether you convert the imaginary limits to real limits or not. It is the median, mode, Histograms and ogive curves which are affected by imaginary limits).

Time in weeks	*Real limits*	*Freq (f_i)*	*Cum freq F_i*
$5 \leq 7$	$4.5 < 7.5$	*5*	*5*
$8 \leq 10$	$7.5 < 10.5$	*20*	*25*
$11 \leq 13$	$10.5 < 13.5$	*45*	*70*
$14 \leq 16$	$13.5 < 16.5$	*10*	*80*
$17 \leq 19$	$16.5 < 19.5$	*6*	*86*
$20 \leq 22$	$19.5 < 22.5$	*4*	*90*

The class containing the median is 10.5 - 13.5 since n=90 and $\frac{90}{2} = 45$ falls in the median class. $L_m = 10.5$, $c_m = 13.5 - 10.5 = 3$, $f_m = 45$, $F_{m-1} = 25$ *and*

$$\begin{aligned} Median &= L_m + \frac{c_m(\frac{n}{2} - F_{m-1})}{f_m} \\ &= 10.5 + \frac{3(\frac{90}{2} - 25)}{45} \\ &= 11.833 \end{aligned}$$

Thus median is 11.833. This means the claim of at most 12 weeks may be true.

Note

You should always check that the median you calculated falls in the median class. If it does not then you know you made an error.

2.4.3 The Mode

The mode is that observation with the largest frequency for ungrouped data. The data is said to have no mode if all the observations are unique, that is they only occur once in the data or the a good number of them occurs with the same high frequency. It is also possible to have more than one mode as in the following data.

With grouped data, we do not have a single most frequently occurring value. However, we can have a class with a high frequency. If indeed a mode exits we expect to find it in the class with the highest frequency.

The mode for grouped data is estimated on the basis if this argument, thus;

$$Mode = L_m + \frac{c_m(f_m - f_{m-1})}{2f_m - (f_{m-1} + f_{m+1}} \tag{2.3}$$

where

L_m = lower end of the modal class
c_m = upper end of modal class lower end of modal class
f_m = frequency of the modal class
f_{m-1} = frequency of the class immediately below the modal class
f_{m+1} = frequency of the class immediately above the modal class

Example 2.22

Using data in Example 2.21, calculate the mode.

Solution

Time in weeks	*Real limits*	*Freq (f_i)*
$5 \leq 7$	$4.5 < 7.5$	*5*
$8 \leq 10$	$7.5 < 10.5$	*20*
$11 \leq 13$	$10.5 < 13.5$	*45*
$14 \leq 16$	$13.5 < 16.5$	*10*
$17 \leq 19$	$16.5 < 19.5$	*6*
$20 \leq 22$	$19.5 < 22.5$	*4*

The modal class is 10.5 - 13.5. that is the class with the highest frequency. $L_m = 10.5$, $c_m = 13.5 - 10.5 = 3$, $f_m = 45$, $f_{m-1} = 20$ *and* f_{m+1}= *10.*

$$\begin{aligned} Mode &= L_m + \frac{c_m(f_m - f_{m-1})}{2f_m - (f_{m-1} + f_{m+1})} \\ &= 10.5 + \frac{3(45 - 20)}{2(45) - (10 + 20)} \\ &= 11.25 \end{aligned}$$

The modal construction time is 11.25 weeks.

For any set of data these three summary measures of the center or location of the distribution are likely to be different; thus we have to decide which is the best one to use in any given situation.

2.4.4 Quantiles/Quartiles/percentiles

Whilst the center of the data, as described by the mean, median and/or mode, it is sometimes useful also to summarise other specific points of location of the data. Suppose we sort or rank the data values in ascending order. The values can then be partitioned into equal sized portions with dividing points called **quantiles**.

Quantiles are used to describe the percentage or proportion of observations lying above or below a certain level. The main quartiles (when the data is divided into four parts ("quarters")) we usually deal with are;

1. 1st quartile/25% percentile:
   The value below which 25% of the observations lie

2. 2nd quartile/50% percentile, the median:
   The value below which 50% of the observations lie and above which the other 50% of the observations lie

3. 3rd quartile/75% percentile:
   The value above which 75% of the observations lie.

For ungrouped data the 1st quartile is calculated by first ordering the data $x_1,\ x_2,\ \cdots,\ x_n$, and then using the formula

$$1^{st}quartile = x_{(|\frac{(n+1)}{4}|)} + \frac{1}{4}\left[x_{(|\frac{(n+1)}{4}|+1)} - x_{(|\frac{(n+1)}{4}|)}\right]$$

The notation $|\quad|$ here means the integer you get after dividing $\frac{(n+1)}{4}$ and discarding the remainder. For example $|\frac{(13)}{4}| = 3$ and $|\frac{(27)}{4}| = 6$.

The 75% percentile is calculated in a similar manner using the formula

$$3^{rd}quartile = x_{(|\frac{3}{4}(n+1)|)} + \frac{3}{4}\left[x_{(|\frac{3}{4}(n+1)|+1)} - x_{(|\frac{3}{4}(n+1)|)}\right]$$

Example 2.23

8 people were searched at a bus stop in search of a pick pocket who had stolen $R550.00$ in weekly wages from another gentleman at the same bus stop. The amounts in Rs they were found with are;

20.12, 20.20, 2000.00, 0.20, 10.00, 122,13, 0, 23.90

Ordering gives the new sample:

0, 0.20, 10.00, 20.12, 20.20, 23.90, 122.13, 2000.00

in this case $n = 8$*, thus*

$$\begin{aligned} 1^{st}\ quartile &= x_{(\lfloor\frac{(8+1)}{4}\rfloor)} + \tfrac{1}{4}\left[x_{(\lfloor\frac{(8+1)}{4}\rfloor+1)} - x_{(\lfloor\frac{(8+1)}{4}\rfloor)}\right] \\ &= x_{(2)} + \tfrac{1}{4}[x_{(3)} - x_{(2)}] \\ &= 0.20 + \tfrac{1}{4}[10.00 - 0.20] \\ &= 2.65. \end{aligned}$$

The 75% *percentile is calculated in a similar manner to give* $R97.57$*.*
There are other techniques used to calculate the quartiles, the formulae may differ slightly.

The quantiles for grouped data are calculated much like we did the grouped data median. The following formula are used to calculate the qth percentile.

$$q^{th}\ percentile = L_p + \frac{c_p(\frac{q\times n}{100} - F_{p-1})}{f_p} \tag{2.4}$$

where

L_p = lower limit of the class containing the qth percentile.
c_p = difference between the upper end and lower end of qth percentile class
f_p = frequency of the kth percentile class
F_{p-1} = cumulative frequency of the class just before the qth percentile class
n = $\sum_{i=1}^{k} f_i$ = the sum of frequencies
k = number of classes

This means to calculate the first quartile we use q=25 so that

$$25^{th}\ percentile = Q_1 = L_q + \frac{c_q(\frac{n}{4} - F_{p-1})}{f_q}$$

where

L_q = lower limit of the class containing Q_1.
c_p = difference between the upper end and lower end of Q_1 class
f_p = frequency of the first quartile class
F_{p-1} = cumulative frequency of the class just before the first quartile class
n = $\sum_{i=1}^{k} f_i$ = the sum of frequencies
k = number of classes

Note

For these quantiles we always use the class boundaries or real limits.

Example 2.24

Using data in Example 2.21, calculate the 75th *percentile.*

Solution

Time in weeks	*Real limits*	*Freq* (f_i)	*Cum freq* F_i
$5 \leq 7$	$4.5 < 7.5$	*5*	*5*
$8 \leq 10$	$7.5 < 10.5$	*20*	*25*
$11 \leq 13$	$10.5 < 13.5$	*45*	*70*
$14 \leq 16$	$13.5 < 16.5$	*10*	*80*
$17 \leq 19$	$16.5 < 19.5$	*6*	*86*
$20 \leq 22$	$19.5 < 22.5$	*4*	*90*

In this case p=75, so we go to Equation 2.4 and make the substitutions shown below. We also note the 75th *percentile is* Q_3 *so that* $\frac{75}{100} \times 90 = \frac{3}{4} \times 90 = 67.5$ *which means our* 75th *percentile lies in the third class.*

$$\begin{aligned} 75^{th}\ percentile = Q_1 &= L_p + \frac{c_p(\frac{75}{100} \times 90 - F_{p-1})}{f_p} \\ &= L_p + \frac{c_p(\frac{3}{4} \times 90 - F_{p-1})}{f_p} \\ &= 10.5 + \frac{3(\frac{90 \times 3}{4} - 25)}{45} \\ &= 13.333 \end{aligned}$$

The other measures of position are **Skewness** and **Kurtosis**. These are explained below.

2.4.5 Skewness

Skewness is a measure of the shape of the data. skewness is a measure of the asymmetry of the probability distribution of a random variable. Roughly speaking, a distribution has **positive skew** (right-skewed) if the right tail is longer and **negative skew** (left-skewed) if the left tail is longer.

If the median is not situated at the center of the distribution, i.e., if the histogram of values is not symmetric about the median, we say the distribution is **skewed**. Skewness is thus, a measure of symmetry. In a perfect situation we expect values to be just as likely to be less than the median, as they are likely to be greater than the median, i.e., symmetrical about the median or mean. Figure 2.15 shows the relationships between the mean, median and mode in skewed and symmetric distributions. Consider the sample $X_1,\ X_2,\ ...,\ X_n$, then one measure of

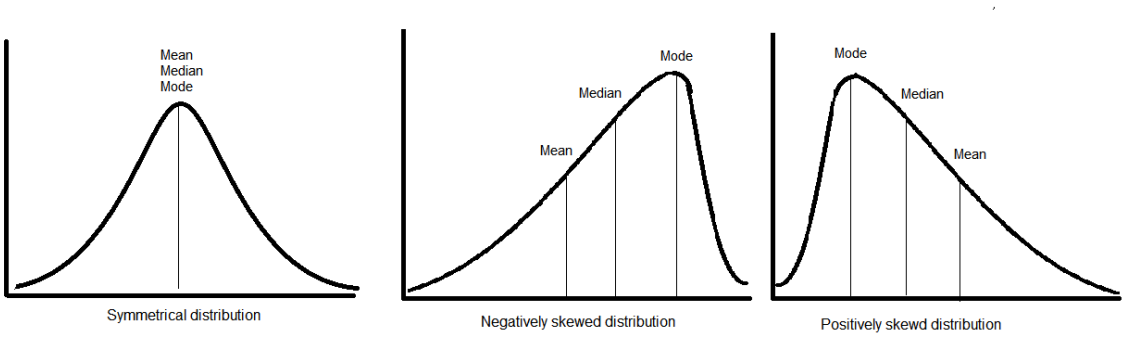


Figure 2.15: Examples of skewed distributions

skewness is the third centered moment m_3 where

$$m_3 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \overline{x})^3, \quad \text{where } \overline{x} \text{ is the mean}$$

If m_3 is negative then we say the data is skewed to the left or negatively skewed, on the other hand, if it is positive we say the data is positively skewed or skewed to the right [has a longer **right tail**].

The actual value of m_3 does not say much because it depends on the scales of measurement. One way of making this value scale free is to divide it by the cubed standard deviation, and use

$$m_3' = \frac{m_3}{s^3},$$

the actual value, also known as the *standardised skewness* measure m_3'. The standardised skewness measures m_3' can be unreliable when an unusual data set is encountered, such as one containing extreme values.

Karl Pearson suggested two simpler calculations as a measure of skewness. These are:

$$P_{s1} = \frac{(mean - mode)}{standard\ deviation} \tag{2.5}$$

$$P_{s2} = \frac{3(mean - median)}{standard\ deviation} \tag{2.6}$$

although there is no guarantee that these will be the same sign as each other they do help us decide on the nature of the data regarding its skewness. As these measures are easier to work with, we will use these for future calculations.

A value of P_s less than 0 indicates negative skewness in the data, while a value of P_s more than 0 indicates positive skewness.

2.4.6 Kurtosis

Kurtosis measures the peakedness of a data set. If a data set has a high kurtosis value [which we shall call m_4] then the frequency histogram will have a high peak. Another way to explain Kurtosis is to say there is a great number of observations around the mode or the mode has a high frequency, if it exists. Mathematically the fourth centered moment measures Kurtosis m_4 and is given by

$$m_4 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^4, \quad \text{where } \overline{x} \text{ is the mean.}$$

We can standardise m_4 in order to get a scale free value which can be used in comparisons, we use

$$m_4' = \frac{m_4}{s^4}.$$

As a guide, data with a kurtosis value equal to 3 is said to have a **Mesokurtic distribution** that is, it is peaked like the normal distribution. Data with a kurtosis greater than 3 is said to have a **Leptokurtic distribution**, that is, it is more peaked than a normal distribution with values concentrated around the mean. When data has a kurtosis less than 3 then it is said to have a **Platykurtic distribution**, which means it is flatter than a normal distribution with a wider peak. The values are wider spread around the mean.

2.5 Box-and-Whisker plot

The last summary statistic that we are going to discuss, is by way of a diagram. This diagram is called a **Box plot** or a **Box-and-Whisker plot**. This diagram was not mentioned earlier, with the other diagrams because, it is constructed using the summary statistics which we have just discussed, as opposed to the former which use the raw data.

This diagram shows all the quartiles, the median, and maximum and minimum values. In this way the plot illustrates where the center of the distribution is and how much variation there is around the center. **Outliers** may also be highlighted here.

The features of the Box-plot are;

1. The box holding [50%] of the observations.
2. The whiskers [they are like cat whiskers] are lines which extend from the box to the maximum and minimum values.
3. The median, quartiles and outliers if any.

Let us look at an example to see how we can construct a Box plot and use it.

Example 2.25

The rate at which accidents occurred at a road junction controlled by a 'yield' signs were recorded, they were also recorded after 'traffic' lights were erected in place of the 'yield' signs. These results are recorded below. Construct a Box plot for this data [before and after the traffic lights were erected] and comment on the distribution of the accidents. Comment on the mean number of accidents before and after the changes.

Year	*Before*	*After*	*Year*	*Before*	*After*	*Year*	*Before*	*After*
1	*395*	*280*	*6*	*548*	*287*	*11*	*600*	*233*
2	*425*	*284*	*7*	*498*	*323*	*12*	*358*	*251*
3	*415*	*271*	*8*	*548*	*275*	*13*	*478*	*244*
4	*369*	*308*	*9*	*355*	*278*			
5	*514*	*268*	*10*	*251*	*296*			

The statistics were calculated as shown earlier and are shown below.

Period	*N*	*Mean*	*Median*	*Std dev*	*Minimum*	*Maximum*	*Q1*	*Q3*
Before	*13*	*276.77*	*278.00*	*24.86*	*233.0*	*323.0*	*259.5*	*291.50*
hline After	*13*	*442.6*	*425.0*	*445.7*	*251.0*	*600.0*	*363.5*	*531.0*

The box plot of accidents before and after the traffic lights were erected are given in Figure 2.16.

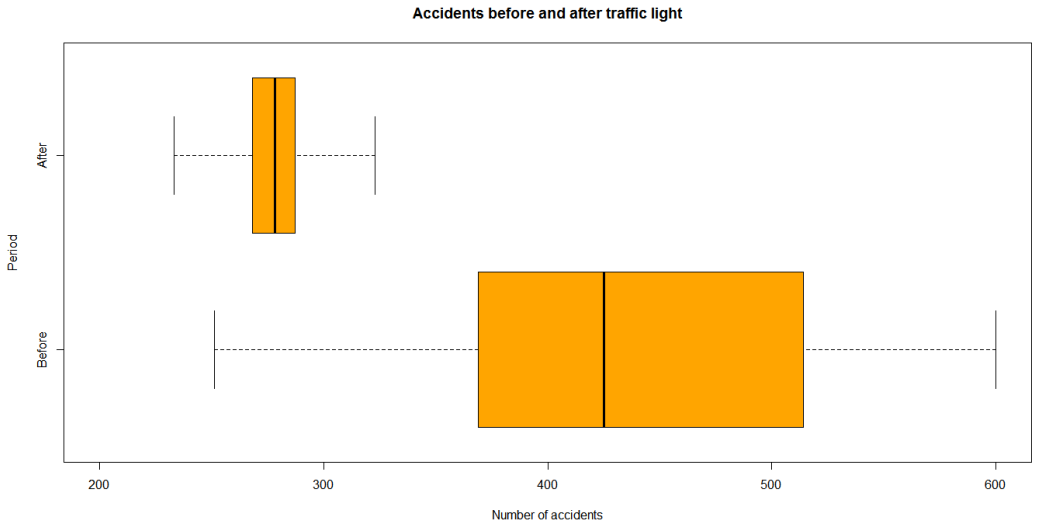


Figure 2.16: Boxplot of Accidents before and after Traffic light

We notice from the two boxplots that the number of accidents after the installation of traffic lights has decreased as shown by the lower values of the mean and median. The variation of the number of accidents is much lower after the installation of the traffic lights. The distribution of the accidents after the traffic lights are installed is symmetric with the mean almost equal to the median. The distribution before the installation was slightly skewed to the left.

Some statisticians prefer to allow the whiskers to extend a maximum length of $1\frac{1}{2}$ times the length of the box. and consider any points outside the bounds of the whiskers as outliers or possible outliers. We generally need to do more tests and analysis before we can conclude that an observation is an outlier.

2.6 Measures of Dispersion

We have just been discussing the central tendency of data, and the measures used to access it. If the values in a data set fluctuate a lot [i.e if at one time we have a high value and the next time we a very low value etc.], then we say that data set has high variability.

When data sets are summarised, the variability of the values is often an important feature of interest. Let us look at a few summary measures, sometimes referred to as measures of variability, dispersion or spread.

2.6.1 The Range (R)

The range [R] is simply the difference between the largest and the smallest value in a data set. Mathematically it is given by

$$R = Maximum\ value - Minimum\ value$$

For example the range of daily takings in *Example 2.13* is $160,000 - 120,0000 = 40,000$ Rands. This measure helps us understand the extent to which our data varies.

This value has one limitation, and that is it depends only on the extreme values. So it is sensitive to outliers, or mistakes in the data or unusual values, which is not desirable in a summary statistic.

2.6.2 Quartile deviation (IQR)

This measure is a slight improvement on the **range**, it is half the difference between the third and first quartile. Mathematically it is given by

$$IQR = \frac{1}{2}[3^{rd}\ Quartile - 2^{nd}\ Quartile]$$

This measure gives half the range of the middle 50% values, which means it is a better statistic than the range. It is not greatly affected by outliers [a property of a statistic referred to as **robustness**, big word!].

2.6.3 Variance s^2

The **variance** is the most commonly used measure of variability in statistical analysis. This measure unlike the other two measures takes into account all the observations in a data set. The greater the variability in the data the higher the value of s^2, on the other hand if all the values are equal or close then the variance is equal to zero or close to zero, respectively.

It is the average of the sum of the squared deviations of each data value from the mean. Mathematically it is given by

$$\begin{aligned} s^2 &= \frac{1}{n-1}\sum_{i=1}^{n}[x_i - \overline{x}]^2 \\ &= \frac{1}{n-1}[\sum_{i=1}^{n} x_i^2 - n\overline{x}^2] \\ &= \frac{1}{n-1}\left[\sum_{i=1}^{n} x_i^2 - \frac{\left[\sum_{i=1}^{n} x_i\right]^2}{n}\right] \end{aligned}$$

If we are dealing with grouped data [e.g several people with the same age] then the formulae above change slightly to give

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{k}\left[x_i^2 f_i - \frac{[\sum_{i=1}^{k} x_i f_i]^2}{n}\right]$$

here the f_i are the k frequencies corresponding to each x_i.

If on the other we are dealing with categorical data or classified data we use the mid point m_i of a class instead of x_i.

Note

1. The positive square root of the variance s^2, is called the **standard deviation** and is denoted by s. i.e., $s = \sqrt{s^2}$
2. In the formulas above we replace $n-1$ by n if we are dealing with the whole population.

2.7 Conclusion

In this Chapter, we looked at a variety of diagramatic representations of data. The diagrams help us to understand the data better and to deduce the structure of the data. We looked at these descriptive statistics but there is however, a bit more we need to learn. How do we collect the data, and how do we make sense out of it, when descriptive statistics are not enough? These questions are addressed by statistical inference.

You have been introduced to Statistics, now you can look at data and come up with a statement about the observations. '*The data has great variation and is skewed to the left*'now means something to you. The bar graphs in most financial articles say something to you.

Exercise 2.3

1. A glass manufacturing company recorded the following profits.(Recorded every four months from January 1985 to December 1987). Measurements are in millions.

   12, 18, 10, 13, 20, 11, 12, 19, 10

   (a) Calculate the mean, median and variance of this data.

   (b) Comment on the median and mean.

   (c) Construct a bar chart and comment on the distribution.

   Construct a box-and-whisker plot for the data in Example 2.12. Comment on the distribution of the data.

Chapter 3

Introduction to Probability

3.1 Introduction

In this Chapter we address the more mathematical side of this course. **Probability** means likelihood or chance. When someone asks the question "what are the chances of me passing the exam?" The answers he or she can get are:

1. None : meaning they have no chance whatsoever.
2. 50-50: meaning they may or may not get the pass.
3. definitely: meaning they are going to get the pass.

Probability is the trying to quantify such situations as above into numbers which we can then use to make decisions. We can define probability as the chances of something happening measured on a scale starting from zero representing **impossibility** to one representing **certainty**. We need to discuss a few terms that will make the ideas used in probability easier to follow.

3.2 Terminology

We start this conversation by defining some terms that we will be using.

Definition 1 **Sample space**

A Sample space Ω is the set of all possible outcomes of an experiment.

You will recall from your Primary/Secondary school that a **set** is collection or class of objects or symbols possessing a property that enables someone to determine whether a given object is in the collection. Intuitively a set is any well-defined list, collection or class of objects.

Definition 2 **event**

An event $\mathbf{E} \in \Omega$ *is a subset of a sample space* Ω*.*

The objects in a set, can be anything, numbers, people, letters, rivers, etc. These objects are called elements or members of a set. Elements of the set belong to or are members of the set. The relationship between objects of the set and the set itself is expressed in the form, "is an element of or is a member of".

Example 3.1

Examples of experiments and events are;

1. *The outcomes from tossing a coin, i.e., head or tail*
2. *writing an examination, you either pass or fail*
3. *playing the LOTTO, you either get the jackpot, a consolation prize or you loose.*
4. *casting a die, you will get a 1, or a 2, or a 3 up to 6*

Example 3.2

Let **X** *be the set of the first 5 letters of the alphabet, that is,* $\mathbf{X} = \{A, B, C, D, E\}$*. Then* $A \in \mathbf{X}$ *means* A *is an element of set* **X***.*

If on the other hand, an object is not an element of a set, then we say **X** does not contain the object. For example $U \notin \mathbf{X}$ means U is not an element of set **X**.

Elements of a set are specified using braces. The set of all possible outcomes when we toss a die can be written as;

$$\{1, 2, 3, 4, 5, 6\}$$

Sets of this nature can also be indicated by a descriptive statement or a rule using the **set builder notation** that encloses within braces a letter or symbol, representing an element of the set followed by a quantifying description of the element, for example,

$$\mathbf{A} = \{n \mid n \; is \; a \; counting \; number \; less \; than \; 100\}$$

The letter or symbol, in this case *n*, used to denote any element of the set and is referred to as the domain of the variable n in this case. Here $\mathbf{A} = \{1, 2, \ldots, 99\}$

Exercise 3.1

Determine whether the following statements are true or false.

1. *Let* $\mathbf{A} = \{a, e, i, o, u\}, then \; a \in \mathbf{A}$*.*

2. *Let* $\mathrm{B} = \{x \mid \ x \ is \ even\}, then \ 3 \in \mathrm{B}$.

3. *Let* $\mathrm{C} = \{2, 4, 6, 8, 10\}, then \ \{2, 3\} \in \mathrm{C}$.

Sample spaces can be finite or infinite. We say a set is finite if it consists of a specific number of elements, that is, if in counting the different members of the set, the counting process can come to an end.

Definition 3 **Empty Set**

A set that contains no elements is called an empty or null set and is denoted by the symbol $\emptyset$ *or* $\{\ \}$.

Definition 4 **Subset**

A set **A** *is said to be a subset of set* **B**, *if and only if, each element of* **A** *is an element of* **B**, *denoted* $\mathrm{A} \subseteq \mathrm{B}$.

Definition 5 **Mutually Exclusive Sets**

If sets **A** *and* **B** *have no elements in common, that is, if no elements of* **A** *is in* **B** *and no element of* **B** *is in* **A**, *then we say that* **A** *and* **B** *are mutually exclusive or disjoint.*

Example 3.3

Let **A** *be the set of even numbers and let* **B** *be the set of odd numbers. Then* **A** *and* **B** *are disjoint since no number is both even and odd.*

Definition 6 **Universal Set**

This is the set in which all the sets under investigation will fall. The universal set is denoted by **U**.

Exercise 3.2

1. *Rewrite the following statement using set notation.*

2. *Let* $\mathrm{A} = \{a, b, c, d\}$. *How many subsets does* **A** *contain? List the subsets.*

A simple way of illustrating the relationship between sets and the universal set is is to use a **Venn Diagram**. The universal set is regarded as the region bounded by a rectangle and the set or sets under consideration as the region bounded by a circle or circles or some other closed figure within the rectangle. In these notes the area of interest is **the shaded part**.

Definition 7 **Complement of a Set**

The complement of a set **A** *is the set of all elements in the universe that are not in set* **A**. *See Figure 3.2 for illustration. The complement of* **A** *can be written in*

$$\overline{\mathbf{A}} \ or \ \mathbf{A}^c \ = \{x \mid x \in \mathbf{U} \ and \ x \notin \mathbf{A}\}$$

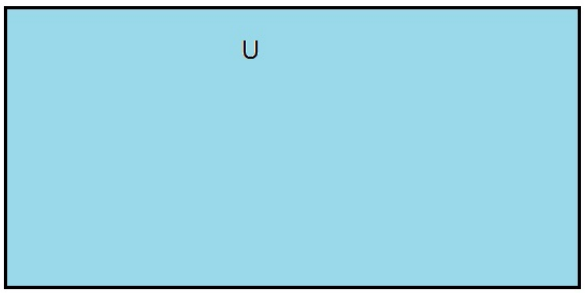


Figure 3.1: Venn Diagram Showing a Universal Set

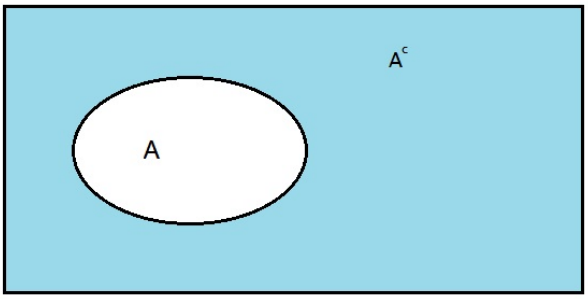


Figure 3.2: Venn Diagram Showing a Complement of s set

3.3 Some Relations from Set Theory

In this section, we define the operations of union, intersections and difference of sets.

Definition 8 **The Union of Sets**

The union of any two sets **A** *and* **B** *(denoted* $\mathrm{A} \cup \mathrm{B}$*) is the set of all elements which belong to set* **A**, *set* **B** *or are both in* **A** *and* **B**. *See Figure 3.3 for illustration.*

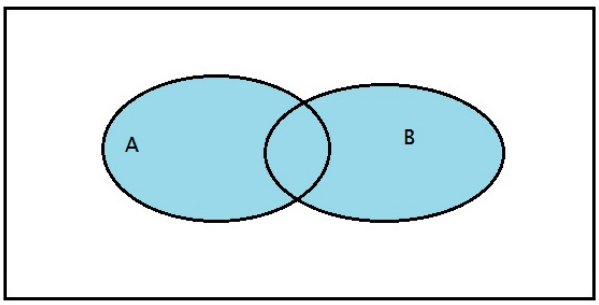


Figure 3.3: Venn Diagram Showing a Union of s set A and B

Example 3.4

If $\mathbf{A} = \{1, 2, 3\}$ *and* $\mathrm{B} = \{3, 4\}$, *then* $\mathbf{A} \cup \mathbf{B} = \{1, 2, 3, 4\}$ The inclusive "or" in this definition means that if $x \in \mathbf{A} \cup \mathbf{B}$, then at least one of the following

condition must be met; $x \in \mathbf{A}$, $x \in \mathbf{B}$ or $x \in \mathbf{A}$ and $x \in \mathbf{B}$. This can also be expressed as, $\mathbf{A} \cup \mathbf{B} = \{x \mid x \in \mathbf{A}\ or\ x \in \mathbf{B}\}$.

Note: If there are elements common to both sets, they are listed only once in the union.

Properties of the Union of Sets

(i) Commutative property $\mathbf{A} \cup \mathbf{B} = \mathbf{B} \cup \mathbf{A}$

(ii) Associative property $(\mathbf{A} \cup \mathbf{B}) \cup \mathbf{C} = \mathbf{A} \cup (\mathbf{B} \cup \mathbf{C})$

(iii) Identity property $\mathbf{A} \cup \emptyset = \mathbf{A}$

Definition 9 The Intersection of Sets

The intersection of sets **A** *and* **B** *(denoted* $\mathbf{A} \cap \mathbf{B}$*) is the set of all elements common to both* **A** *and* **B** *(see Figure 3.4 for illustration, that is,*

$$\mathbf{A} \cap \mathbf{B} = \{x \mid x \in \mathbf{A}\ and\ x \in \mathbf{B}\}$$

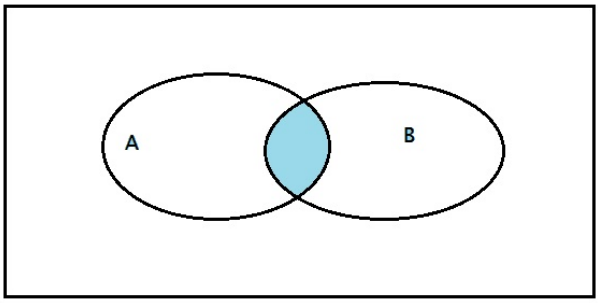


Figure 3.4: Venn Diagram Showing a Intersection of s set A and B

Note

(i) If **A** and **B** are disjoint, $\mathbf{A} \cap \mathbf{B}$ is the null set, that is, $\mathbf{A} \cap \mathbf{B} = \emptyset$

(ii) If $\mathbf{A} \subset \mathbf{B}$, then the elements common to both **A** and **B** are the elements in set **A** and so $\mathbf{A} \cap \mathbf{B} = \mathbf{A}$

(iii) If **A** and **B** are equal then $\mathbf{A} \cap \mathbf{B} = \mathbf{A} = \mathbf{B}$

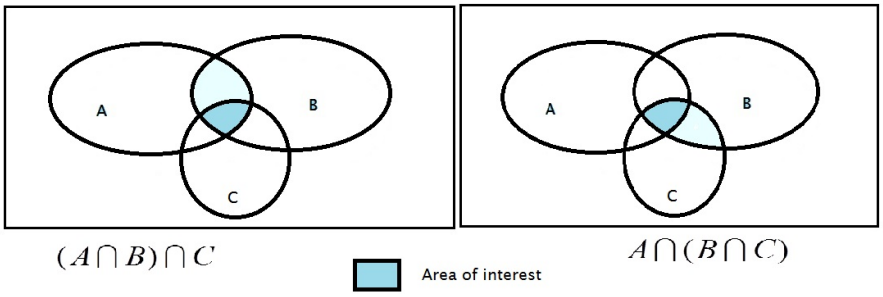
Example 3.5

If $\mathbf{A} = \{1, 2, 3, \ldots, 100\}$ *and* $\mathbf{B} = \{60, 61, \ldots, 100\}$*, then* $\mathbf{A} \cap \mathbf{B} = \{60, 61, \ldots, 100\}$

Properties of the Intersection of Sets

(i) Commutative property $\mathbf{A} \cap \mathbf{B} = \mathbf{B} \cap \mathbf{A}$.

(ii) Associative property $(\mathbf{A} \cap \mathbf{B}) \cap \mathbf{C} = \mathbf{A} \cap (\mathbf{B} \cap \mathbf{C})$. The venn diagram below shows this relationship.



(iii) Identity property $\mathbf{A} \cap \mathbf{U} = \mathbf{A}$.

Exercise 3.3

Let the universal set $\mathbf{U} = \{a, b, c, d, e, f, g\}$ *and let* $\mathbf{A} = \{a, b, c, d, e\}$, $\mathbf{B} = \{a, c, e, g\}$ *and* $\mathbf{C} = \{b, e, f, g\}$. *Find*

1. $\mathbf{A} \cup \mathbf{C}$
2. $\overline{\mathbf{B}} \cup \mathbf{C}$
3. $\overline{(\mathbf{A} \cap \overline{\mathbf{A}})}$

3.3.1 De Morgan's Law

De Morgan's Law states that

(i) $\overline{(\mathbf{A} \cup \mathbf{B})} = \overline{\mathbf{A}} \cap \overline{\mathbf{B}}$

(ii) $\overline{(\mathbf{A} \cap \mathbf{B})} = \overline{\mathbf{A}} \cup \overline{\mathbf{B}}$

This will help us solve some sticky probability problems.

Exercise 3.4

Use venn diagrams to prove De Morgan's Law.

3.4 Counting Methods: Permutations and Combinations

In order to calculate probabilities we will often need to count the number of possible outcomes of an event in a sample space. **Permutations and combinations** allow us to do this in an efficient manner especially where there are several possible outcomes.

3.4.1 Multiplication Principle

If two operations A and B are performed and there are n_1 possible outcomes of A and n_2 possible outcomes of B, then the combined operation of A and B has $n_1 \times n_2$ possible outcomes.

Example 3.6

In a chemistry experiment to study the effects of temperature and pressure, 5 temperature levels and 3 different pressure levels were used. How many experiments should be run to have used all the combinations of temperature and pressure?

Solution

Total number of experiments $= 5 \times 3 = 15$

Solution using a tree diagrams

A tree diagram is a pictorial way of being able to represent the possible outcomes in a problem. The tree diagram for Example 3.6 is given below in Figure 3.5, also shows 15 possible experiments.

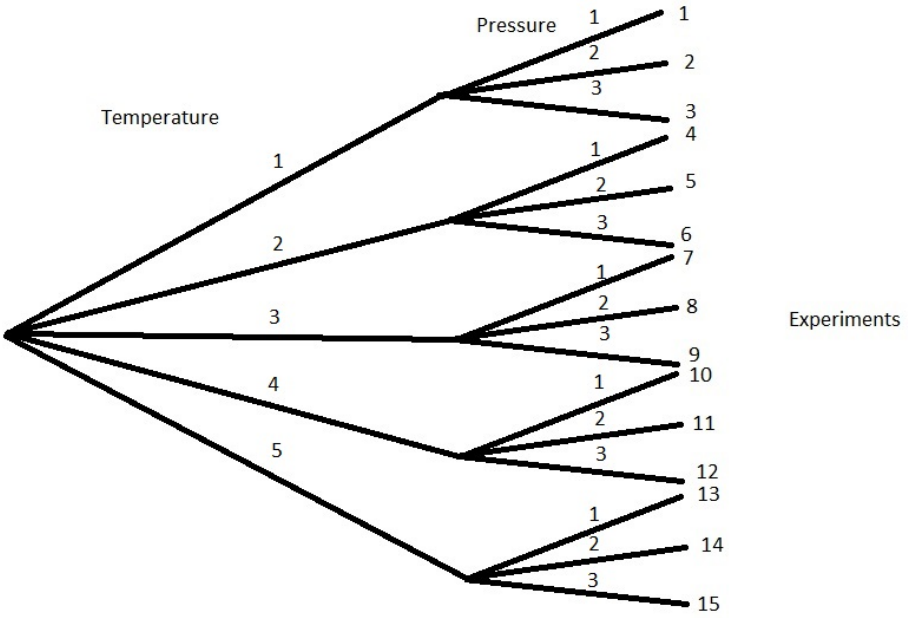


Figure 3.5: Number of possible experiments

The multiplication principle can be extended to where there are more than two operations. If the operations are labelled $1,\ 2, \ldots, k$ with $n_1,\ n_2, \ldots, n_k$ possible outcomes respectively then the combined operation has $n_1 \times n_2 \times \cdots \times n_k$ possible outcomes. This may also be represented using a tree diagram.

3.4.2 Permutation of n objects

The number of arrangements of n different objects, taken all together is $n! = n \times (n-1) \times (n-2) \cdots \times 2 \times 1$ and is called a **permutation** of n objects.

Example 3.7

Sarah has 4 new books. She takes a week to read each book. If she reads the books in any order over the next four weeks, how many different ways can she read the books?

Solution

Week 1 she has 4 choices, week 2, 3 choices, week 3, 2 choices and 1 book left in week 4.

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

Example 3.8

A group of 6 friends who all study statistics, always sit together during their lectures. In how many different ways can the group sit together in a row if there are no other students who sit in between them?

Solution

The number of ways they can arrange themselves in a row

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$$

3.4.3 Permutation n objects taken r at a time

The number of ordered arrangements or **permutations** of n different objects taken r at a time is

$$\begin{aligned} {}^nP_r &= n \times (n-1) \times (n-2) \times \cdots \times (n-r+1) \\ &= \frac{n!}{(n-r)!} \end{aligned} \tag{3.1}$$

Note: In this case **AB** is treated as different from **BA**, i.e., order matters.

Example 3.9

How many 2 letter words can be made out of the letters in **ABC**, *assuming any arrangement of the letters is a word?*

Solution

$$\mathbf{AB},\ \mathbf{AC},\ \mathbf{BC},\ \mathbf{BA},\ \mathbf{CA}\ \textit{and}\ \mathbf{CB}$$

or simply

$$^3P_2 = \frac{3!}{(3-2)!} = \frac{3!}{1!} = 6$$

Example 3.10

How many 3 letter words can be made out of the letters in the word **TABLE**, *assuming any arrangement of the letters is a word?*

Solution

$$^5P_3 = \frac{5!}{2!} = 60 \tag{3.2}$$

Example 3.11

A college gives students a registration number the first time that they register to study. The registration number consists of a letter and 3 digits and the year they first registered. The digits can be in any order and only occur once. What is the maximum number of students that may be registered each year?

Solution

$$26 \times {}^{10}P_3 = 18720$$

3.4.4 Combination of n objects taken r at a time

The number of different unordered selections of r objects made from n different objects is called a **combination** and is given by

$$^nC_r = \binom{n}{r} = \frac{n!}{r!\,(n-r)!} \tag{3.3}$$

Note: In this case **AB** is considered to be the same as **BA**, i.e., order does not matter.

Example 3.12

There are 4 types of fruit in a fruit bowl: Apple (A), Banana (B), Pear (P) and Orange (O). You are allowed to pick two fruits. How many different ways can you pick two different fruits out of the bowl?

Solution

AB AP AO PO BO *and* PB

or simply

$$^4C_2 = \binom{4}{2} = \frac{4!}{2!2!} = 6$$

Example 3.13

How many ways can we select three letters from the word ATMOSPHERIC? **(Order is not important)**.

Solution

$$^{11}C_3 = \binom{11}{3} = \frac{11!}{3!8!} = 165$$

Note here that since order is not important selecting ATM, AMT, TMA, TAM, MTA, MAT are counted as just one selection. Compare this to a **permutation** **where order is important**.

3.4.5 Permutations of n different objects when there are groups of identical items

Suppose that in a set of n objects there are k subgroups with n_1 in the first group, n_2 in the second group, $\ldots$, and n_k in the kth group such that $n_1 + n_2 + \cdots + n_k = n$ then the number of distinguishable permutations is

$$\binom{n}{n_1, n_2, \cdots, n_k} = \frac{n!}{n_1! n_2! \cdots n_k!} \tag{3.4}$$

Example 3.14

A boy has 20 marbles of which 5 are red, 4 are blue, 7 are yellow and 4 are green. He places these marbles in a row. If we consider only the colours of the marbles how many different line ups can we have of the marbles?

$$\binom{20}{5, 4, 7, 4} = \frac{20!}{5!4!7!4!}$$

Exercise 3.5

1. At a frozen yoghurt bar, you have a choice of 4 yoghurt bases, 10 toppings, and 3 syrups.
   (a) How many frozen yoghurt combos are possible using a base, 2 toppings and syrup?
   (b) How many frozen yoghurt combos are possible using a base and 0 to 3 toppings?
2. A padlock requires a 5 digit code to be unlocked
   (a) How many 5 digit codes can start with an odd number and end with an even number?
   (b) How many codes can be less than 1000?

3. How many five letter words can be made from the letters in the word MENTAL
   (a) if letters may not be repeated
   (b) if letters may be repeated
4. Five students stay together in a commune. They share supper cooking duties during the week (Monday to Friday). How many ways can they allocate duties for next week if one of them can only do Monday or Friday and only one person is on duty each day?

Now that we have gone through some basic set theory let us discuss probability and see how we calculate it.

3.5 Rules Governing Probability (Axioms)

It is very important to remember the following rules which govern probability also know as **Probability Axioms**. Probability can be seen as a number between 0 and 1, with 0 meaning there is no chance of an event happening and 1 meaning the event is definitely going to occur.

This means;

1. You never get a probability which is less than 0. Put mathematically let $\mathscr{F}$ be the class of all events Ω. A probability measure P on $(\Omega, \mathscr{F})$ is a real valued function satisfying $P(E) \geq 0$ for every $E \in \mathscr{F}$
2. The sum of the probabilities of all events in a sample space should add up to one, $P(\Omega) = 1$. (You never get a probability which is more than 1)
3. Suppose E_1, E_2, E_3,... are mutually exclusive (that is $E_i \cap E_j = \emptyset$ for $i \neq j$) then ;

$$P(E_1 \cup E_2 \cup E_3 \cup \dots) = \sum_{i=1}^{\infty} E_i$$

On the basis of these axioms many questions concerning probability can be answered. For example you can prove the following.

1. $P(E^c) = 1 - P(E)$.
2. $P(\emptyset) = 0$.

Below we have some examples of probability.

Example 3.15

1. *The probability that the sun will rise tomorrow is 1, since the sun will always rise even on a cloudy day the sun will have risen.*
2. *The probability that a computer will work without being plugged to a power source is 0, since a computer needs power. (What about a laptop?)*

3.6 Approaches to Probability

There are three main approaches to probability and these are:

1. the subjective approach.
2. the relative frequency approach.
3. the classical approach.

3.6.1 The Subjective Approach

Subjective probability is a type of probability which is based on an individuals' personal experiences. There is no scientific basis in determining such probability. This is not a very popular approach to determining probability (although some experts in some specific areas give it a good try).

Example 3.16

A football commentator announces that there is a 80% *chance that team A will beat team B.*

3.6.2 Relative Frequency Approach to Probability

The relative frequency approach to probability is based on the number of times an event (E) has occurred over all the possible number of occurrences it could have occurred (n). It states that;

$$P(E) = \frac{Number\ of\ times\ E\ has\ occurred}{Number\ of\ times\ E\ could\ have\ occurred} \tag{3.5}$$

Example 3.17

The number of students studying for the CIS qualification in a particular year who are enrolled in a certain college can be classified as shown below;

Gender	*Financial sector*	*manufacturing*	*agricultural*	*Mining*	*Total*
Male	*200*	*140*	*50*	*10*	*400*
Female	*220*	*60*	*20*	*0*	*300*
Total	*420*	*200*	*70*	*10*	*700*

Find the probability that a randomly selected student is;

1. *a man.*
2. *a female from the manufacturing sector.*

3. *from the financial sector.*

4. *a male from the education sector.*

solution

1. *The probability that it is a man is;*

$$\begin{aligned} P(man) &= \frac{number\ of\ man}{Total\ number\ of\ students} \\ &= \frac{400}{700} \\ &= 0.5714 \end{aligned}$$

2. *The probability that it a female from the manufacturing sector is;*

$$\begin{aligned} &P(female\ from\ the\ manufacturing\ sector) \\ &= \frac{number\ of\ ladies\ from\ the\ manufacturing\ sector}{Total\ number\ of\ students} \\ &= \frac{60}{700} \\ &= 0.0857 \end{aligned}$$

3. *The probability that they are from the financial sector is;*

$$\begin{aligned} P(financial\ sector) &= \frac{from\ the\ financial\ sector}{Total\ number\ of\ students} \\ &= \frac{420}{700} \\ &= 0.6 \end{aligned}$$

4. *The probability that it is a male from the education sector is;*

$$\begin{aligned} &P(male\ from\ the\ education\ sector) \\ &= \frac{male\ from\ the\ education\ sector}{Total\ number\ of\ students} \\ &= \frac{0}{700} \\ &= 0 \quad no\ information\ people\ from\ education \end{aligned}$$

3.7 The Classical Approach

The **classical approach** is applicable when the the sample space is known, each event is equally likely to occur or known.

- Let **E** represent the event you are interested in, this could be number of heads in coin toss, number successes of some experiment, etc.
- let $n(x)$ (also known as the **cardinality** of an event) represent number of events of interest that can occur.
- Let $n(S)$ represent the number of all possible outcomes. (Or the number of items in the sample space).

Then the classical approach to probability states that:

$$P(E) = \frac{n(x)}{n(S)} \tag{3.6}$$

i.e.,

$$The\ probability\ of\ an\ event\ occurring = \frac{number\ of\ all\ events\ of\ interest}{number\ of\ objects\ in\ the\ sample\ space}$$

Example 3.18
A car dealer has 6 trucks, 7 sedans and 4 station wagons in his lot. A man comes a purchases a vehicle. What is the probability that he purchases a;

1. *truck*
2. *sedan*
3. *station wagon*

solution

1. *Probability that a truck is purchased can be written as*

$$\begin{aligned} P(truck) &= \frac{number\ of\ trucks\ in\ the\ lot}{number\ of\ cars\ in\ the\ lot} \\ &= \frac{6}{17} \\ &= 0.3529 \end{aligned}$$

2. *Probability that a sedan is purchased is;*

$$\begin{aligned} P(sedan) &= \frac{number\ of\ sedans\ in\ the\ lot}{number\ of\ cars\ in\ the\ lot} \\ &= \frac{7}{17} \\ &= 0.4118 \end{aligned}$$

3. *Probability that a station wagon is purchased is;*

$$\begin{aligned} P(station\ wagon) &= \frac{number\ of\ station\ wagons\ in\ the\ lot}{number\ of\ cars\ in\ the\ lot} \\ &= \frac{4}{17} \\ &= 0.2353 \end{aligned}$$

Exercise 3.6
A company has 100 employees, 60 of them male and the other 40 female. If two employees are absent from work, what is the probability that,

1. *the two are men?*
2. *the two are female?*

There are generally two laws used in probability, these are;

- Addition of events
- Multiplication of events

3.7.1 Addition Rule of Probability

The **Addition Rule** of probability of the union of events A and B is that the sum of the probabilities of events A and B minus the probability of events A and B occurring together, i.e.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \tag{3.7}$$

This can be expanded to

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ &\quad + P(A \cap B \cap C) \end{aligned} \tag{3.8}$$



Example 3.19
Consider the set of numbers N={1, 2, 3, 4, 5, 6, 7, 8, 9 }. Let A be the event of picking an even numbers and let B be the event of picking a number less than 5. Find the;

1. *probability of A, i.e., $P(A)$.*
2. *probability of B, i.e., $P(B)$.*
3. *probability of $A \cap B$, i.e., $P(A \cap B)$.*
4. *probability of A$\cup B$, i.e., $P(A \cup B)$.*

Solution
The set of all possible out comes is N={1, 2, 3, 4, 5, 6, 7, 8, 9, 10} and we have been given events A and B, this means we can use the classical approach to probability.

1. *probability of A is*

$$\begin{aligned} P(A) &= \frac{number\ of\ even\ numbers}{total\ number\ of\ observations} \\ &= \frac{5}{10} \\ &= 0.5 \end{aligned}$$

2. *probability of B is*

$$\begin{aligned} P(B) &= \frac{numbers\ less\ than\ 5}{total\ number\ of\ observations} \\ &= \frac{numbers\ 4\ or\ less}{total\ number\ of\ observations} \\ &= \frac{4}{10} \\ &= 0.4 \end{aligned}$$

3. *probability of $A \cap B$ is*

$$\begin{aligned} P(A \cap B) &= \frac{even\ numbers\ and\ numbers\ less\ than\ 5}{total\ number\ of\ observations} \\ &= \frac{even\ numbers\ less\ than\ 5}{total\ number\ of\ observations} \\ &= \frac{2}{10} \\ &= 0.2 \end{aligned}$$

4. *probability of* $A \cup B$ *is*

$$\begin{aligned} P(A \cup B) = & \ P(A) + P(B) - P(A \cap B) \ \ From\ equation\ 3.7 \\ = & \ 0.4 + 0.5 - 0.2 \\ = & \ 0.9 - 0.2 \\ = & \ 0.7 \end{aligned}$$

3.7.2 Mutually or Disjoint Exclusive Events and Exhaustive Events

Mutually exclusive events or disjoint events are events which cannot occur at the same time. The occurrence of one event precludes the occurrence of the other events.

Consider two events A and B. Events, A and B are said to be mutually exclusive if they cannot occur simultaneously. i.e., $A \cap B$ is empty. This means P(A∩B)=0. Therefore when two event are mutually exclusive (can not occur together) we have;

$$P(A \cup B) = P(A) + P(B) \tag{3.9}$$

A set of events is said to be **Mutually exhaustive** if they include all the possible outcomes of an experiment, i.e., they form the universal set.

Example 3.20
Two gamblers have made a bet, they agree to cast a fair die once and that an outcome of 4 or 3 means you win the bet. Find the probability;

1. *you get a 4*
2. *you get a 3*
3. *you win the bet*
4. *you loose the bet*

Solution
When you cast a die you either get 1, 2, 3, 4, 5 or 6. We again use the classical approach.

1. *Probability of getting a 4 is*

$$\begin{aligned} P(4) = & \ \frac{numbers\ of\ 4s}{total\ number\ of\ possible\ outcomes} \\ = & \ \frac{1}{6} \\ = & \ 0.1667 \end{aligned}$$

2. *Probability of getting a 3 is*

$$\begin{aligned} P(4) &= \frac{numbers\ of\ 3s}{total\ number\ of\ possible\ outcomes} \\ &= \frac{1}{6} \\ &= 0.1667 \end{aligned}$$

3. *winning the bet means getting a 3 or a 4, or means* $\cup$

$$\begin{aligned} P(4 \cup 3) = &\ P(4) + P(3) \\ &\quad since\ getting\ a\ 3\ or\ 4\ are\ mutually\ exclusive\ events. \\ = &\ \frac{1}{6} + \frac{1}{6} \\ = &\ 0.333 \end{aligned}$$

4. *losing the bet means getting a 1, 2, 5 or a 6, (*$\cup$ *means "or")*

$$\begin{aligned} P(1 \cup 2 \cup 5 \cup 6) = &\ P(1) + P(2) + P(5) + P(6) \\ &\quad since\ these\ mutually\ exclusive\ events. \\ = &\ \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} \\ = &\ 0.667 \end{aligned}$$

3.7.3 Multiplicative rule of Probability

Suppose in a big organisation with 400 people made up of 240 men and 160 women two people are selected to go on a holiday abroad. The probability that the second person was a lady can not be easily calculated. This is because we need to know if the first one selected was a man of a woman. The two events dependent on each other. How do we handle such situations?

3.7.4 Independent Events

Two or more events are said to be independent if they are unrelated. This means the occurrence of one event has no effect on the occurrence of another event.

If two events A and B are independent, then the probability that both A and A occur simultaneously, is the product of their probabilities.

$$P(A \cap B) = P(A)P(B) \tag{3.10}$$

Example 3.21
The probability that a man gets a promotion at his work place is 0.3 and the probability that his wife who works elsewhere gets a promotion there is 0.7. Find the probability that both the wife and husband get promotions.

Solution
The two events of a husband and his wife getting a promotion are independent events, therefore we can use Equation 3.10. The "and" in the question means $\cap$.

P(man is promoted and wife is promoted)
=P(Man promoted $\cap$ *Wife promoted)*
=0.3$\times$*0.7*
=0.21

3.7.5 Complementary Events

Complementary events are two mutually exclusive events such that the union of these events include all the outcomes. Examples of complementary events are passing or failing and exam, defective and not defective and heads or tails.

If two events A and B are complementary, the sum of their probabilities is equals to 1,

$$P(A) + P(B) = 1 \tag{3.11}$$

The complement of event A is often written as $\overline{A}$. So that

$$\begin{aligned} P(A) + P(\overline{A}) &= 1 \\ P(\overline{A}) &= 1 - P(A) \end{aligned} \tag{3.12}$$

Example 3.22
The probability that the bus comes on time to the pick-up point is 0.95. Find the probability that it comes late.

Solution
Since coming on on time and coming late are complementary events it follows that.

$$\begin{aligned} P(Bus\ comes\ late) &= 1 - P(\overline{\textit{Bus comes late}})\ by\ Equation\ 3.12 \\ &= 1 - P(Bus\ comes\ on\ time) \\ &= 1 - 0.95 \\ &= 0.05 \end{aligned}$$

3.7.6 Conditional Probability

Conditional probability is the probability of some event A, given that some other event, B, has already occurred. Conditional probability (also

known as the **posterior probability**) is written P(A|B), and is read "the probability of A, given B has occurred".
Let A and B be any two events where $P(A \geq 0)$, then

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \tag{3.13}$$

and similarly

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \tag{3.14}$$

provided P(B) $>$ 0.
This means we can write
$P(A \cap B) = P(A|B)P(B)$ and $P(B \cap A) = P(B|A)P(A)$

If two events A and B are independent then from equation 3.10 we know that P(A $\cap$ B)= P(A)P(B). This means for two independent events A and B,

$$\begin{aligned} P(B|A) &= \tfrac{P(A \cap B)}{P(A)} \\ P(B|A) &= \tfrac{P(A)P(B)}{P(A)} \\ P(B|A) &= P(B) \end{aligned}$$

Which makes sense since if events are independent it should not matter whether the other has occurred or not.

Let us return to the question we discussed earlier and could not answer.

Example 3.23
Suppose in a big organisation with 400 people made up of 240 men and 160 women two people are selected to go on a holiday abroad. Find the probability that the second person was a lady given that the first one was a man.

Solution
The probability that a man is picked first is given by $\frac{240}{400}$ *= 0.6*
The probability that a man is picked fist and lady second is $\frac{240}{400} \times \frac{160}{399}$ *=0.241. We multiply these since they are at this stage independent events.*

$$\begin{aligned} P(\textit{pick lady second}|\textit{man first}) &= \tfrac{P(pick\ man\ first\ and\ lady\ second)}{P(pick\ man\ first)} \\ &= \tfrac{0.241}{0.6} \\ &= 0.401 \end{aligned}$$

You can also do this problem by looking how many women and people are left after the first one is chosen. In this case we have 399 people left and 160 women. Therefore the probability is $\frac{160}{399} = 0.401$ the same answer.

There two practical ways you can use solve probability problems. These are Venn diagrams and Probability tree diagrams. They give you the opportunity to clearly lay out all the possible outcomes and hence solve the problem.

Example 3.24
Consider letters of the alphabet a, b, c, d, e, f, g and h. Let the A be the set of letter a, b, c, d and e. Represent this information in a venn diagram and find the probability that a randomly selected letter will belong to A.

Solution*Figure 3.6 represent the venn diagram the circle set A.*

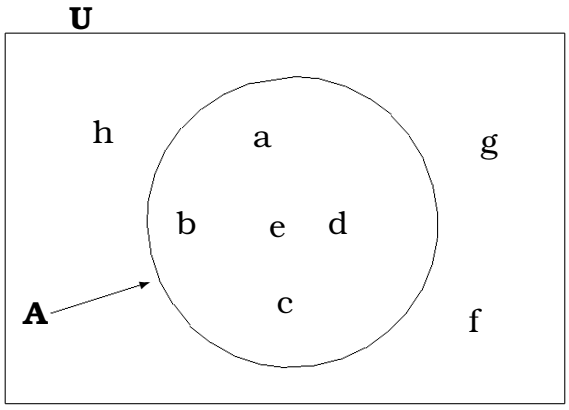


Figure 3.6: The Venn Diagram and the a Universal Set

$$\begin{aligned} P(A) &= \frac{number\ of\ letters\ in\ A}{total\ number\ of\ letters} \\ &= \frac{5}{8} \\ &= 0.625 \end{aligned}$$

The venn diagram is a practical way of solving problems if you are dealing with situations involving intersections and counts. When the problem involve conditional probability, probability trees usually produce easy solutions.

3.7.7 Probability Tree Diagrams

When you are dealing with a problem which has multiple outcomes a probability tree diagram will present the information in a clear and convenient way. A **Probability tree diagram** is a diagram with lines indicating all possible outcome and their probabilities.

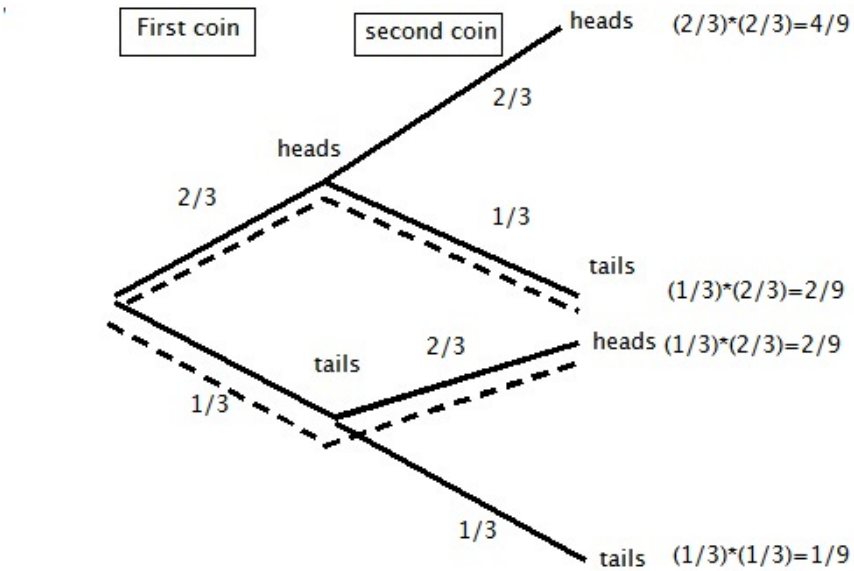
The tree diagram below represents the experiment of tossing a coin twice. The probabilities of each outcomes can be shown on the tree diagram and the probabilities of interest calculated see Example 3.16.

Example 3.25
Suppose two coins were tossed at the same time and it is known that the

coins are not fair, thus the probability of a head = $\frac{2}{3}$. Let F be the event that the result of the first toss was a head and let S be the event that the second toss was a tail.

1. *Draw a tree diagram.*
2. *Find the probability that the result of the first toss was a tail.*
3. *Find the probability of getting a head and a tail.*

Solution



1.
2. *The Probability that the first toss was a tail =* $P(Tail) = \frac{1}{3}$.
3. *Using the indicated path on the tree diagram there are two ways to get a a head and a tail. i.e., you get a head first then a tail or you get a tail first then a head.*

$$\begin{aligned} P(Head\ and\ a\ Tail) &= [P(1^{st}\ is\ Head)][P(2^{nd}\ is\ Tail)] \\ &\quad +[P(2^{nd}\ is\ Tail)][P(1^{st}\ is\ Head)] \\ &= \frac{2}{3}\times\frac{1}{3}+\frac{1}{3}\times\frac{2}{3} \\ &= \frac{4}{9} \end{aligned}$$

3.7.8 Bayes' Theorem

A useful result in computing the total probability based on conditional probabilities is the **Law of Total Probability**.

Definition 10 The law of total probability

Let A_1, A_2, A_3, ..., A_k be mutually exclusive and exhaustive events. The law of total probability states that the probability of any event B is given by;

$$\begin{aligned} P(B) &= P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \cdots + P(B|A_k)P(A_k) \\ &= \sum_{i=1}^{k} P(B|A_i)P(A_i) \end{aligned} \tag{3.15}$$

In order to appreciate this result let us consider a situation where we want to calculate the probability of an event B given several **prior probabilities** $P(A_i)$ of mutually exhaustive events A_i see Figure 3.7.

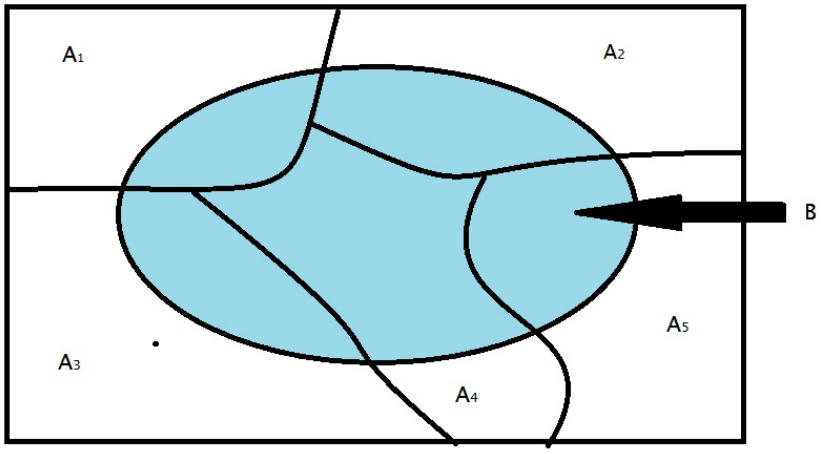


Figure 3.7: Partition of B by mutually exclusive and exhaustive events A_i

To calculate the probability of B (n Figure 3.7) using the conditional probability $P(A_j|B)$ from a **prior probability** $P(A_j)$ we use 3.15, i.e.,

$$P(B) = P(B|A_1)P(A_1)+P(B|A_2)P(A_2)+P(B|A_3)P(A_3)+P(B|A_4)P(A_4)+P(B|A_5)P(A_5)$$

A useful result for calculating **posterior probability** $P(A_i|B)$ is Bayes's theorem.

Definition 11 Bayes' Theorem

Let A_1, A_2, A_3, ..., A_k be mutually exclusive and exhaustive events, with $P(A_i) > 0$ for $i = 1, 2, \ldots, k$. Bayes' theorem states that the posterior probability of any event $A_j|B$ is given by;

$$P(A_j|B) = \frac{P(A_j \cap B)}{P(B)} = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^{k} P(B|A_i)P(A_i)} \quad j = 1, 2, \ldots, k \tag{3.16}$$

Example 3.26

Suppose there are 3 tyre manufacturers: A (supplying 3000), B (supplying 4000), and C (supplying 1000). Suppose it is known that $P(D|A) = 0.01$, $P(D|B) = 0.015$ *and* $P(D|C) = 0.02$ *(where D represents a defective tyre). Suppose a randomly selected tyre was defective tyre:*

1. *What is the probability that it is from factory B.*
2. *What is the probability that it is from factory C.*
3. *What is the probability that it is from factory A.*

Solution
1.

$$\begin{aligned} P(D) &= P(D|A)P(A) + P(D|B)P(B) + P(D|C)P(C) \\ &= 0.01 \times \left(\frac{3000}{3000+4000+1000}\right) + 0.015 \times \left(\frac{4000}{8000}\right) + 0.02 \times \left(\frac{1000}{8000}\right) \\ &= 0.014 \end{aligned}$$

$$P(B|D) = \frac{P(B \cap D)}{P(D)} = \frac{P(D|B)P(B)}{P(D)} = \frac{0.015 \times \frac{4000}{8000}}{0.014} = 0.536$$

2.

$$P(C|D) = \frac{P(C \cap D)}{P(D)} = \frac{P(D|C)P(C)}{P(D)} = \frac{0.02 \times \frac{1000}{8000}}{0.014} = 0.179$$

3.

$$P(A|D) = \frac{P(A \cap D)}{P(D)} = \frac{P(D|A)P(A)}{P(D)} = \frac{0.01 \times \frac{3000}{8000}}{0.014} = 0.268$$

3.7.9 Using counting methods in Probability

Example 3.27
There are 12 girls and 10 boys in a class. Two children are randomly selected to represent the class. What is the probability that 2 girls are chosen?

Solution

$$P\left(2\ girls\right) = \frac{\binom{12}{2}\binom{10}{0}}{\binom{22}{2}} = \frac{66}{231}$$

Note: This is an example of the **Hypergeometric distribution** see Section 4.6.5.

Example 3.28

In a standard pack of cards there are 52 cards with 4 suits. What is the probability that when 3 cards are dealt

1. *that all of them are hearts?*
2. *that two of the cards are either a king or a queen?*

Solution

1. $P\left(all\ heart\right) = \frac{\binom{13}{3}}{\binom{52}{3}} = \frac{286}{22100} = 0.0129.$
2. $P\left(2\ either\ king\ or\ queen\right) = \frac{\binom{8}{2}\binom{44}{1}}{\binom{52}{3}} = \frac{1232}{22100} = 0.0557.$

Or

$$P\left(2\ either\ king\ or\ queen\right) = \frac{\binom{4}{1}\binom{4}{1}\binom{44}{1} + 2\binom{4}{2}\binom{44}{1}}{\binom{52}{3}} = \frac{1232}{22100} = 0.0557.$$

Example 3.29

Four fair dice are tossed. Find the probability that

1. *The sum of the scores is 4.*
2. *The only even numbers.*
3. *There are no 3's.*

Solution

1. P(sum of scores is 4) =P(every dice has a 1) = $\frac{1}{6^4} = 0.00077.$
2. $P\left(Only\ even\ numbers\right) = \frac{3\times3\times3\times3}{6^4} = \frac{3^4}{6^4} = 0.00625.$
3. $P(no\ 3's) = \frac{5^4}{6^4} = 0.48225$

Example 3.30

What is the probability that you have words that start and end with a consonant when making 7 letter words using the letters in the word GRAVITY? (Assuming that any order of letters makes a word)

Solution

P(start and end with consonants) = $\frac{\binom{5}{2}2!5!}{7!} = \frac{2400}{5040} = 0.47619.$

Exercise 3.7

1. *It has been found that the chances of an outbreak of the army worm in any year is 5%. A farmer buys the army worm pesticide at the beginning of the growing season. What is the probability that he will not use the pesticide.*

2. *In a certain region, the probability of having a male baby is $\frac{89}{100}$. A family has already 2 boys. What is the probability of having a baby girl.*

3. *A shop orders toys in boxes each holding 10 toys. Suppose 10% of all boxes opened are found to have a problem toy. If 4 boxes are opened what is the probability that they all will have problem toys.*

3.8 Summary

In this unit we discussed probability concepts. We define a set and its relationship to venn diagrams we also defined a sample space and events.

We calculated the probability of an events using the classical approach and the the relative frequency approach. We went of to draw and interpret probability tree diagrams and used them to compute probabilities.

Chapter 4

Random Variables and Distributions

4.1 Introduction

We have been looking at variables, describing them through samples and populations. Now, we will proceed to look at the distributions of data.

There is a set/class of distributions where most data fall into. We will discuss these distributions, for **continuous** and **discrete** data. This discussion should help us understand the structure of populations better.

4.2 Discrete random variables

These are variables that takes on distinct values, e.g. the number of games won by a team 0, 1, 2, e.t.c., up to the total number of games played.

4.2.1 Probability Distribution

Let X be a random variable and let x_i, $i = 1, 2, \cdots, k$ denote the k distinct values that X may assume. Since each X corresponds to a basic outcome of a random trial, a **probability distribution** for the sample space will associate a probability value with each x_i. The probability that random variable X will assume value x_i will be denoted by $P(X = x_i)$ for example, $P(X = 0)$ is the probability that $X = 0$.

Example 4.1

In the game problem mentioned above earlier, suppose a team plays 3 *games, and has the following probabilities of winning the games.*

Prob(Winning 0 games)=0.65,
Prob(winning 1 game)=0.15,
Prob(Winning 2 games)=0.10,

Prob(winning 3 games)=0.10.

X	*0*	*1*	*2*	*3*
P(X=X)	*0.65*	*0.15*	*0.10*	*0.10*

This represents a probability distribution, i.e., the probability associated with each possible outcome.

Some properties

Any probability distribution for a discrete random variable X has the properties.

1. $0 < P(X = x_i) < 1$ for $i = 1, 2, \cdots, k$,
2. $\sum_{i=1}^{k} P(X = x_i) = 1$.

The distribution function can also be referred to as a **probability function** or **probability mass function**.

4.2.2 Cummulative Probability Distribution

This is the probability that a random variable X takes a value less than or equal to some specified value x, denoted by $P(X \leq x)$.

$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$ or
$P(X < 2) = P(X = 0) + P(X = 1)$

In Example 4.1, the cumulative distribution is given by;

X	$P(X = x)$	$P(X \leq x)$
0	0.65	0.65
1	0.15	0.80
2	0.10	0.90
3	0.10	1.00

4.2.3 Expected value of a descrete random variable

The mean value of a random variable in many trials is known as the **expected value**. For a discrete random variable X, the expected value denoted by;

$$E[X] = \sum_{i=1}^{k} x_i P(x_i) \text{ where } P(x_i) = P(X = x_i)$$

4.2.4 Variance of a Discrete Random Variable

The outcomes of a discrete random variable will vary, and as such it is useful to have a measure of their variability. As we discussed earlier a key measure of variation is the **variance**, defined;

$$\begin{aligned} \sigma_x^2 = & \ Var[X] \\ = & \ \sum_{i=1}^{k}(x_i - E[X])^2 P(x_i) \\ & \ or \\ = & \ E[X - E[X]]^2 \\ = & \ E[X^2] - [E[X]]^2 \end{aligned}$$

The standard deviation of a random variable X is given by

$$\sigma_x = \sqrt{Var[X]}.$$

Example 4.2

Find E[X] and Var[X] in the above example.

Solution

$$\begin{aligned} E[X] = & \ 0 \times 0.65 + 1 \times 0.15 + 2 \times 0.10 + 3 \times 0.10 \\ = & \ 0.15 + 0.20 + 0.30 \\ = & \ \underline{0.65} \end{aligned}$$

$$\begin{aligned} Var[X] = & \ 0^2 \times 0.65 + 1^2 \times 0.15 + 4^2 \times 0.1 + 3^2 \times 0.1 - 0.65^2 \\ = & \ \underline{1.03} \end{aligned}$$

Exercise 4.1

Consider a man who tosses a coin once, the probability of getting a head is $\frac{1}{2}$, the probability of getting a tail is 1-$\frac{1}{2}$=$\frac{1}{2}$. Let $X = 1$, if he gets a head and $X = 0$ if he gets a tail. What is the distribution function of X? Find E[X].

4.3 Moments and Moment Generating Functions of Discrete Random Variables

We have so far introduced the idea of a **moment** when we discussed the mean and variance. Moments summarise a distribution function. The kth moment of a random variable X about 0 is;

$$\mu'_k = E(X^k)$$

for $k = 1, 2, \ldots, k$ and the kth moment of a random variable X about the mean μ is;

$$\mu_k = E(X - \mu)^k$$

for $k = 2, 3, \ldots, k$ where $\mu = E(X) = \mu'_1$ is the mean of X and $\mu_2 = Var(X) = E(X - \mu)^2$ is the variance of X as previously defined.

A useful function we can use to obtain moments is the and **moment generating function** (mgf) defined below.

Definition 12 **mgf**

The moment generating function (mgf) of a discrete random variable X is defined to be;

$$M_X(t) = E(e^{tX}) = \sum_{x \in \mathscr{X}} e^{tx} P(x)$$

where $\mathscr{X}$ is the set of possible X values. The mgf() exists if $M_X(t)$ is defined for the $t \in [-h, h]$ and $h > 0$.

Note:

Some of the merits of using the mgf include the following properties.

1. If the mgf exists then, when $t = 0$, $M_X(0) = 1$ for any random variable.

2. If the mgf exists and is the same for two distributions then the two distributions are the same. This means mgf uniquely identify probability distributions.

3. If $mgf()$ exist then;

$$E(X'_r) = E(X^r) = \frac{d^r}{dt^r} M_X(t)|_{t=0} = M_X^{(r)}(0)$$

   So we can say

$$\mu = M'(0) \text{ and } \sigma^2 = M''(0) - (M'(0))^2$$

4. Let X have mgf $M_X(t)$ and let $Y = aX + b$. Then $M_Y(t) = e^{bt} M_X(at)$

4.4 Continuous Random Variables

A **Continuous random variable** on the other hand is a random variable that can take on a continuum (Any value in an interval). An example of this is the length of your finger. Remember a finger can have an infinite number of possible lengths.

Here we talk of values on an interval as opposed to particular values. Thus the probability density function of a continuous random variable

X is a mathematical function for which the area under the curve corresponding to any interval is equal to the probability that X will take a value in that interval. The **probability density function** is denoted by $f(x)$. The value $f(x)$ is called the probability density at x.
For example

$$f(x) = \begin{cases} 12x(1-x)^2 & 0 \leq x \leq 1, \\ 0 & \text{elsewhere} \end{cases}$$

For instance, the probability density at $x = 0.5$ is

$$f(0.5) = 12(0.5)(1-0.5)^2 = 1.5.$$

Example 4.3 *Consider the continuous random variable X, which represents the yield of a crop in tons per acre. Suppose the yield can take any value between 0 and 1 ton, and that X has the density above.*

Then

$$\begin{aligned} P(0.5 \leq X \leq 0.7) &= \int_{0.5}^{0.7} 12(x)(1-x)^2 dx \\ &= [6x^2 - 8x^3 + 3x^4]_{0.5}^{0.7} \\ &= 6 \times 0.7^2 - 8 \times 0.7^3 + 3 \times 0.7^4 - 6 \times 0.5^2 - 8 \times 0.5^3 + 3 \times 0.5^4 \\ &= \underline{0.2288} \end{aligned}$$

see figure 4.1. This means the probability that the crop yield will be between 0.5 and 0.7 is 0.2288

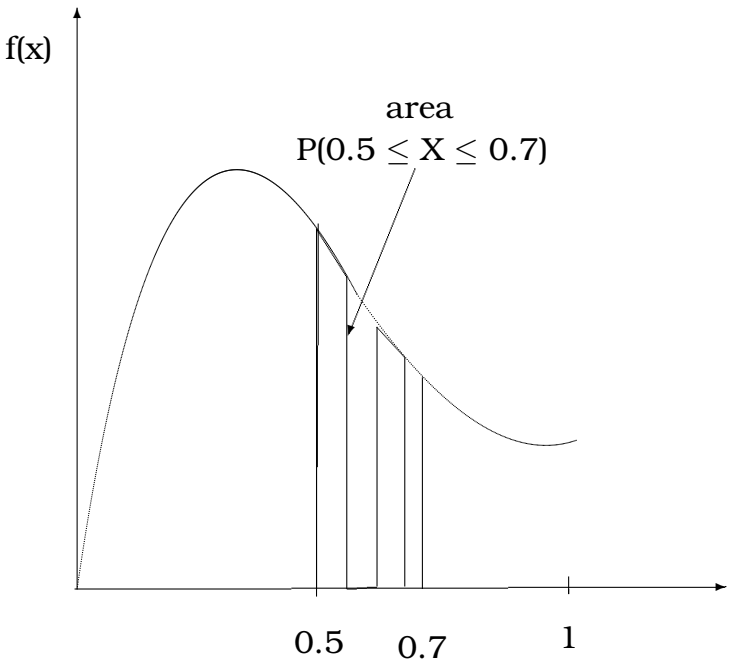


Figure 4.1: The area representing the probability

Some properties

As in the Discrete case above;

1. $f(x) \geq 0$ for x ϵ $[a, b]$ interval of definition.
2. $\int_a^b f(x) = 1$.

Note that, in this particular case,
$P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b) = P(a < X < b)$.

4.4.1 Cummulative Probability Function

The **cumulative Probability function** of a continuous random variable is denoted by $F(x)$ and defined by; $F(x)$=P($X < x$) for x $\epsilon(-\infty, \infty)$ The Probability that $X < x_0$ = P($X < x_0$)=F(x_0) is the area under the curve $f(x)$ to the left of x_0. i.e., using calculus $F(x_0)$=$\int_{-\infty}^{x_0} f(x)dx$. This is the cumulative distribiton function.

4.4.2 The expected value of a continuous random variable

The **expected value of a contionous variable** is defined as

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx.$$

We continue with our example.

Example 4.4

In example 4.3*, we calculate the expected value from,*

$$\begin{aligned} E[X] = & \int_0^1 xf(x)dx \\ = & \int_0^1 x \times 12x(1-x)^2dx \\ = & 6 \times 1^2 - 8 \times 1^3 + 3 \times 1^4 - 6 \times 0^2 - 8 \times 0^3 + 3 \times 0^4 \\ = & 6 - 8 + 3 \\ = & \underline{1.} \end{aligned}$$

4.4.3 The variance of continuous random variable

The variance, as in the discrete case is calculated with the summation replaced by the integral, as follows;

$$\begin{aligned} Var[X] = & \int_{-\infty}^{\infty}(x - E[X])^2 f(x)dx \\ = & \int_{-\infty}^{\infty} x^2 f(x)dx - E[X]^2 \\ = & E[X^2] - [E[X]]^2 \end{aligned}$$

Lets go through an example.

Example 4.5

$$\begin{aligned} Var[X] &= \int_0^1 (x - E[X])^2 f(x) dx \\ &= E[X - E[X]]^2 \\ &= E[X^2] - [E[X]]^2 \\ &= \int_0^1 x^2 \times 12x(1-x)^2 dx - 1^2 \\ \\ &= \tfrac{12}{4}x^4 - \tfrac{24}{5}x^5 + \tfrac{12}{6}x^6 \\ &= \underline{0.2.} \end{aligned}$$

Exercise 4.2

In example 4.3, find the cumulative distribution function of X.

Definition 13 **mgf**

The moment generating function (mgf) of a continuous random variable X is defined to be;

$$M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f(x)$$

The mgf() exists if $M_X(t)$ is defined for the $t \in [-h, h]$ and $h > 0$.

4.5 Empirical Distributions

There are is wide variety of probability distributions. However, only a limited number of types, or families, of probability distributions are used in practical applications. These fall under the two categories discrete and continuous.

Let us discuss these starting with some discrete probability distributions namely:

1. Bernoulli,
2. Binomial,
3. Poisson,
4. Discrete Uniform.

And some continuous density functions

1. Normal,
2. Continous Uniform,
3. Exponential.

There are many other distributions. However, the few distributions mentioned above fall under a class of distributions we call empirical distributions. We will keep meeting these distributions in Statistics.

4.6 Some discrete distributions

We will mention a few discrete empirical distributions. You will meet these and more during your study.

4.6.1 Bernoulli trials

A random variable X which has two possible outcomes say 0 and 1 is called a Bernoulli random variable.

The probability distribution of X is,

$$\begin{aligned} P(X=1) &= p \\ P(X=0) &= 1-p \\ i.e., \quad P(X=0) &= 1-P(X=1) \\ &= \underline{1-p} \end{aligned}$$

Example 4.6

Tossing a fair coin, you get a head or a tail each with probability $p = \frac{1}{2}$. Thus if a head is labelled 1 and a tail 0, the random variable X representing the outcome takes values 0 or 1 if the probability that $X = 1$ is p, then we have that

$$\begin{aligned} p = P(X=1) &= \tfrac{1}{2} \\ P(X=0) &= 1-p = 1-\tfrac{1}{2} \\ &= \tfrac{1}{2} \end{aligned}$$

Since events $X = 1$ and $X = 0$ are mutually exclusive.

4.6.2 Binomial Distribution

Suppose in an experiment there are two possible outcomes (failure and success) and that the probability of success is p. Suppose also that the experiment is repeated n times, the probability of x successes follows a Binomial distribution.

Let $X = X_1 + X_2 + \cdots + X_n$ where X_i are independent and identically distributed Bernoulli random variables, then X is called a binomial random variable. Thus;

$$\begin{aligned} P(x) &= P(X=x) \\ &= \tbinom{n}{x} p^x (1-p)^{n-x} \end{aligned}$$

i.e., for $x = 0, 1, \cdots, n$ and $0 < p < 1$, $\binom{n}{x} = \frac{n!}{x!(n-x)!}$

The quantities n and p are called **parameters** and they specify the distribution. Let us look at one application of the Binomial with parameters n and p i.e Bin(n, p).

Example 4.7
Bin(n, p) here means Binomial distribution with parameters n and p.

Find
(i) probability of getting 4 heads in 6 tosses of a fair coin.
(ii) E[X] and Var[X].

Solution

$$\begin{aligned} P(X=4) &= \binom{6}{4}\tfrac{1}{2}^4(1-\tfrac{1}{2})^{6-4} \\ &= 15 \times \tfrac{1}{16} \times \tfrac{1}{4} \\ &= \underline{\tfrac{15}{64}}. \end{aligned}$$

$$\begin{aligned} E[X] &= \sum x\binom{n}{x}p^x(1-p)^{n-x} \\ &= np \\ Var[X] &= \sum x^2\binom{n}{x}p^x(1-p)^{n-x} - (np)^2 \\ &= \underline{np(1-p)} \end{aligned}$$

4.6.3 Poisson Distribution

A Poisson random variable is a discrete random variable that can take integer values from 0 up to ∞. The parameter for this distribution is λ i.e., $P_0(\lambda)$.

An example of the application of the Poisson distribution follows, The number of individuals arriving at a bank teller per quarter hour X is a poisson random variable.

The Poisson probability function is

$P(x) = \frac{\lambda^x e_{-\lambda}}{x!}$, where $P(x) = P(X = x),\ x = 0,\ 1,\ \cdots, \infty$, and $0 < \lambda < \infty$

Example 4.8
The number of students arriving at a take away every 15minutes is a poisson random variable with parameter $\lambda = 0.2$. Find the probability that zero, one and two students arrive at the take away.

$$\begin{aligned} P(X=0) &= \frac{(0.2)^0 e^{-0.2}}{0!} \\ &= 0.8187 \textit{ (no students arrive)} \end{aligned}$$

$$\begin{aligned} P(X=1) &= \frac{(0.2)^1 e^{-0.2}}{1!} \\ &= 0.1637 \end{aligned}$$

$$\begin{aligned} P(X=2) &= \frac{(0.2)^2 e^{-0.2}}{2!} \\ &= 0.0164 \end{aligned}$$

Some properties

$$\begin{aligned} E[X] &= \sum \frac{x\lambda^x e^{-\lambda}}{x!} \\ &= \sum \frac{\lambda^x e^{-\lambda}}{(x-1)!} \\ &= \lambda \sum \frac{\lambda^{x-1} e^{-\lambda}}{(x-1)!} \\ &= \lambda \times 1 \\ &= \lambda. \end{aligned}$$

$$\begin{aligned} \text{Similarly } Var[X] &= \sum \frac{x^2\lambda^x e^{-\lambda}}{x!} - \lambda^2 \\ &= \lambda^2 + \lambda - \lambda \\ &= \lambda. \end{aligned}$$

4.6.4 Poisson Distribution's link with the Binomal Distribution

The Poisson distribution can be viewed as a limiting case of a Binomial distribution, when the number of trials n gets large and the the probability p the probability of success gets small.

In the Binomial distribution $P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$ we denote, $E(X) = np = \lambda$ so that

$$p = \frac{\lambda}{n}$$

which means

$$(1 - p) = 1 - \frac{\lambda}{n}$$

Thus, $P(X = x) = \binom{n}{x} \left(\frac{\lambda}{n}\right)^x (1 - \frac{\lambda}{n})^{n-x} \to \frac{e^{-\lambda}\lambda^x}{x!}$ as $n \to \infty$

Exercise 4.3

Show that the limiting distribution of the Binomial distribution is the Poisson distribution.

$$\lim_{n \longrightarrow \infty} P(x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

4.6.5 The Hypergeometric Distribution

We will at this stage mention a distribution related to the Binomial distribution. Suppose again we are looking at a population consisting of N items. Suppose there are two possible outcomes Success (S) and Failure (F) and that k of out comes are successes. If a random sample of n items is drawn and x of the items are successes, then x follows a **Hypergeomtric distribution** given by;

$$P(X = x) = \frac{\binom{S}{x}\binom{N-S}{n-x}}{\binom{N}{n}}$$

See Example 3.27.

4.6.6 Discrete Uniform distribution

The discrete Uniform random variable is a variable that can take on integer values within a given interval with equal probabilities.

For example let x be a discrete random variable which can assume s values.

Thus, the discrete Uniform Probability function is defined

$$P(X = x) = P(x) = \frac{1}{s}$$

$x = a,\ a + 1,\ ...,\ a + (s - 1)$ where $s > 0$ is the number of terms and a the first term.

Some properties

E[X]= a+ $\frac{s-1}{2}$
Var[X]= $\frac{s^2-1}{12}$

Example 4.9
Let x be a discrete random variable which can assume 5 values. Then

$X = \quad 0 \quad 1 \quad 2 \quad 3 \quad 4$

in this case, $s = 5$, therefore,

X	=	*0*	*1*	*2*	*3*	*4*
P(X)	=	*0.2*	*0.2*	*0.2*	*0.2*	*0.2*

That is, the discrete Uniform Probability function. One way is to use the formula,

$$P(X = x) = P(x) = \frac{1}{5}$$

$x = 0,\ 0 + 1,\ 0 + 2,\ 0 + 3,\ 0 + (5 - 1) = 4$, *thus* $s = 5$.

Exercise 4.4

Show that for the discrete Uniform distribution

E[X]=$a + \frac{s-1}{2}$, and Var[X]=$\frac{s^2-1}{12}$

4.7 Some Common Continuous Empirical distributions

Let us now look at some common continuous distributions used in Statistics. We will not discuss these distributions in detail, since you will have occasion to meet them in the module on probability.

4.7.1 The Continuous Uniform probability distribution

This distribution is also known as the **Rectangular distribution**. This is the continuous equivalent to the discrete Uniform distribution, discussed above. A continuous Uniform variable has a Uniform probability density over an interval. This distribution is of the form,

$$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b, \\ 0 & \text{elsewhere.} \end{cases}$$

Example 4.10
The marks from a certain exam are uniformity distributed over to 50 to 75. The density function for the marks is given by

$$f(x) = \begin{cases} \frac{1}{25} & 50 < x < 75, \\ 0 & \textit{elsewhere.} \end{cases}$$

Some properties

E[X]=$\frac{1}{2}(b+a)$

Var[X]=$\frac{(b-a)^2}{12}$

For the example above E[X]=62.5.

Exercise 4.4

For the Continuous discrete Uniform distribution, show that

E[X]=$\frac{1}{2}(a+b)$,
and Var[X]=$\frac{(b-a)^2}{12}$.

4.7.2 The Normal distribution

This is one of the most important probability distributions in ststistics. The normal probability density function is

$$f(x) = \begin{cases} \frac{1}{\sqrt{2\pi\sigma^2}} e^{\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]} & \text{where } -\infty < x < \infty,\ -\infty < \mu < \infty,\ \sigma > 0 \\ 0 & \text{elsewhere} \end{cases}$$

We shall use the normal distribution to a great extent in inference. This is a two parameter distribution, usually denoted by $N(\mu, \sigma^2)$. A random variable z is said to be standard normal if $Z = \frac{X-\mu}{\sigma}$ has mean $\mu = 0$ and variance $\sigma^2 = 1$, for X a normal random variable with mean μ and variance σ^2. Tables which give normal cumulative probabilities are widely available.

Some properties

Let X be a normally distributed random variable with mean μ and variance σ^2, i.e., $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma}$ is normally distributed with mean 0 and variance 1. We say Z has a standard normal distribution and $\frac{X-\mu}{\sigma}$ is called standardisation.

4.7.3 The Exponential probability distribution

An exponential random variable is a continuous random variable that can take on any positive value and is usually used to describe the time between events.

Example 4.11
Let X be the length of a long distance telephone call. Then X has an exponential distribution.

The density function is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } 0 \geq x < \infty, \ \& \ \lambda > 0 \\ 0 & \text{elsewhere.} \end{cases}$$

Some properties

$$E[X] = \frac{1}{\lambda}, \ and.$$

$$Var[X] = \frac{1}{\lambda^2}.$$

The parameter here is λ.

4.8 Memoryless Property

The memoryless property of a given probability distribution is mostly associated with the distribution of times. Consider tossing a coin, if your outcome was a head (H) and you toss the coin again does it mean at the second toss it is going to be a tail (T)? i.e., $P(T|H) = P(T)$ in other words ;

$$P(X_{t+1}|X_i, X_2, \ldots, X_x) = P(X_{t+1})$$

since these events are independent. The **memoryless** property means the future is independent of the past. There are certain probability distributions which have this property.

We will mention two in this course, in the discrete case a product of independent Bernoulli trials see coin tossing example.

The only memoryless continuous probability distribution is the exponential distribution. Suppose $X \in [0, \infty)$ is a continuous random variable, the probability distribution of X is said to be memoryless if for any real numbers t, $a \in [0, \infty]$

$$P(X > t + a | X > t) = P(X > a)$$

Exercise 4.5

1. *Let X be Exponentially distributed. Show that $E(X) = \frac{1}{\lambda}$.*
2. *Prove that the Exponential distribution is memoryless.*

4.9 Conclusion

We have briefly looked at some types of distributions. Some kinds of data tend to follow certain distributions. These distributions are not so many, rarely do we met data with an unknown distribution, especially when the sample size is large.

The number of people arriving at a certain bus stop in specified periods of time will generally follow a Poisson distribution. The time taken by radio active material to decay generally follows the exponential distribution. Such natural phenomenon make it necessary for us to study Empirical distributions.

Exercises

1. The heights (in metres) of children aged between 10 and 14 are recorded below

   $$1.4, \quad 1.5, \quad 1.6, \quad 1.2, \quad 1.63$$

   .

   (a) Find $\overline{x}$ and s^2, and use them to estimate μ and σ^2 respectively.

   (b) If the height is normally distributed, find $P(X < 1.5)$.

2. Find the cumulative probability distributions of the Exponential, Poisson and the continuous uniform distribution.

3. Suppose that X, the number years of schooling a student completes beyond the age of 14, is distributed normally with a mean of 4 and a standard deviation of 2.

   (a) What is the probability that a student completes more than 8 years of schooling beyond the age 14?

(b) What is the probability that a student completes 2 to 8 years of schooling beyond the age 14?

4. The probability that an individual gets a loan from a bank is 0.25. If 12 people applied for loans what is the probability that

   (a) at least 10 get loans,

   (b) more than 2 but less than 9 get loans,

   (c) exactly 5 get loans?

5. A new typist makes on average 1 error per page on her typing. What is the probability that she will make

   (a) no errors on a page?

   (b) at least 4 errors on at page?

   (c) makes no errors on a 5 page document? [hint: her error rate in now 5 per document]

Chapter 5

Estimation and Hypothesis Testing

5.1 Introduction

The objective of statistics is to make inferences about a population based on information contained in a sample. Statistical inference is mainly concerned with making inferences about population parameters. Methods of making inferences about parameters fall into two categories, making decisions concerning the value of a parameter or estimating/predicting the value of the parameter. The relevant information in a sample can be used to estimate the likely values of their associated population parameters.

We will often need to test the truth of some claims made about a population this will be covered under hypothesis testing.

5.2 Estimation

When a single statistic is used to estimate a population parameter we call it **A point estimator**. A good estimator of a population parameter should at least be an unbiased estimator. The next question is what is an unbiased estimator?

5.2.1 Point Estimation

A statistic is called an **unbiased estimator** *of a parameter if the expected value of the statistic, for all samples equals the parameter.*

Although this technical definition sounds complicated it simply means, if we were to take a sample from a population and compute a statistic, and continue taking samples and calculating the corresponding statistics, the average of these statistics should be equal to the population parameter.

Although estimation can be applied to an many population parameters,

in this Chapter we will concentrate on the population mean and standard deviation.

5.2.2 Point Estimate of the Mean

A point estimator of the population mean μ is given by the statistic $\overline{x}$. Suppose we have a random sample of observations of size $n, x_1, x_2, ` \ldots, x_n$. The point estimator of μ is given by $\hat{\mu}$ where;

$$\hat{\mu} = \overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{5.1}$$

One of the most important properties of the sample mean $\overline{x}$, is that it is an **unbiased** estimator of the population mean.

5.2.3 Point Estimate of the Population Standard Deviation

The point estimate of the population variance is given;

$$\hat{\sigma}^2 = s^2 = \frac{1}{n-1}\left[\sum_{i=1}^{n} x_i^2 - \frac{\left[\sum_{i=1}^{n} x_i\right]^2}{n}\right] \tag{5.2}$$

This means if you have a representative sample you can estimate the variance of the population. Similarly, the point estimate of population standard deviation is given by,

$$\hat{\sigma} = s = \sqrt{\frac{1}{n-1}\left[\sum_{i=1}^{n} x_i^2 - \frac{\left[\sum_{i=1}^{n} x_i\right]^2}{n}\right]} \tag{5.3}$$

5.2.4 Point Estimate of a Population Proportion π

Suppose we wish to estimate the proportion of the population that have a cellphone in a large city. There will not be enough time to go to every individual to gather the information in order to estimate the population proportion π.

If a representative sample of size n is taken and the number x of individuals with cellphones in the sample is recorded. We can estimate the population proportion using;

$$\hat{\pi} = p = \frac{x}{n} \tag{5.4}$$

Example 5.1
A company which manufactures and bottles chemicals, collected a bottle

from each batch of 20, which they dispatched for quality control purposes. They measured the volumes and found the following results.

350, 351, 348, 352, 350, 356, 348, 347, 348, 352, 354

1. *Estimate the mean volume of the whole consignment.*

2. *Estimate the standard deviation of the volume of chemicals.*

3. *Estimate the proportion of bottles which have less than 350ml in the whole consignment.*

Solution

1. *The point estimate of the consignment mean is*

$$\begin{aligned}\hat{\mu} &= \overline{x} \\ &= \frac{1}{n}\sum_{i=1}^{n} x_i \\ &= \frac{1}{11} \times 3856 \\ &= 350.545\end{aligned}$$

2.

$$\begin{aligned}\hat{\sigma} &= s \\ &= \sqrt{\frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}\right)} \\ &= \sqrt{\frac{1}{10} \times \left(1351782 - \frac{3856^2}{11}\right)} \\ &= 2.8058\end{aligned}$$

3.

$$\begin{aligned}\hat{\pi} &= p \\ &= \frac{x}{n} \\ &= \frac{4}{11} \\ &= 0.3636\end{aligned}$$

5.3 Confidence Intervals

Once we have calculated a point estimate, we can not be sure or state with any confidence the degree of accuracy. The sample mean is one point in the sample space and may not accurately estimate the true mean of the population. For this reason, the interval estimation technique is more useful and appropriate for providing estimates of population values.

When computing the confidence intervals for our population parameters, we need to consider the following;

1. the size n of the sample we are dealing with
2. whether we know the population standard deviation σ or not.

We can define a **confidence interval** as a range of values in which probability that the population parameter will lie in it is known. In confidence estimation we will be using the symbol $z_{\frac{\alpha}{2}}$, this value is found using normal distribution tables. $\frac{\alpha}{2}$ is the value of z for which P($Z > z_{\frac{\alpha}{2}}$)=$\frac{\alpha}{2}$

You will get the value of α from the question. For instance, if you are asked to construct a 95% confidence interval, then

$$\begin{aligned} 95\% &= 0.95 \times 100\% \\ &= (1 - 0.05) \times 100\% \end{aligned}$$

so that α = 0.05. So that $\frac{0.05}{2}$ = 0.025. We now do the opposite of what we did in the last chapter, i.e., we want to find the value of k that gives the $P(Z > k) = 0.025$. Looking down column $\phi(-z)$ (depending on the tables) until we get the probability closest to 0.025 we see it is 1.96. So in this case $z_{\frac{\alpha}{2}} = z_{\frac{0.05}{2}} = z_{0.025} = 1.96$

z	0.00	$\cdots$	0.06	$\cdots$	0.09
-3.4	0.0003	$\cdots$	0.0003	$\cdots$	0.0002
$\vdots$	$\vdots$		$\vdots$		$\vdots$
-1.9	0.0287	$\cdots$	0.025	$\cdots$	0.0233
$\vdots$	$\vdots$		$\vdots$		$\vdots$
3.4	0.9997	$\cdots$	0.9997	$\cdots$	0.9998

5.3.1 Case 1: Confidence Interval for μ, when σ^2 is known

A $(1 - \alpha)100\%$ confidence interval for μ, when σ known, is given by

$$\overline{x} - z_{\frac{\alpha}{2}} \times \frac{\sigma}{\sqrt{n}} \leq \mu \leq \overline{x} + z_{\frac{\alpha}{2}} \times \frac{\sigma}{\sqrt{n}} \tag{5.5}$$

where $\overline{x}$ is the mean of a sample of size n from a population with known variance σ^2 and $z_{\frac{\alpha}{2}}$ is a value in the standard normal distribution that leaves an area of $\frac{\alpha}{2}$ to the right of the normal curve.

5.3.2 Case 2: Confidence Interval for μ, when sample size is large, (i.e. $n > 30$) and σ^2 is unknown

A $(1-\alpha)100\%$ confidence interval for μ, when σ unknown and the sample large is given by

$$\overline{x} - z_{\frac{\alpha}{2}}\frac{s}{\sqrt{n}} \leq \mu \leq \overline{x} + z_{\frac{\alpha}{2}}\frac{s}{\sqrt{n}} \tag{5.6}$$

where $\overline{x}$ is the mean of a sample of size n from a population with sample variance s^2 and $z_{\frac{\alpha}{2}}$ is the standard normal distribution leaving an area of $\frac{\alpha}{2}$ to the right.

5.3.3 Case 3: Confidence Interval for μ, when the sample size is small, (i.e. $n < 30$) and σ^2 is unknown

If sample size is small and σ^2 is unknown we use a t-distribution.

A $(1-\alpha)100\%$ confidence interval for μ, when σ unknown and the sample is small is given by

$$\overline{x} - t_{n-1}(\alpha/2)\frac{s}{\sqrt{n}} \leq \mu \leq \overline{x} + t_{n-1}(\alpha/2)\frac{s}{\sqrt{n}} \tag{5.7}$$

where $\overline{x}$ and s are the mean and standard deviation respectively of a sample of size $n < 30$ from an approximate normal population, and $t_{n-1}(\alpha/2)$ is the value of the t-distribution, with $n-1$ degrees of freedom, leaving an area of $\alpha/2$ to the right.

5.3.4 Using Student's t Tables

You will recall from the last Chapter that if X is normally distributed with mean μ and standard deviation σ then $Z = \frac{\overline{X}-\mu}{\sigma/\sqrt{n}}$ will also be normally distributed with mean 0 and standard deviation 1.

If σ is unknown then we can not standardise X to get Z, instead, we create a new statistic called the student's t, where we substitute s for σ

$$T = \frac{\overline{X} - \mu}{\frac{s}{\sqrt{n}}} \tag{5.8}$$

This statistic follows what is called the **students' t** distribution with $n-1$ degrees of freedom. This is the value being used in Equation 5.7. We look up this value from our *students'* t tables which look like Table 5.1 below

Table 5.1: *Student's* t distribution Tables

	Amount of α in one tail					
				↓		
df	0.25	0.10	0.05	0.025	0.01	0.005
1	1.000	3.08	6.31	12.7	31.8	63.7
2	0.816	1.98	2.92	4.30	6.97	9.92
3	0.765	1.64	2.35	3.18	4.54	5.84
4	0.741	1.53	2.13	2.78	3.75	4.60
5	0.727	1.48	2.02	2.57	3.37	4.03
⇝6	0.718	1.44	1.94	2.45	3.14	3.71
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
29	0.583	1.31	1.70	2.05	2.46	2.76
For n> 30 z	0.0674	1.28	1.65	1.96	2.33	2.58

Suppose you want to look up $t_{n-1}(\alpha/2)$ where n=7, and $\alpha = 0.05$ then we have;

$$\begin{aligned} t_{n-1}(\alpha/2) &= t_{n-1}(\alpha/2) \\ &= t_{7-1}(0.05/2) \\ &= t_6(0.025) \\ &= 2.45 \qquad \text{See arrows} \end{aligned}$$

As the sample size n increases the t-distribution tends to the standard normal distribution. So we can read of the values of $Z_{\frac{\alpha}{2}}$ here. For example $z_{\frac{\alpha}{2}} = z_{\frac{0.05}{2}} = z_{0.025} = 1.96$, (see last row column 5). Let us look at a few examples. Each of these examples should help you appreciate when to apply which formula.

Example 5.2

A sample of 36 people at a rave night club revealed an average age of $\overline{x} = 19.38\ years$, *sample standard deviation* $s = 4.760$ *years. Determine a* 95% *confidence interval for the true age of individuals at the rave night club.*

Solution
n *is large (*$n > 30$*)*
σ *is unknown, so we use case 2.*

A 95% *confidence interval for* μ *is given by*

$$\overline{x} - z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} \leq \mu \leq \overline{x} + z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

$$19.38 - 1.96 \times \frac{4.760}{\sqrt{36}} \leq \mu \leq 19.38 + 1.96 \times \frac{4.760}{\sqrt{36}}$$

$$17.83 \leq \mu \ \leq 20.93$$

We are 95% confident that the true mean lies between 17.83 and 20.83.
Example 5.3
The scores below where recorded after a statistics examination was marked. The standard deviation of the marks is known to be 7.93%

73	*52*	*67*	*53*	*51*	*61*	*49*	*66*	*41*	*48*
52	*47*	*65*	*46*	*71*	*67*	*48*	*66*	*47*	*44*
63	*65*	*44*	*46*	*61*	*52*	*55*	*54*	*51*	*56*
49	*62*	*57*	*56*	*47*	*45*	*56*	*59*	*59*	*47*
48	*57*	*48*	*52*	*53*	*52*	*51*	*63*	*68*	*53*

Determine the 98.44 confidence interval of the average weight of the marks.

Solution

$n = 50,\ \overline{x} = 54.78\%,\ s = 7.73\%$

σ *is known, and* n *is large, so we use case 1.*

Then a 98.44% *confidence interval for the mean mark is*

$$\overline{x} - z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} \leq \mu \leq \overline{x} + z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

$$54.78 - 2.42 \times \frac{7.93}{\sqrt{50}} \leq \mu \leq 54.78 + 2.42 \times \frac{7.93}{\sqrt{50}}$$

$$52.07 \leq \mu \ \leq 57.49$$

We are 98.44% confident that the true mean mark lies between 52.07 and 57.49.
Example 5.4
The weights of seven similar containers of a chemical are recorded below.

277.83 289.17 294.84 277.83 283.5 289.17 272.16

Find a 95% confidence interval for the mean of such containers, assuming an approximate normal distribution.

Solution
$\overline{x} = 283.5, n = 7, s = 8.0186$ *from calculator.*

σ is unknown, n is small, so we use case 3.
$t_6(0.025) = 2.45$

Then a 95% confidence interval for μ is

$$\begin{aligned}
&\left(\overline{X} - t_{n-1}(\alpha/2) \times \tfrac{s}{\sqrt{n}} \quad , \quad \overline{X} + t_{n-1}(\alpha/2) \times \tfrac{s}{\sqrt{n}}\right) \\
&\left(283.5 - 2.45 \times \tfrac{8.0186}{\sqrt{7}} \quad , \quad 283.5 + 2.45 \times \tfrac{8.0186}{\sqrt{7}}\right) \\
&(283.5 - 7.4253 \quad , \quad 283.5 + 7.4253) \\
&(276.0747 \quad , \quad 290.9253)
\end{aligned}$$

Therefore, a 95% confidence interval for μ is $(276.0747 \leq \mu \leq 290.9253)$.
Confidence interval estimation can be extended to cover other parameters like proportions, the standard deviation as well as differences between two means and a ratio of standard deviations.

The ideas used in the construction of confidence intervals can be extended to help determine the size of a sample that will lead us to estimate the mean to any desired degree of accuracy.

5.4 Sampling to a Desired Precision

Let $x_1, x_2, \ldots, x_n$ be a random sample from $N(\mu, \sigma^2)$ and let $\overline{x}$ be sample mean and and σ be population standard deviation respectively. Then the minimum sample size required for $\overline{X}$ to be within ϵ of the true mean μ with a $100(1-\alpha)\%$ probability is

$$n = \left[\frac{Z_{\frac{\alpha}{2}\times\sigma}}{\epsilon}\right]^2$$

Example 5.5
The Manager of a department wants the margin of error of the mean number of calculation errors in reports in his department to be within $\pm$ 3 points for the year 2006 trainees. The extent to which this error is likely to occur is 0.95. What sample size should he take of reports if the standard deviation is known to be 9.2?

Solution

$$\begin{aligned} n &= \left[\frac{z_{\frac{\alpha}{2}} \times \sigma}{\epsilon}\right]^2 \\ &= \left[\frac{1.96 \times 9.2}{3}\right]^2 \\ &= 36.128 \end{aligned}$$

Therefore, 36 observations is the minimum needed sample.

Exercise 5.1

1. *Show that $\overline{x}$ is an unbiased estimator of μ.*
2. *Show that $var(\overline{x}) = \sigma^2/n$.*
3. *Show that $\widehat{s^2}$ is an unbiased estimator of σ^2.*
4. *Suppose*

   22.4 31.6 -3.8 34.0 -35.2 -21.5 23.0 10.8 23.2

   is a random sample from $N(\mu, \sigma^2)$. Construct a 90% confidence interval for the mean μ in each of the following cases.

   (a) $\sigma^2 = 7.24$ is known.

   (b) σ^2 is unknown.

5. *A sample of 44 retired gentlemen was drawn from a group of pensioners and their weekly cigarette expenses recorded in order to estimate the population mean cigarette expenses. A mean of $325.00 and a standard deviation of $125.00 was found from the sample.*

   (a) Find a 99% confidence interval of the true mean.

   (b) Estimate μ, the mean amount spent on cigarettes during the past year.

   (c) How large a sample is need so that one is

      i. 95% confident that the sample mean will be within 0.1 of the true mean?

      ii. 98% confident that the sample mean will be within 0.5 of the true mean?

We can use confidence intervals to test the validity of a claim. Tests of claims are best handled by hypothesis testing.

5.5 Hypothesis Testing

Often before you engage in any study or eventually make any recommendations. You will need supporting evidence. It has become generally acceptable (among NGOs, private and government institutions) that recommendations and findings be accompanied by statistical evidence. This is often in the form of statistical tests.

The testing of a statistical hypothesis is perhaps the most important part of decision making.

5.5.1 What is a Statistical Hypothesis

A statistical hypothesis is an assumption or statement, which may or may not be true, concerning one or more populations.

The truth or falsity of a statistical hypothesis is never known with certainty unless we examine the entire population. A random sample is taken from the population of interest and the information contained in this sample is used to decide whether the hypothesis is likely to be true or false. Evidence that is inconsistent with the stated hypothesis leads to a rejection whereas evidence supporting the hypothesis leads to its acceptance. For example, One might want to test the the hypothesis that the pass rate for an exam is always 40%, or test the claim that women live longer than man. There are two types of hypothesis, a null and alternative hypothesis.

The Null Hypothesis H_0

The null hypothesis usually denoted H_0 is a claim about a population parameter or characteristic.

The Alternative Hypothesis H_1

The alternative hypothesis usually denoted by H_1, is expresses the way in which the value of a population parameter or characteristic in a statistical model may deviate from that specified in the null hypothesis.

We accept the null hypothesis if we have insufficient evidence to reject it. It may not in fact be true. This leads us to two possible outcomes, we can reject the null hypothesis when in fact it is correct or we can accept it when in fact it is wrong. A wrong decision could lead to wrong conclusions. Which makes it necessary to study these types of errors. There are two main types of errors we can make, these are;

1. **type I error**

2. **type II error**.

The rejection of the null hypothesis when in fact it is true is called a **Type I error** while the acceptance of the null hypothesis when in fact it is false is called a **Type II error**.

The probability of committing a Type I error is known as the **the level of significance** of the test, it is denoted by α. This is the value we were using in confidence interval construction. The value of α can vary at the researcher's discretion. However a value of α= 0.05 is a common benchmark for reasonable doubt

The probability of accepting the null hypothesis when in fact it is false is denoted by β. This probability helps us determine the power of our test. The **Power of a test** can be defined as the probability of a test to make a correct decision. So we can say the power of a test = 1- P(making a type two error)=1-β

5.5.2 How Do You Formulate a Hypothesis test

When testing a hypothesis we follow the steps below;

1. **State the Hypothesis**
   Null Hypothesis (H_o): There is no difference between population parameter and given parameter.
   Alternative Hypothesis (H_a): There is a difference between population parameter and given parameter.

   Note: The alternative hypothesis will indicate whether a 1-tailed or a 2-tailed test is utilized to reject the null hypothesis.

2. **Set the Rejection Criteria**
   This determines how different the parameters and/or statistics must be before the null hypothesis can be rejected. This "region of rejection" is based on α the significance level. The point of rejection is known as the **critical value**.

3. **Compute the Test Statistic**
   The collected data are converted into standardized scores for comparison with the critical value.

4. **Decide Results of Null Hypothesis**
   If the test statistic equals or exceeds the region of rejection bracketed by the critical value(s), the null hypothesis is rejected.

5.5.3 Terms Used In Hypothesis Testing

A **critical value** is a value that separates the acceptance region from the rejection region.

An **acceptance region** is a range of values of the sample statistics centered about the null hypothesised population parameter that would lead to the acceptance of the null hypothesis for values of the sample statistic which fall within its limits.

A **rejection region** is the range of values of sample statistic value that would lead to the rejection of the null hypothesis for values of the sample statistic which fall within its limits.

A **two tailed test** has an area of rejection both below and above the hypothesised value.

A **one sided and right tailed test** has an area of rejection which lies above the null hypothesised value of the population parameter

A **one sided and left tailed test** has the area of rejection which lies below the null hypothesised value of the population parameter.

A **test statistic** T is a value calculated from the sample data which is used to decide whether or not H_0 should be rejected.
In this Chapter we will confine our discussion of hypothesis testing to tests concerning the population mean. The hypotheses we will discuss are;

1. $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$.
2. $H_0 : \mu \leq \mu_0$ versus $H_1 : \mu > \mu_0$.
3. $H_o : \mu \geq \mu_0$ versus $H_1 : \mu < \mu_0$.

5.5.4 Hypothesis Testing for a Single Population Mean

Let us now proceed to test a hypothesis. When calculating the test statistic of a single mean, one has to ascertain whether the variance is known or not and also have an appreciation of the sample size.

The following test statistics are used depending on whether σ is known.

Case 1.: When σ^2 is known: Use Z is, where;

$$z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \tag{5.9}$$

Case 2.: When σ^2 is unknown and n is < 30: Use T where;

$$T = \frac{\overline{x} - \mu}{\frac{s}{\sqrt{n}}} \tag{5.10}$$

Case 3.:When σ^2 is unknown and n large Use Z where;

$$z = \frac{\overline{x} - \mu}{\frac{s}{\sqrt{n}}} \tag{5.11}$$

5.5.5 Testing $H_0 : \mu = \mu_0$ Versus $H_1 : \mu \neq \mu_0$ When σ^2 is Known

The test statistic is

$$Z_{calc} = \frac{\overline{X} - \mu_o}{\frac{\sigma}{\sqrt{n}}} \tag{5.12}$$

Then $|Z_{calc}|$ tends to be large if H_0 is false and small otherwise. An α-size test is to reject H_0 if $|Z_{calc}| > z_{\frac{\alpha}{2}}$

5.5.6 Testing $H_0 : \mu = \mu_0$ Versus $H_1 : \mu \neq \mu_0$ When σ^2 is Unknown $n < 30$

The test statistic is

$$T_{calc} = \frac{\overline{X} - \mu_o}{\frac{s}{\sqrt{n}}} \tag{5.13}$$

Then $T \sim t_{n-1}$ An α-size test is to reject H_0 if $|T_{calc}| > t_{n-1}(\frac{\alpha}{2})$

5.5.7 Testing $H_0 : \mu = \mu_0$ Versus $H_1 : \mu \neq \mu_0$ When σ^2 is Unknown, n Large

The test statistic is

$$Z_{calc} = \frac{\overline{X} - \mu_o}{\frac{s}{\sqrt{n}}} \tag{5.14}$$

An α-size test is to reject H_0 if $|Z_{calc}| > z_{\frac{\alpha}{2}}$

Example 5.7
The ages of a sample employees who come very early to work were observed in one financial institute. The following information from a sample was obtained. $\overline{x} = 26.3$, $\sigma^2 = 9$ and $n = 36$. You are required to test the hypothesis that the mean age is 25.

Solution
Since σ is known, we are using Case 1, the Z distribution. The test is a two tailed test.
$\alpha = 0.05 \Rightarrow Z_{0.025} = 2.81$

$H_0 : \mu = 25$ *versus*
$H_1 : \mu \neq 25$
The critical region is $Z < -2.81$ *and* $Z > 2.81$*. Thus, reject* H_0 *if* $|Z_{calc}| > 2.81$*. The test statistic is*

$$\begin{aligned} Z_{calc} &= \frac{\overline{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \\ &= \frac{26.3 - 25}{\frac{3}{\sqrt{36}}} \\ &= 2.6 \end{aligned}$$

Since $|Z_{calc}| < 2.81$*, that is,* $2.6 < 2.81$*, thus, we fail to reject* H_0 *and conclude that the average age of early comers is equal to 25.* **Example 5.8**
It is desired that a statistician test the hypotheses given below and make correct recommendations based on the information gathered from a sample of measurements of 40mm pipes which were supplied by a new supplier.
$H_0 : \mu = 40$*,* $H_1 : \mu \neq 40$*,* $n = 18$*,* $\overline{x} = 34$*, and* $s^2 = 64$*.*

Solution
Since n *is small and* σ *unknown we use the t-distribution. The test is a two tailed test.*
$H_0 : \mu = 40$ *versus*
$H_1 : \mu \neq 40$
$\alpha = 0.01, t_{n-1}(\frac{\alpha}{2}) = t_{0.005}(17) = 2.90$
The critical region is $T < -2.90$ *and* $T > 2.90$*. Thus reject* H_0 *if* $|T_{calc}| > 2.90$*. The test statistic is*

$$\begin{aligned} T_{calc} &= \frac{\overline{X} - \mu_o}{\frac{s}{\sqrt{n}}} \\ &= \frac{34 - 40}{\frac{8}{\sqrt{18}}} \\ &= -3.182 \end{aligned}$$

Since $|T_{calc}| > 2.90$*, that is,* $3.182 > 2.90$*, we reject* H_0 *and conclude that the true mean is not equal to 40. The pipes are not 40mm.*

Exercise 5.3

1. *A geologist is testing the hypothesis that the melting point of an unusual carbon substance is* 1946^oC*. He makes 7 determinations and obtained the values of 1944, 1947, 1945, 1947, 1949, 1946 and* 1944^oC*. What conclusions can you draw at a significance level of 0.05?*

2. *An insurance broker claims that teachers spend an average of R65.00 per month life insurance. To test this hypothesis a random sample of 124 teachers was taken and it was observed that on average teachers spend about R52.24 with a standard deviation of R11.64. Using a 0.01 significance level, does this support the broker's claim.*

3. *A baker stated that on average the number of loaves bread sold daily is 3 000 with a standard deviation of 300. An employer want to test the accuracy of this statement. A random sample of 36 days showed the average daily sales were 3 150. Test at the* 1% *level of significance if the bakery's statement can be accepted.*

Let us discuss one sided or one tailed tests.

5.5.8 Testing $H_0 : \mu = \mu_0$ Versus $H_1 : \mu > \mu_0$

The ideas discussed in Section 5.5.4, cases 1, 2 and 3 can be extended to testing this one sided test. The test statistic to be used in this test will be determined by whether we know σ or not and by the sample size n (whether its a small sample or a large sample), see Section 5.5.4.

We wish to test the hypotheses
$H_0 : \mu = \mu_0$ Versus
$H_1 : \mu > \mu_0$

Depending on the sample size and knowledge of σ, we reject H_0 if $test\ statistic > tabulated\ value(\alpha)$

1. If σ is known, then use $Z = \frac{\overline{x}-\mu}{\frac{\sigma}{\sqrt{n}}}$ and reject H_0 if $z > z_\alpha$
2. If σ is unknown and $n < 30$, then use $t = \frac{\overline{x}-\mu}{\frac{s}{\sqrt{n}}}$ and reject H_0 if $t > t_\alpha(n-1)$
3. If σ is unknown and $n > 30$, then use $z = \frac{\overline{x}-\mu}{\frac{s}{\sqrt{n}}}$ and reject H_0 if $Z > z_\alpha$

Example 5.9
The average distance traveled by a small engined vehicle on 10 litres of petrol is 162.5 kms with a standard deviation of 6.9 kms. Is there reason to believe that adding a new additive to petrol increases the distance travelled on 10 litres if a random sample of 50 small cars has an average of 165.2 kms per 10 litres, at the 5% *level of significance.*

Solution

$$H_o : \mu = 162.5$$

$$H_1 : \mu > 162.5$$

n = 50, $\overline{x} = 165.2$ *and* σ *= 6.9. Since* σ *is known we use the Z distribution. The test is a one tailed test.*
$\alpha = 0.05, z_\alpha = z_{0.05} = 1.645$

The critical region is $Z > 1.645$*. Thus reject* H_0 *if* $Z_{calc} > 1.645$*.*

The test statistic is

$$\begin{aligned} Z_{calc} &= \frac{\overline{X} - \mu_o}{\frac{\sigma}{\sqrt{n}}} \\ &= \frac{165.2 - 162.5}{\frac{6.9}{\sqrt{50}}} \\ &= 2.7669 \end{aligned}$$

Since $Z_{calc} > z_{0.05}$*, that is,* $2.7669 > 1.645$*, thus we reject* H_o *and conclude that the the additive increases the distance travelled on 10 litres of petrol.*

Exercise 5.2

1. *A supermarket claims that customers to its stores spend on average 25 minutes carrying out their purchases. A consumer body wants to verify this claim. They observed then entry and departure times from supermarkets in the chain of 24 random selected customers. The sample average time was half an hour with a standard deviation of 14.1 minutes. Test the validity of the supermarket's belief at the 2.5% level of significance.*

2. *It is claimed that an automobile is driven on average less than 12 000kms per year. To test this claim a random sample of 100 automobile owners are asked to keep a record of the kilometres they travel. Would you agree with this claim if the random sample showed an average of 14 500 kilometres and a standard deviation of 2 400kilometres? Use a 0.01 level of significance.*

5.5.9 Testing $H_0 : \mu = \mu_0$ Versus $H_1 : \mu < \mu_0$

The test statistic to be used in this test will be determined by whether we know σ or not and by the sample size n (whether its a small sample or a large sample), as we did in Section 5.5.8.

We wish to test the hypotheses
$H_0 : \mu = \mu_0$ Versus
$H_1 : \mu < \mu_0$

Depending on the sample size and knowledge of σ, we reject H_0 if test statistic <tabulated value(α)

The critical values, for particular tests, sample sizes and significance levels, are available available in tables. Remember, sometimes, we have to extrapolate the critical values from tables.

1. If σ is known, then use $z = \frac{\overline{x}-\mu}{\frac{\sigma}{\sqrt{n}}}$ and reject H_0 if $z < z_\alpha$
2. If σ is unknown and $n < 30$, then use $t = \frac{\overline{x}-\mu}{\frac{s}{\sqrt{n}}}$ and reject H_0 if $t < -t_\alpha(n-1)$
3. If σ is unknown and $n > 30$, then use $z = \frac{\overline{x}-\mu}{\frac{s}{\sqrt{n}}}$ and reject H_0 if $z < -z_\alpha$

Example 5.10
The average length of time spent in a bank queue has been 50 minutes with a standard deviation of 10 minutes. A new banking system is testing a new software. If a random sample of 12 clients had an average banking time of 42 minutes with a standard deviation of 11.9 minutes under the new system. Test the hypothesis that the population mean is now less than 50 using a level of significance of

(i) 0.05 and

(ii) 0.01.

Assume the population of times to be normal.

Solution.
Let $H_o : \mu = 50\ minutes$ *versus* $H_1 : \mu < 50\ minutes$.

(i) Using t-distribution at $\alpha = 0.05 \Rightarrow t_{n-1}(0.05) = t_{11}(0.05) = 1.80$. *Reject* H_0 *if* $T < -1.80$, *that is, critical region is* $[-\infty, -1.80]$. *The test statistic is*

$$\begin{aligned} T &= \frac{\overline{X} - \mu_o}{\frac{s}{\sqrt{n}}} \\ &= \frac{42 - 50}{\frac{11.9}{\sqrt{12}}} \\ &= -2.3288 \end{aligned}$$

Since $-2.3288 < -1.80$, *that is, the test statistics is in the critical region, we reject the null hypothesis at* 5% *level of significance*

(ii) Using t-distribution at $\alpha = 0.01 \Rightarrow t_{11}(0.01) = t_{11}(0.01) = 2.72$ Reject H_o if $T < -2.72$, that is critical region is $[-\infty, -2.72]$. Since $-2.33 > -2.72$, we do not reject H_o at 1% level of significance.

Note*: We did note use $\sigma = 10$, since the sample partains to a new banking system with new software so the population is not the same.*

This essentially means that the true mean is likely to be less than 50 minutes but does not differ significantly to warrant the high cost that would be required to overhaul the current banking system.

Exercise 5.3

1. *A cement manufacturer packs 50kg bags. To see if the manufacturer puts enough cement in the bags, the contents of random sample of 200 such bags were weighed. The average contents turned out to be 48kgs with a standard deviation of 0.12. State the null and research hypothesis for this problem. Hence test at the 0.05 significance level, whether or not the manufacturer satisfies the requirement?*

2. *A scientist claimed that mice with an average life span of 28 months will live to be about 43 months old when 45% of the calories in their food are replaced by vitamins and proteins. Is there any reason to believe that the mean age is less than 43 months if 54 mice that are placed on this diet have an average life of 38 months with a standard deviation of 5.8 months. Use a 0.025 level of significance.*

3. *Redo number 2, and use a sample of size 20 instead 54 are placed on this diet.*

Note

In inference we use samples to make inferences about a population. You will never have to calculate the population standard deviation. You are either given its value or you do not know it.

5.5.10 The p-value

What is the p-value? You will find this value under a variety of names, some of these are; **critical level**, **the probability value** and the **associated probability**. Suppose, for a given null hypothesis H_0, we calculate a test statistic, say k_{test}, then

$$p - value = P(k \geq k_{test}|H_0).$$

where k is a real number. Generally we reject H_0, at the α level, if the $p - value < \alpha$.

You can can also say the $p - value$ is the smallest value of α for which test results are statistically significant.

The probability value is commonly used in statistical computer packages. Once calculated you do not need statistical tables.

Exercise 5.4

1. *Suppose it is known that a variable X is a normally distributed with a mean of 340 minutes. If a random sample of 20 observations has an average of 332 with a standard deviation of 43 minutes. Test the hypothesis at the 0.025 level of significance at that $\mu = 340$ minutes against the alternative $\mu < 340$ minutes.*

2. *Suppose the observations*

   1.4 -2.6 1.3 2.1 -2.2 -3.6 3.2 1.8 2.4

   are independent and identically distributed as $N(\mu, \sigma^2)$. Test the hypothesis at 5% significance level that the mean of the population is 2.2.

3. *A random sample of size 14 from a normal distribution has a mean $\overline{X}$ = 33.2 and a standard deviation of $s = 5.41$. Does this suggest, at the 0.05 level of significance, that the population mean is greater than 32?*

4. *A manufacturer claims that the average life of batteries produced by his firm is at least 30 months. You disagree, contending that the average life of the batteries is less than 30 months. A random sample of 12 batteries has a mean of 38.7 months and a standard deviation of 18 months. Perform the appropriate hypothesis test. Use a significance level of 0.05.*

5. *The manufacturer of an over the counter pain reliever claims that its product brings pain relief to headache sufferers in less than 3.5 minutes on average. To be able to make this claim in its television advertisements the manufacturer was required by a particular television network to present statistical evidence in support of the claim. The manufacturer reported that for a random sample 50 headache sufferers, the mean time to relief was 3.3 minutes and the standard deviation was 66 seconds. Does this data support the manufacturer's claim. Test using $\alpha = 0.05$.*

6. *What is the advantage of using the p-value over the critical value?*

5.6 Summary

In this chapter we discussed the concept estimation and hypothesis testing. These two concepts fall under the topic statistical inference. In statistical inference we use sample data to make inferences about the population from which the sample came.

You should now be able to estimate the point and interval estimate of the population. You should also be able to test a variety of hypotheses, depending on the question.

Chapter 6

Introduction to Correlation and Regression Analysis

6.1 Introduction

The **Correlation** analysis is the study of the strength of the relationship between **any** pair of variables. Thus, correlation measures how strongly a pairs of variables variables are related. The term **Regression analysis** on the other hand describes a collection of statistical techniques that quantifies how one variable depends on another (or several other variables). Regression analysis is now perhaps the most widely used method of modelling relationships.

6.2 Scatter Plots

The easiest way to visualise the nature of the relationship between variables is to use a scatter plot. A **scatter plot** is a plot of one variable against another. You may use three variables to get a three-dimensional plot. By looking at the scatter plot, you can get a feel of the relationship between variables, that is, whether the variables are **linearly** related or related in some other way.

Note

A scatter plot is made up of the axes and the points where values meet. The points should not be joined together. An example of a scatter plot is shown in Figure 6.1.

Example 6.1

The incomes and amounts spent on entertainment by a sample of individuals at a bar was recorded and a scatter plot constructed.

The scatter plot shows that as income increases the amount spent on entertainment also increases.

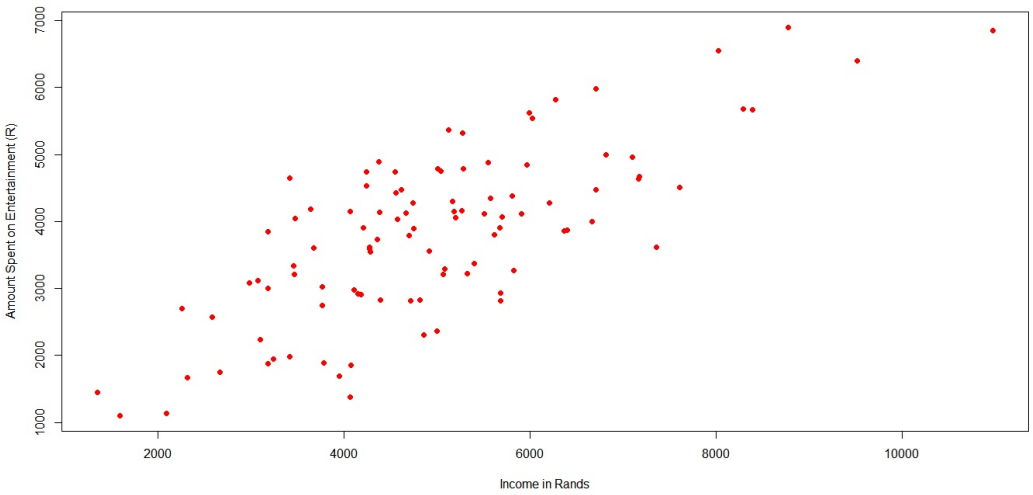


Figure 6.1: Scatter plot of incomes versus entertainment

It is important to construct a scatter plot in order to get a feel of what you will expect the relationship to be.

6.3 Correlation

Correlation is the **intensity** or **strength** of the relationship between two variables. It is a measure of the extent to which variables are related or associated. If the correlation between two variables is zero, then the two variables are not related. On the other hand, a correlation of 1, means that there is a perfect linear relationship between the two variables. Here, "perfect" means an exact relationship. How is the correlation calculated?

6.3.1 Correlation Coefficient

Symbolically, the population correlation between two variables X and Y is given by the correlation coefficient ρ_{xy}. This is defined by:

$$\rho_{xy} = \frac{cov(X,Y)}{\sqrt{var(X)var(Y)}}.$$

We rarely deal with population parameters as they stand. We often estimate the population parameters on the basis of samples. In this particular case, ρ_{xy} is estimated from a sample, say (x_1, y_1), (x_2, y_2), ..., (x_n, y_n), to give $\hat{\rho}_{xy}$.

This quantity is calculated using the formula given in Equation 6.1:

$$\hat{\rho}_{xy} = \frac{\sum_{i=1}^{n}[X_i - \bar{X}][Y_i - \bar{Y}]}{\sqrt{\sum_{i=1}^{n}[X_i - \bar{X}]^2 \times \sum_{i=1}^{n}[Y_i - \bar{Y}]^2}} \tag{6.1}$$

There are other measures of association, but for now we shall work with the one above. ρ_{xy} is confined to the interval $[-1 < \rho_{xy} < 1]$. If ρ_{xy}= -1, then there is an exact inverse relationship between x and y. If x increases, then y decreases and vice-versa.

Example 6.2
Consider the number of expensive goods n sold by a company (which got the goods at a very cheap cost price) and the profit p. $\rho_{np} \approx 1$ in this case. This means that the more the sales, the higher the profit. ("$\approx$" here means approximately) If $\rho_{np} \approx 0$, then we would conclude that, there is no relationship between the profit and number of sales.

On the other hand, $\rho_{np} \approx -1$ suggests that the more the sales the lower the profit. The product is probably being sold at less than the cost price.

6.3.2 Correlation Matrix

You will notice that the correlation coefficient is defined for two variables. Now let us assume we have many variables. Take, for instance, p variables, then how do we handle the correlation problem? Let these variables be $X_1,\ X_2,\ \cdots,\ X_p$. Then we can express the different combinations of correlation coefficients in a matrix, known as a **correlation matrix**, as follows:

$$\rho = \begin{pmatrix} 1 & \rho_{x_1x_2} & \cdots & \rho_{x_1x_p} \\ \rho_{x_2x_1} & 1 & \cdots & \rho_{x_2x_p} \\ . & . & . & . \\ . & . & . & . \\ . & . & . & . \\ \rho_{x_px_1} & \rho_{x_px_2} & \cdots & 1 \end{pmatrix}$$

$\rho_{x_ix_j}$ for $i,\ j = 1,\ 2,\ \cdots,\ p$, represents the correlation between variables X_i and X_j. Therefore by looking at the correlation matrix, we can tell which variables are related. Notice that the correlation between X_k and itself is always 1. For example, let us look at the sample correlation

coefficient for any k, $\hat{\rho}_{x_k x_k}$. Then:

$$\begin{aligned}
\hat{\rho}_{x_k x_k} &= \frac{\sum_{i=1}^{n}[X_{ki}-\overline{X_k}][X_{ki}-\overline{X_k}]}{\sqrt{\sum_{i=1}^{n}[X_{ki}-\overline{X_k}]^2 \times \sum_{i=1}^{n}[X_{ki}-\overline{X_k}]^2}} \\
&= \frac{\sum_{i=1}^{n}[X_{ki}-\overline{X_k}]^2}{\sqrt{[\sum_{i=1}^{n}[X_{ki}-\overline{X_k}]^2]^2}} \\
&= \frac{\sum_{i=1}^{n}[X_{ki}-\overline{X_k}]^2}{\sum_{i=1}^{n}[X_{ki}-\overline{X_k}]^2} \\
&= 1
\end{aligned}$$

This gives us the matrix:

$$\hat{\rho} = \begin{pmatrix} 1 & \hat{\rho}_{x_1 x_2} & \cdots & \hat{\rho}_{x_1 x_p} \\ \hat{\rho}_{x_2 x_1} & 1 & \cdots & \hat{\rho}_{x_2 x_p} \\ . & . & . & . \\ . & . & . & . \\ . & . & . & . \\ \hat{\rho}_{x_p x_1} & \hat{\rho}_{x_p x_2} & \cdots & 1 \end{pmatrix}$$

Let us have an example to "internalise" this concept.

Example 6.3
Suppose you are given three variables z_1, z_2 and z_3 representing the coded tensile strength, melting point and amount of Titanium in a new alloy respectively. Use the following data to calculate the correlation matrix.

z_1	z_2	z_3
12	3	0.2
23	7	0.8
9	2	0.1
30	10	1.0

$$\begin{aligned}
\hat{\rho}_{z_1 z_2} &= \frac{\sum_{i=1}^{n}[z_{1i}-\overline{z_1}][z_{2i}-\overline{z_2}]}{\sqrt{\sum_{i=1}^{n}[z_{1i}-\overline{z_1}]^2 \times \sum_{i=1}^{n}[z_{2i}-\overline{z_2}]^2}} \\
&= 0.999 \\
\hat{\rho}_{z_1 z_3} &= \frac{\sum_{i=1}^{n}[z_{1i}-\overline{z_1}][z_{3i}-\overline{z_3}]}{\sqrt{\sum_{i=1}^{n}[z_{1i}-\overline{z_1}]^2 \times \sum_{i=1}^{n}[z_{3i}-\overline{z_3}]^2}} \\
&= 0.993 \\
\hat{\rho}_{z_2 z_3} &= \frac{\sum_{i=1}^{n}[z_{2i}-\overline{z_2}][z_{3i}-\overline{z_3}]}{\sqrt{\sum_{i=1}^{n}[z_{2i}-\overline{z_2}]^2 \times \sum_{i=1}^{n}[z_{3i}-\overline{z_3}]^2}} \\
&= 0.988
\end{aligned}$$

$$\hat{\rho} = \begin{pmatrix} 1 & 0.999 & 0.993 \\ 0.999 & 1 & 0.988 \\ 0.993 & 0.988 & 1 \end{pmatrix}$$

The correlation between any two of the variables is very high meaning that the variables are highly related. They either increase or decrease

together. Addition of Titanium increases the tensile strength and melting point of the alloy. The process seems to produce an alloy whose strength can be increased or decreased, by changing the amount of Titanium.

Exercise 6.1

Consider the following data collected on two variables which are suspected to be related. The variable x represents the grade scores of students in a class and y the number of students with the same grade.

Grade x	*2*	*5*	*9*	*3*	*6*	*7*
Frequency y	*34*	*45*	*59*	*40*	*50*	*48*

The data above shows the cross-tabulation relating the two variables x and y. From this illustration, it's clear that there is a linear relationship between the two variables x and y. As one variable increases, the other variable also increases. This means that as the grade increases the number of students with high grades also increases.

Calculate the correlation coefficient of the grade and the frequency. Comment on your result.

When we calculated the correlation coefficient we quantified the strength of the relationship between two variables, in regression analysis we study how one variable (**dependent** variable) depends on other variables (**independent** variable). In this course we will introduce the case of one independent variable.

In real life situations we are usually interested in the relationship between variables. For example, it is difficult to study the impact or effect of salary increments, food price increases, etc. on inflation by using descriptive techniques. Thus, Regression Analysis equips you with a way of studying such situations.

Note: Descriptive techniques generally look at **one variable** at a time, while Regression analysis looks at the relationship **between variables**.

6.4 Simple Linear Regression Analysis

Simple regression analysis is seldom used in applied research because the workings of most socio-economic systems cannot be adequately represented by such a simple formulation. However, knowledge of simple regression analysis is a good foundation for understanding multiple regression analysis [which is usually used in applied research].

The term **simple** implies that a single independent variable x is involved and the term **linear** implies linearity in the parameters.

Consider the following equation which relates Y and X:

$$Y = f(X) \tag{6.2}$$

where f is a function showing how Y is related to X. In literature, the response variable, i.e., the variable of interest Y, and the explanatory or independent variable X, which is used to explain Y, have several names. We give these names in the table below;

Table 6.1: Names given to response and explanatory variables in Regression Analysis

	Y	X
(a)	Predictand	Predictor
(b)	Regressand	Regressor
(c)	Dependent variable	Independent variable
(d)	Effect variable	Causal variable
(e)	Endogenous variable	Exogenous variable
(f)	Target variable	Control variable

Each pair of the above terms is appropriate for a particular use of regression analysis. For example, the terminology in (a) is often used if the purpose of the regression is prediction; pairs (b), (c) and (d) are used by different applied researchers in their discussion of **regression models**; (e) is usually used in studies of causation or causality; while pair (f) is more appropriate in control problems.

6.4.1 Types of Relationship

There are two main types of relationship between or among variables, namely exact and statistical relationships.

Exact Relationships

An exact relationship is a relationship of the form:

$$y_i = \beta_0 + \beta_1 x_i \tag{6.3}$$

where the subscript i (for $i = 1, 2, \cdots, n$) refers to the ith observation. What does this mean? Well, for any value of x_i the y_i value will be equal to some constant value β_0 added to the $\beta_1 \times x_i$. β_0 and β_1 are constants. y_i is determined by x_i, i.e., a unit change in X causes a change equal to β_1 in Y.

Note: here, the variables X and Y can be random or deterministic i.e., non-random.

Generally this equation can be expressed as follows:

$$y = \beta_0 + \beta_1 x \tag{6.4}$$

where x and y are possible values of X and Y respectively. An example of an exact relationship follows.

Exercise 6.2
The relationship between the area of a square A and the length of one side L is given by:

$$Area = \beta_0 + \beta_1 \times length^2$$

where $\beta_0 = 0$ and $\beta_1 = 1$ This is an exact relationship Figure 6.2 gives a plot of the relationship between Area [A] and the square of the length [L^2].

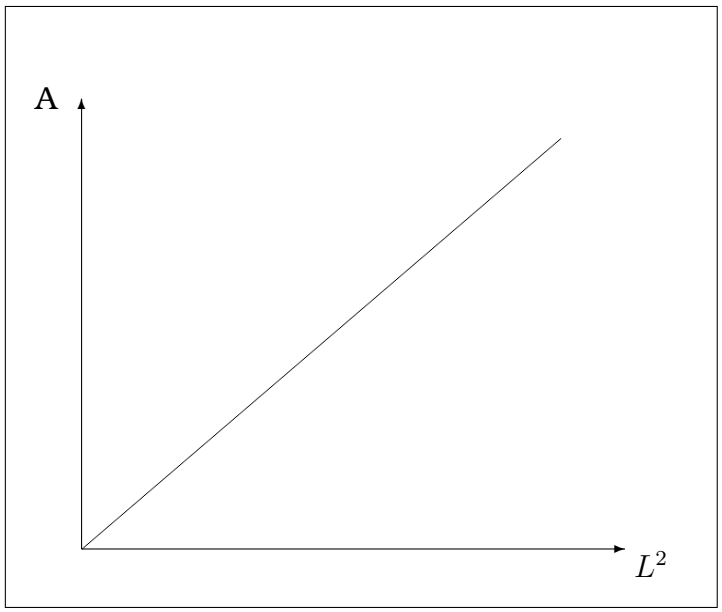


Figure 6.2: The relationship between A and L^2

If we had plotted A against L, we would have obtained the relationship shown in Figure 6.3

These are both exact relationships. In Figure 6.2, we plotted A versus L^2 which gives a linear relationship of the form $y = \beta_0 + \beta_1 x$. On the other hand, in Figure 6.3 we plotted A versus L, we notice that, this time, there is a quadratic relationship.

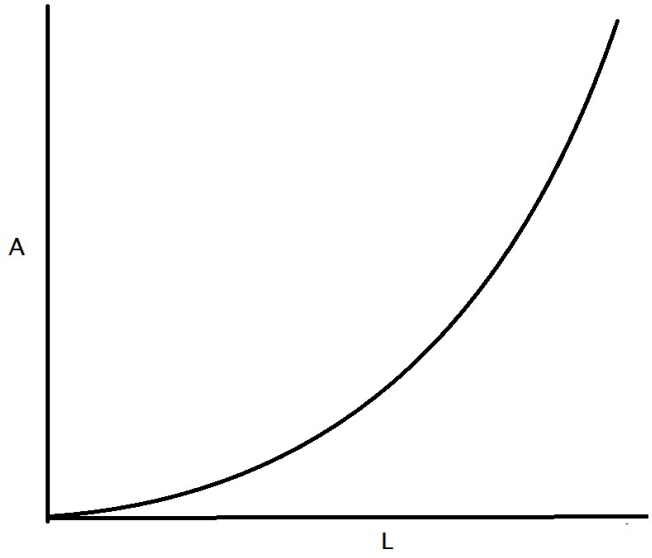


Figure 6.3: The relationship between A and L

Statistical Relationships

A statistical relationship, unlike an exact relationship, is not a perfect one, that is, it does not give unique values of Y for a given value of X, but can be described exactly in probabilistic terms. For instance, consider the following **regression model** showing a statistical relationship between Y and X which is no longer exact because of the error term ϵ_i:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i. \tag{6.5}$$

The variable ϵ_i is a value added to the equation to make the two sides of the equation equal. The term ϵ_i is called the **error term**. The error term is usually assumed to have a normal distribution with mean 0 and variance σ^2. The relationship between Y and X in Equation 6.5 is called a **stochastic** or **statistical** relationship because of the presence of the random error term needed to make the equation exact.

Definition 14 *A regression model is a statistical relationship between a dependent variable, say Y, and some explanatory variable(s), say X. The model is said to be deterministic if the explanatory variable X is non-random.*

Note: The following term from Equation 6.5:

$$\beta_0 + \beta_1 X_i \tag{6.6}$$

is the **deterministic component** of Y_i, and β_0 and β_1, are called **regression coefficients** or **regression parameters**. ϵ_i is the **stochastic** or **random disturbance term**, it takes care of all the variation of

Y which is not explained by the deterministic component $\beta_0 + \beta_1 X_i$. We will discuss how to estimate the regression coefficients from a given data later.

Let us look at a practical example of a statistical relationship in order for us to appreciate the differences between an exact relationship and a statistical relationship.

Exercise 6.3
[Statistical relationship] A group of students are interested in evaluating the advantages and disadvantages of different study patterns and their effect on their performance. Consider Y, the mark a student gets after an examination, and X_1, the number of hours the student puts into reading for the examination.

The variable X_1 was chosen by the students because it seemed [appeared] to contribute a lot to the examination mark. A possible equation to represent the relationship between Y and X_1 is given as:

$$y = \beta_0 + \beta_1 x_1 + \epsilon \tag{6.7}$$

β_0 and β_1 are unknown constants or regression parameters. If a student puts 0 hours into studying for the examination, then we expect him/her to get β_0 marks. On the other hand, if a student increases his/her study time by one hour, the model suggests that the mark should change by β_1. Please note that, as in Equations 6.7, we will index y and x to y_i and x_i respectively, when we have the actual observations $x_1, x_2, \cdots, x_n$ and $y_1, y_2, \cdots, y_n$ at hand.

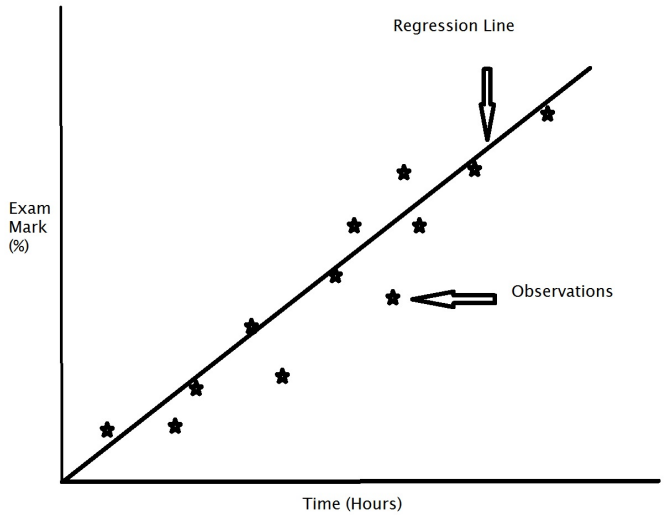


Figure 6.4: Examination mark versus time spent studying

Notice how the points in Figure 6.4 are not always on the line. This is because the relationship between examination mark and the time spent

on studying is not exact.

The students could have added x_2, the number of books the student consulted as a variable, since this appears to have an impact on the final examination mark. This would give Equation 6.8.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon \tag{6.8}$$

Procedures embraced by regression analysis concern themselves with drawing conclusions about these coefficients. An example of the implications of these coefficients follows from the fact that a positive coefficient means that the more the hours spent studying, the higher the examination mark etc. The term ϵ in the equation is added to account for the fact that the equation is not exact. If there are p explanatory variables, a regression equation can be expressed more generally as in Equation 6.9.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon \tag{6.9}$$

This is referred to as **multiple regression**.

6.5 When To Apply Regression Analysis

There are conditions which must be satisfied before we can apply regression analysis. The first condition to be met is that the variables of concern should be related to each other, otherwise the idea of regression collapses. The second condition is that one variable should change in response to the other, i.e there should be a dependence relationship. How do we check on these requirements? This is often done by:

1. constructing a **scatter plot**.
2. calculating the correlation of the variables.

Exercise 6.4
Figure 1.4, shows examples of a scatter plots showing the relationship between an independent variable and a dependent variable. The variable y represents the dependent variable, while x represents the independent variable. In the first plot we have a linear relationship. This could be the relationship between Intelligence Quotient (IQ) x and the mark obtained in an achievement test y. In the second plot, we have a quadratic relationship. This could be the relationship between time x and the distance y travelled (from the source) by an object thrown up into the air.

So, by constructing a scatter plot, we are guided in our decision or choice of the equation to use.

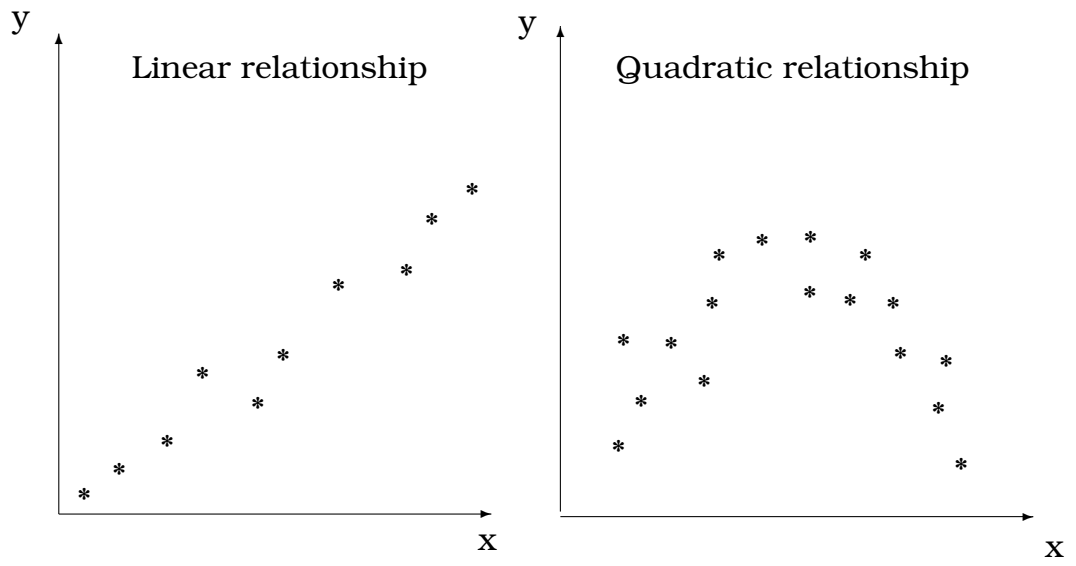


Figure 6.5: Scatter plots showing two possible relationships between x and y

All procedures and conclusions drawn in regression analysis depend, at least indirectly, on the assumptions of the **regression model**. A model is what the data analysts perceive as the mechanism that generates the data on which the regression analysis is conducted.

The term **fitting the model** to a set of data involves estimation of the regression coefficients and formulation of a **fitted regression model** [i.e the model with the estimated coefficients]:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x \tag{6.10}$$

6.5.1 Some Uses of Regression Analysis

Uses of regression analysis fall into the following categories:

1. Prediction: using the data availabe in conjunction with the model to predict/estimate future outcomes.
2. Variable screening: removing unnecessary variables from the a multiple linear regression model. In this way we can find variables that are important to a dependent variable.
3. Explaining: system explanation, which variables contribute the most and how they contribute to the dependent variable.
4. Inference: estimation of the parameters in the model.
5. Planning and control: if we have an appropriate model we can explain the physical system and thus plan ahead and control the system.

Regression techniques are also widely used in econometrics.

6.6 Regression Assumptions

We shall use the **Least Squares** (LS) procedure to estimate the parameters in the model given by Equation 6.5. Although there are other procedures available for estimating the parameters, we shall, however, only use the LS procedure. We will also make the following assumptions (these are necessary for inference):

- There is a linear relationship between the independent variable x_i and the dependent variable y_i.
- The x_is are non-random and are observed with negligible error.
- The ϵ_is are random variables with mean zero and constant variance. This is called the **homogeneous** variance assumption. Mathematically, this assumption is:

$$E(\epsilon_i) = 0 \text{ and } Var(\epsilon_i) = E(\epsilon_i^2) = \sigma^2.$$

- The ϵ_is are uncorrelated, i.e.

$$E(\epsilon_i, \epsilon_j) = \begin{cases} 0 \text{ if } i \neq j \\ \sigma^2 \quad if \quad i = j \end{cases}$$

  fo $i, j = 1, 2, \cdots, n$

- The normal theory assumption is imposed on the ϵ_is. This is the assumption that the ϵ_is are normally distributed with mean zero and variance σ^2. Mathematically, this is stated as

$$\epsilon_i \sim N(0, \sigma^2).$$

From above, it follows that:

$$E(y_i) = \beta_0 + \beta_1 x_i \qquad i = 1, 2, \ldots, n, \tag{6.11}$$

since $E(\epsilon_i) = 0$. Thus, we can use the notation:

$$E(y|x) = \beta_0 + \beta_1 x. \tag{6.12}$$

This is the expected value of y for a given value of X, i.e.. given $X = x$.

6.7 The Least Squares Technique

The method of least squares, attributed to K. Gauss, a German scientist, is perhaps the most extensively used technique for estimating the parameters β_0 and β_1, to give the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, respectively. These parameters have to be estimated because we do not have the actual values available. When we use these estimates, we get the fitted model given by:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i. \tag{6.13}$$

We call this the fitted model because the model now has estimated parameters. Generally in Statistics, the 'hat' notation is used to indicate an estimate. Notice that we don't have the error term in Model 6.13. The relationship between x and $E(Y|x)$ is now an exact one.

Definition 15 ***(Residual)***
Let $r_i = y_i - \hat{y}_i$, This difference is called a **residual**.

The distinction between the residual r_i and the error term ϵ_i is important. The former measures the deviation of y_i from $\hat{y}_i$. Since ϵ is usually unknown, it is estimated by r_i. The residuals are needed not only for estimating the magnitude of the random variation in the y_is, but also for assessing the appropriateness of the regression model employed. We shall discuss this later.

6.7.1 Least Squares Estimation

We want the fitted values $\hat{y}_i$ to be as close as possible to y_i. To achieve this, let us consider $\sum_{i=1}^{n} r_i^2$, the **Residual Sum of Squares** (RSS). It makes good sense to minimise RSS, since a good fit should produce the smallest possible sum of squares. This is the basis of the least squares technique.

Figure 6.6 illustrates what is really happening, the points marked by the '★'s represent the observed values while the fitted values lie on the line indicated by '◊'s. The residual r_i is shown clearly as the difference between the observed value y_i and the fitted value $\hat{y}_i$.
To minimise the RSS, $\hat{\beta}_0$ and $\hat{\beta}_1$ must satisfy the conditions:

$$\frac{\partial}{\partial \hat{\beta}_0}\left(\sum_{i=1}^{n} r_i^2\right) = 0, \text{ and}$$

$$\frac{\partial}{\partial \hat{\beta}_1}\left(\sum_{i=1}^{n} r_i^2\right) = 0.$$

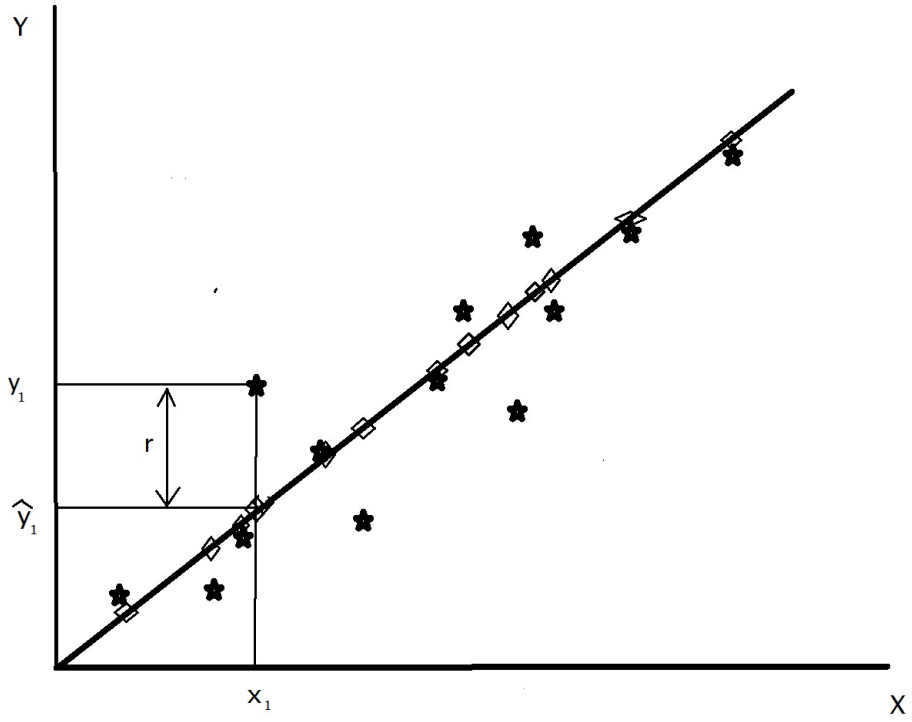


Figure 6.6: The observed values y_i (marked '★'), residuals r_i and fitted values $\hat{y}_i$ (marked '◊')

Thus:

$$\begin{aligned}\frac{\partial}{\partial\hat{\beta}_0}\left(\sum_{i=1}^n r_i^2\right) &= \frac{\partial}{\partial\hat{\beta}_0}\sum_{i=1}^n\left(y_i-\hat{\beta}_0-\hat{\beta}_1x_i\right)^2\\ &= -2\sum_{i=1}^n\left(y_i-\hat{\beta}_0-\hat{\beta}_1x_i\right)\\ &= -2\sum_{i=1}^n y_i+2n\hat{\beta}_0+2\hat{\beta}_1\sum_{i=1}^n x_i\\ &= 0.\end{aligned}$$

So that:

$$\sum_{i=1}^n y_i - n\hat{\beta}_0 - \hat{\beta}_1\sum_{i=1}^n x_i \;= 0.$$

Simplification of Equation 2.6, (by dividing by n) leads us to:

$$\overline{y} \;= \hat{\beta}_0 + \hat{\beta}_1\overline{x} \tag{6.14}$$

For β_1 we have:

$$\begin{aligned}\frac{\partial}{\partial\hat{\beta}_1}\left(\sum_{i=1}^n r_i^2\right) &= \frac{\partial}{\partial\hat{\beta}_1}\sum_{i=1}^n\left(y_i-\hat{\beta}_0-\hat{\beta}_1x_i\right)^2\\ &= -2\sum_{i=1}^n x_i\left(y_i-\hat{\beta}_0-\hat{\beta}_1x_i\right)\\ &= -2\sum_{i=1}^n y_ix_i+2\hat{\beta}_0\sum_{i=1}^n x_i+2\hat{\beta}_1\sum_{i=1}^n x_i^2\\ &= 0.\end{aligned}$$

This simplifies to,

$$\sum_{i=1}^{n} x_i y_i = \hat{\beta}_0 \sum_{i=1}^{n} x_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i^2 \tag{6.15}$$

Equations 6.14 and 6.15 are called normal equations. Solving for $\hat{\beta}_0$ in Equation 6.15 gives us the following estimate for β_0 in terms of β_1. $\sum_{i=1}^{n} y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_i$ can be solved to give:

$$\begin{aligned} \hat{\beta}_0 &= \frac{1}{n}\sum_{i=1}^{n} y_i - \hat{\beta}_1 \frac{1}{n}\sum_{i=1}^{n} x_i \\ &= \overline{y} - \hat{\beta}_1 \overline{x} \end{aligned} \tag{6.16}$$

Substituting for $\hat{\beta}_0$ in equation 2.8 gives us

$$\begin{aligned} 0 &= \textstyle\sum_{i=1}^{n} x_i y_i - \left[\frac{1}{n}\sum_{i=1}^{n} y_i - \hat{\beta}_1 \frac{1}{n}\sum_{i=1}^{n} x_i\right] \sum_{i=1}^{n} x_i - \hat{\beta}_1 \sum_{i=1}^{n} x_i^2 \\ &= \hat{\beta}_1 \left[\frac{1}{n}\left(\sum_{i=1}^{n} x_i\right)^2 - \sum_{i=1}^{n} x_i^2\right] + \sum_{i=1}^{n} x_i y_i - \frac{1}{n}\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right) \end{aligned}$$

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum_{i=1}^{n}(x_i y_i) - \frac{1}{n}\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{\sum_{i=1}^{n} x_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n} x_i\right)^2} \\ &= \frac{\sum_{i=1}^{n}(x_i y_i) - n\overline{x}\overline{y}}{\sum_{i=1}^{n} x_i^2 - n\overline{x}^2} \\ &= \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^{n}(x_i - \overline{x})^2} \end{aligned}$$

We often state the above as:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \tag{6.17}$$

where $S_{xy} = \sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})$ and $S_{xx} = \sum_{i=1}^{n}(x_i - \overline{x})^2$. Note that we first find $\hat{\beta}_1$ and then find $\hat{\beta}_0$, using $\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}$.

The fitted line is $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$. This fitted line is referred to by many different names in statistics. Some of the names are: the least squares line, fitted regression line, estimated regression line or just the fitted model.

We shall study the properties of the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ in Part II of this degree course.

Example 6.4
The starting salary S per year of people of different educational background (Ed) has always been of interest to people going to university. They have always tried to find out the relationship between these. We expect the starting salary to be directly related to the educational level,

i.e., as the educational level increases, so does the salary.

As this may not be the case, we shall investigate this suspicion using regression analysis. Suppose that an individual's educational level is given a score, then an appropriate model is given by:

$$S = \beta_0 + \beta_1 Ed + \epsilon$$

The data on S and Ed were collected and recorded as shown below. Find the estimates for β_0 and β_1 and discuss your results.

Number	*Annual salary ($)*	*Educational level score*
1	*20 000*	*2.8*
2	*24 500*	*3.4*
3	*23 000*	*3.2*
4	*25 000*	*3.8*
5	*20 000*	*3.2*
6	*22 500*	*3.4*

Solution:

Calculations give

Number	S_i	Ed_i	$S_i Ed_i$	Ed_i^2
1	*20 000*	*2.8*	*56 000*	*7.84*
2	*24 500*	*3.4*	*83 300*	*11.56*
3	*23 000*	*3.2*	*73 600*	*10.24*
4	*25 000*	*3.8*	*95 000*	*14.44*
5	*20 000*	*3.2*	*64 000*	*10.24*
6	*22 500*	*3.4*	*76 500*	*11.56*
Total	*135 000*	*19.8*	*448 400*	*65.88*

$$\begin{aligned} \hat{\beta}_1 &= \frac{S_{Ed,S}}{S_{S,S}} \\ &= \frac{448400 - \frac{(19.8)(135000)}{6}}{65.88 - \frac{19.8^2}{6}} \\ &= \frac{2900}{0.54} \\ &= \$5.370. \end{aligned}$$

Therefore, from (2.8)

$$\begin{aligned} \hat{\beta}_0 &= \frac{135000}{6} - 5370\left(\frac{19.8}{6}\right) \\ &= \$4779 \end{aligned}$$

Thus, we have the fitted model $\hat{S}_i = 4779 + 5370Ed_i$. How do we interpret this model? The starting salary is predicted to be \$4 779, when the Educational level score is zero. This may not say much since an educational score of zero does not apply to this group of people by virtue of their being at University.

Perhaps, of primary interest is the slope (coefficient) which indicates that for a one-unit increase in educational score, the predicted salary increases by \$5 370. For example, for an educational level score of 2.8, the predicted salary is $\hat{P} = 4779 + 5370 \times 2.8 = \$19815$.

Exercise 6.5
1. In Example 6.4, remove the last (sixth) number and estimate β_0 and β_1.
2. Predict the salary for someone with an educational score of 2.8.

6.8 Properties of Estimates

We shall briefly mention, without proving, some of the properties of the estimates we derived earlier. We shall then use these properties in assessing how **good** our model is.

The Least Squares estimates of β_0 and β_1 are unbiased so that we have the following properties.

The Sampling Distribution of β_0 is given by:

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2\left[\frac{1}{n} + \frac{\overline{x}^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2}\right]\right), \tag{6.18}$$

where σ^2 is the variance of the error term.

The sampling distribution of $\hat{\beta}_1$ is given by;

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2}\right). \tag{6.19}$$

Now, using the properties of the sampling distributions of $\hat{\beta}_0$ and $\hat{\beta}_1$, inferences about β_0 and β_1 can be made. First, however, an estimate of one other unknown parameter in the regression model is needed. This is an estimate of σ^2. This estimate is given by:

$$\hat{\sigma}^2 \approx s^2 = \frac{1}{n-2}\sum_{i=1}^{n}(r_i - \overline{r})^2 \tag{6.20}$$
$$= \frac{1}{n-2}\sum_{i=1}^{n}(r_i)^2$$
$$= \frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y})^2.$$

Since $\sum_{i=1}^{n} r_i = 0$ and $\overline{r} = 0$.

Note that s^2=MSE, the Mean Square Error and SSE=$\sum_{i=1}^{n}(y_i - \bar{y}_i)^2$ the Error Sum of Squares.

We will use s^2 to estimate σ^2 in our expressions of $Var(\hat{\beta}_0)$ and $Var(\hat{\beta}_1)$ to get:

$$\widehat{Var(\hat{\beta}_1)} = \frac{s^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2} \tag{6.21}$$

and

$$\widehat{Var(\hat{\beta}_1)} = \frac{s^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2}, \tag{6.22}$$

respectively.

Let discuss how we can make inferences about the regression parameters before proceeding to investigate how well the fitted line fits the data.

Exercise 6.6

1. Deduce the variance of y_i.
2. Assuming $\hat{\beta}_1$ and $\hat{\beta}_2$ are independent, find the variance of $\hat{y}_i$.

6.9 Making Inferences about β_0 and β_1

The following procedure can be used to test hypotheses about the slope and the intercept.

1. Establish the null and alternative hypotheses.
   (a) The null hypothesis is that: There is no linear relationship between Y and X, that is, $H_0 : \beta_1 = 0$.

(b) The alternative hypothesis is that: There is a linear relationship between Y and X, i.e. $H_1 : \beta_1 \neq 0$.

2. Determine the tolerance α for a type I error probability.

3. Identify an appropriate test-statistic. Using $H_0 : \beta_1 = 0$, the test statistic is given by:

$$t = \frac{\hat{\beta}_1}{\sqrt{s^2 / \sum_{i=1}^{n}(x_i - \overline{x})^2}} \tag{6.23}$$

Note that t has a student's t-distribution with $n - 2$ degrees of freedom.

4. State the assumptions under which this test statistic has a t-distribution. The assumptions about the error must be valid for the conclusions or inferences to be valid.

5. Determine whether these assumptions are satisfied.

6. Find the values for the test statistic that allow a rejection of the null hypothesis. We find the critical values $t_{n-2}(\frac{1}{2}\alpha)$, such that we reject H_0 if $t < -t_{n-2}(\frac{1}{2}\alpha)$. or if $t > t_{n-2}(\frac{1}{2}\alpha)$

7. Compute the t-value based on sample data.

8. Interpret the result from a statistical viewpoint.

Example 6.6
We assume that a two-tailed test is appropriate and we use the data in Example 6.4 to test the hypothesis
$H_0 : \beta_1 = 0$ *versus*
$H_1 : \beta_1 \neq 0$
at $\alpha = 0.05$.

First we calculate s *as follows:*

$$s = \sqrt{\frac{\sum_{i=1}^{n}(S_i - \hat{S}_i)^2}{n-2}}$$

Number	S_i	$\hat{S}_i$	$S_i - \hat{S}_i$	$(S_i - \hat{S}_i)_i^2$
1	*20 000*	*19 815*	*185*	*34.225*
2	*24 500*	*23 037*	*1,463*	*2 140 369*
3	*23 000*	*21 963*	*1,037*	*1 075 369*
4	*25 000*	*25 185*	*-185*	*34 225*
5	*20 000*	*21 963*	*-1,963*	*3 853 369*
6	*22 500*	*23 037*	*-537*	*288 369*
Total			*0*	*7 425 926*

$$s = \sqrt{\frac{7425926}{4}} \approx 1363$$

Calculation of the variance estimate:

Number	Ed_i	$Ed_i - \overline{Ed}$	$(Ed_i - \overline{Ed})^2$
1	*2.8*	*-0.5*	*0.25*
2	*3.4*	*0.1*	*0.01*
3	*3.2*	*-0.1*	*0.01*
4	*3.8*	*0.5*	*0.25*
5	*3.2*	*-0.1*	*0.01*
6	*3.4*	*0.1*	*0.01*
Total			*0.54*

We calculate $\widehat{Var(\hat{\beta}_1)}$ *as follows:*

$$\widehat{Var(\hat{\beta}_1)} = \frac{1363}{\sqrt{0.54}} = 1854$$

Our test statistic t *is found to be:*

$$t = \frac{5370 - 0}{1854} = 2.896$$

The critical value is $t_{n-2}(\frac{1}{2}\alpha) = t_4(0.025) = 2.78$*. Therefore, we reject* H_0 *since* t *exceeds the critical value* $t_{n-2}(\frac{1}{2}\alpha)$*. We therefore conclude that there is a significant statistical relationship between the starting salary and the educational level score. See Figure 6.7 for the decision-making diagram.*

A similar approach can be used to test the significance of the intercept.

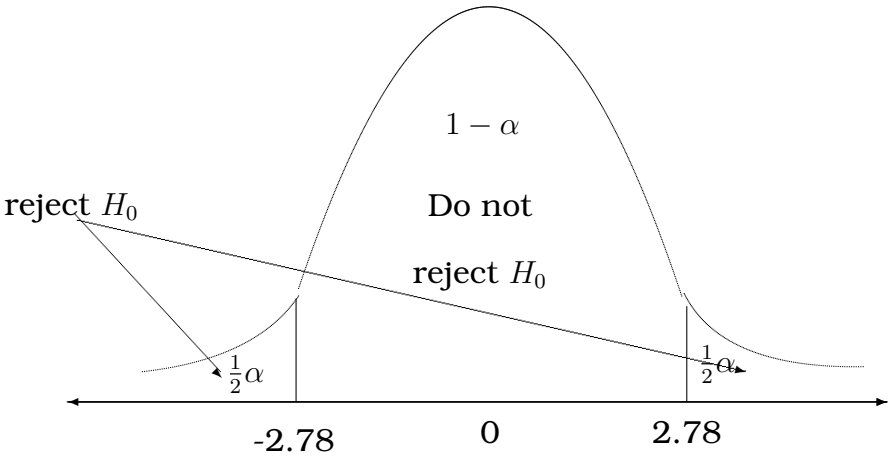


Figure 6.7: t-Distribution: when to reject or accept H_0

Exercise 6.7
1. If $\hat{\beta}_1$ had been -0.32, what conclusions would you have drawn?
2. Test the null hypothesis that the intercept is zero.

6.10 Analysis of Variance of Simple Linear Regression

Analysis of variance (ANOVA) is a highly useful and flexible mode of analysis for regression models. We will use ANOVA to compute σ^2 and to measure the degree of linear association between X and Y in the sample data.

6.10.1 Partitioning the Total Sum of Squares

The uncertainty associated with a prediction is related to the variability of the Y observations around their mean, as measured by the following deviations:

$$Y_i - \overline{Y}$$

The greater the variability in the data, the larger will be the deviations, $Y_i - \overline{Y}$, and the greater is the uncertainty associated with a prediction Y_i, without utilising knowledge of X_i.

Conventionally, the measure of variability of the observations is expressed in terms of the sum of squares of the observations $Y_i - \overline{Y}$ and is denoted by:

$$SST = \sum_{i=1}^{n} (Y_i - \overline{Y})^2 \tag{6.24}$$

where SST stands for Total Sum of Squares. If there is a lot of variability in the Y_i, then SST is large.

Error Sum of Squares

The uncertainty associated with a prediction is related to the variability of the Y_i around the fitted regression line as measured by the following deviations:

$$r_i = Y_i - \hat{Y}_i$$

If all the Y_i values fall on the regression line, all the deviations r_i, will be zero. The larger the deviations r_i the greater the uncertainty associated with a prediction utilising knowledge of the independent variables X_i.

The conventional measure of variability around the fitted regression is the Error Sum of Squares (SSE) which is calculated as follows:

$$SSE = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 = \sum_{i=1}^{n} r_i^2 \tag{6.25}$$

If all the Y_i values fall on the regression line, SSE will be zero.

Regression Sum of Squares

The reduction in the variability associated with the utilisation of the knowledge of the independent variable X_i is another sum of squares known as Regression Sum of Squares (SSR). Fig. 3.2 illustrates how each of these components arises.

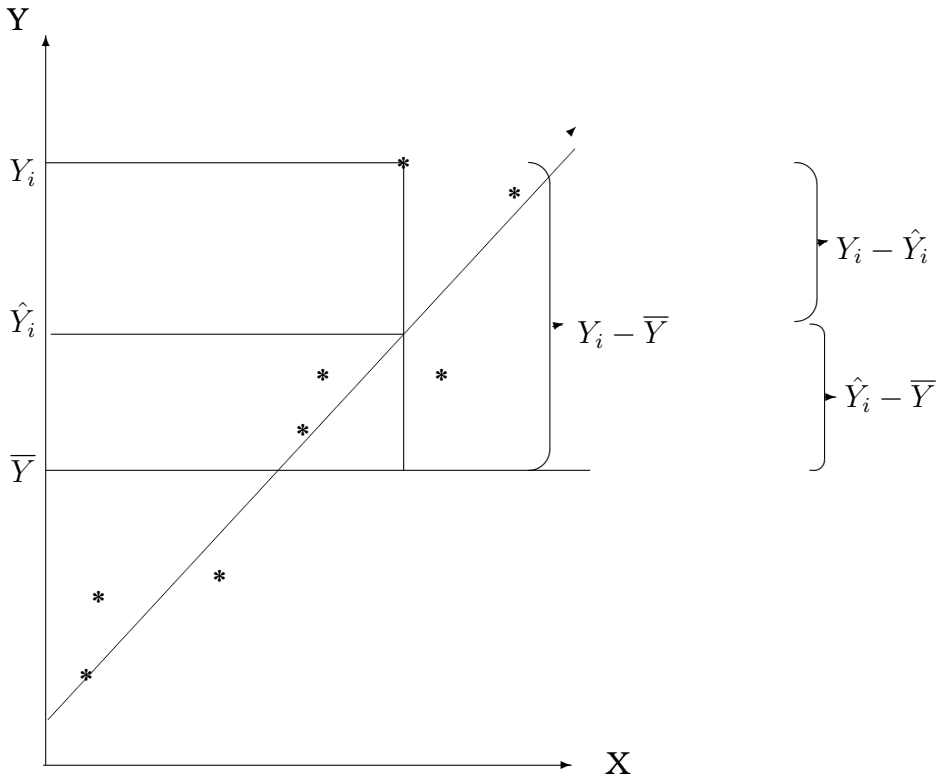


Figure 6.8: Variability on the regression line

$$SSR = SST - SSE \tag{6.26}$$

We can show that SSR is the sum of squares involving the deviations:

$$\hat{Y}_i - \overline{Y},$$

which represent the fitted value and the mean of the fitted value.

$$SSR = \sum_{i=1}^{n} (\hat{Y}_i - \overline{Y})^2 \tag{6.27}$$

SSR can be viewed as a measure of the effect of the regression relation in reducing the variability of Y_i. If SSR = 0, the regression calculation will not reduce variability at all. SSR can be interpreted as the proportion of variation in Y explained by the regression.

A more mathematical approach is to say we are partitioning the variability of the Y_is. Thus, for Simple Linear Regression, the decomposition of the SST into two components is achieved as follows:

$$SST = SSR + SSE \tag{6.28}$$

$$\sum_{i=1}^{n} (Y_i - \overline{Y})^2 = \sum_{i=1}^{n} (\hat{Y}_i - \overline{Y})^2 + \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 \tag{6.29}$$

The computational formulas are put as follows:

$$SST = \sum_{i=1}^{n} Y_i^2 - n\overline{Y}^2 \tag{6.30}$$

$$SSR = \frac{(\sum_{i=1}^{n} Y_i X_i - n\overline{YX})^2}{\sum_{i=1}^{n} X_i^2 - n\overline{X}^2} \tag{6.31}$$

and

$$SSE = SST - SSR \tag{6.32}$$

Partitioning of Degrees of Freedom

A sum of squares has an associated number of degrees of freedom. Recall that the variance estimate s^2 in Equation 3.3 has a denominator $n - 2$. These are the degrees of freedom associated with the numerator sum of squares in s^2.

Corresponding to the partitioning of the sum of squares is a partitioning of the degrees of freedom. SST has $n - 1$ degrees of freedom (*df*) associated with it. Why? SST has n deviations, namely $Y_i - \overline{Y}$. However, there is one constraint on these deviations, namely $\sum(Y_i - \overline{Y}) = 0$, so we lose one degree of freedom, to remain with $n - 1$ degrees of freedom in the n deviations.

SSE has $n - 2$ degrees of freedom, since we imposed constraints on the r_is during the estimation of β_0 and β_1. As SSR has 1 *df*, there are two

parameters in the regression function, but the deviations $\hat{Y}_i - \overline{Y}$ are subject to the constraint $\sum(\hat{Y}_i - \overline{Y}) = 0$. Thus, the degrees of freedom are additive and given by $n - 1$ = 1 + $n - 2$.

Mean Squares

A sum of squares divided by the degrees of freedom is called a mean square. For example, $s^2 = MSE$. The two important mean squares are the *regression mean square* denoted by MSR and the *error mean square* denoted by MSE.

Thus:

$$MSR = \frac{SSR}{1} \tag{6.33}$$

and

$$MSE = \frac{SSE}{n-2} = s^2 \tag{6.34}$$

Some Properties of Mean Squares

It can be shown that the expectations of the mean squares are given by:

E[MSE] = σ^2

It can also be shown that: E[MSR] = σ^2 + $\beta_1^2 \sum(X_i - \overline{X})^2$

Thus, when $\beta_1 = 0$, E[MSR] = σ^2, both MSE and MSR have the same expected value under this condition. On the other hand, when $\beta_1 \neq 0$, the term σ^2 + $\beta_1^2 \sum[X_i - \overline{X}]^2$ will be positive and E[MSR] $>$ nE[MSE]. Hence, if $\beta_1 \neq 0$, MSR will tend to be larger than MSE.

Exercise 6.8

1. *What does it mean if the fitted model gives you SSE = 0.*
2. *If SSR is zero, what does it tell you about the model?*
3. *In the Simple Linear Regression model, suppose that SST has 14 degrees of freedom. Deduce the SSE and SSR degrees of freedom.*

6.11 The Basic ANOVA Table

It is useful to collect the sum of squares, degrees of freedom and mean squares in an ANOVA table for regression analysis. Table 3.1 gives the

structure and the appearance of the basic ANOVA table.

Source of variation	SS	df	MS	F
Regression	$SSR = \sum(\hat{Y}_i - \overline{Y})^2$	1	$MSR = \frac{SSR}{1}$	$F = \frac{MSR}{MSE}$
Error	$SSE = \sum(Y_i - \hat{Y}_i)^2$	$n-1$	$MSE = \frac{SSE}{n-2}$	
Total	$SST = \sum(Y_i - \overline{Y})^2$	$n-1$		

Table 6.2: The basic ANOVA table for simple linear regression

From the ANOVA table, we can get the variance s^2 and test the hypothesis that there is a regression relationship. How do we do this? The ratio F in the ANOVA table has what we call the Fisher's distribution with 1 and $n-2$ degrees of freedom if the assumptions of the model hold.

If F is near 1, then MSR and MSE are approximately equal. $F > 1$, suggests that $\beta_1 \neq 0$. Thus, an upper-tail test is appropriate.

The hypotheses we are testing here are as follows:
$H_0 : \beta_1 = 0$ versus
$H_1 : \beta_1 \neq 0$
at level α. Our decision rule here is as shown in Fig. 3.3

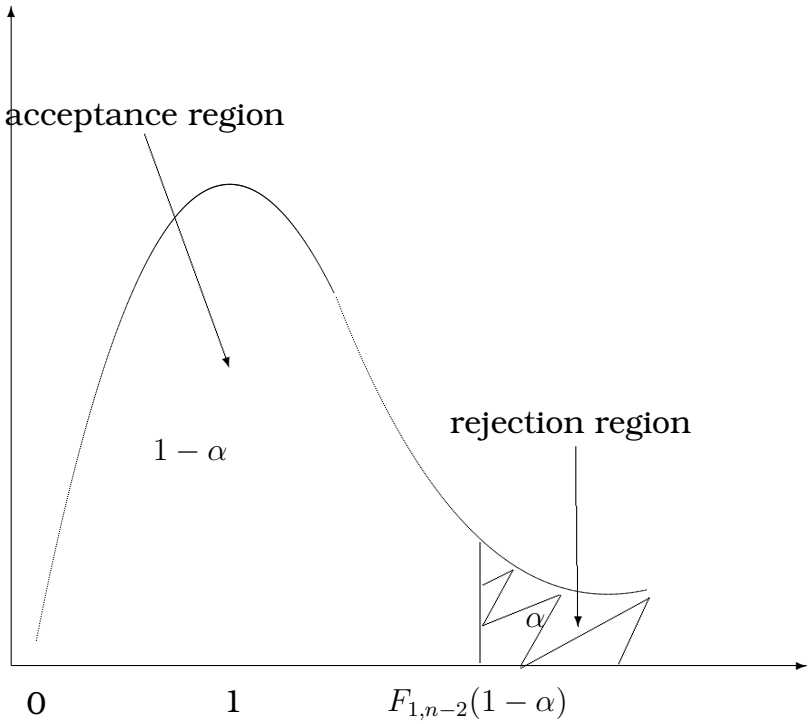


Figure 6.9: The general form of the statistical decision rule for an F-test

The decision rule is given by:
Accept H_0 if $F \leq F_{1,n-2}(1-\alpha)$
Reject H_0 if $F > F_{1,n-2}(1-\alpha)$.

6.12 The Coefficient of (simple) Determination

The **coefficient of determination** R^2 is a measure of the effect an independent variable X has in the regression model in explaining the total variability in Y.

The coefficient of determination, denoted by R^2, is defined as follows:

$$R^2 = \frac{SSR}{SST} \tag{6.35}$$

$$= 1 - \frac{SSE}{SST} \tag{6.36}$$

Thus, R^2 measures the proportionate reduction in SST associated with the use of an independent variable.

In the Simple Linear Regression case, we usually refer to the coefficient of Determination as the **Coefficient of Simple Determination** (R^2). Note that R is the simple correlation coefficient of the independent and dependent variables.

R^2 takes values between 0 and 1. We obtain $R^2 = 0$ when $\beta_1 = 0\%$, and $R^2 = 100\%$ when the Y_is fall directly on the regression line. A value of $R^2 > 80\%$ or sometimes 70%, suggests that the model has a good fit.

6.12.1 Adjusted R^2

One phenomenon found on adding terms to a regression model is that the R^2 increases. Although this may be an indication that the extra terms improve the regression equation, it is may also be a reflection of the fact that one is using more variables to predict the same number of data points. This problem may be taken into account by examining not only the actual value of R^2, but also the value of the adjusted R^2. This statistic takes into account the number of data points and variables in the regression equation, by replacing the SSE and $\sum_{i=1}^{n}(y_i - \overline{y})^2$ by the corresponding MSE's, giving

$$\overline{R}^2 = 1 - \frac{SSE/(n-2)}{\sum_{i=1}^{n}(y_i - \overline{y})^2/(n-1)} \tag{6.37}$$

which can also be written as $\overline{R}^2 = 1 - \frac{(n-1)}{(n-2)}(1 - R^2)$.

It is so scaled that, if a second added variable results in a non-significant improvement in the regression fit, the adjusted R^2 will decrease.

Example 6.7
The price of a kilogram of flour (Y) at a market place in a particularly busy township seems to vary according to what the vendor thinks is your salary X (in thousands of Rands) Use the data supplied to investigate if this suspicion is true.

induvidual	X	Y
1	*2*	*8.74*
2	*2*	*10.53*
3	*2*	*10.99*
4	*2*	*11.97*
5	*3*	*12.83*
6	*3*	*14.69*
7	*3*	*14.69*
8	*3*	*15.30*
9	*4*	*16.11*
10	*4*	*16.31*
11	*4*	*16.46*
12	*4*	*17.69*
13	*5*	*19.65*
14	*5*	*18.86*
15	*5*	*19.93*
16	*5*	*20.51*

Solution
The model is given by: $y = \beta_0 + \beta_1 x + \epsilon$ *(*$price = \beta_0 + \beta_1 salary + \epsilon$*)*

Fitting the model gives us;

$$\begin{aligned} \widehat{price} &= \hat{\beta}_0 + \hat{\beta}_1 salary \\ &= 4.8970 + 2.9805 salary \end{aligned}$$

Let us now construct the ANOVA table. First we calculate SSR, SST, and SSE.

$$\begin{aligned} SST = &\sum y_i^2 - n\overline{y}^2 \\ = &\ 197.195, \end{aligned}$$

$$\begin{aligned} SSR = &\sum_i^{16} (\hat{y})_i - n\overline{y}^2 \\ = &\ 177.668, \end{aligned}$$

$$\begin{aligned} \textit{and } SSE = &\ SST - SSR \\ = &\ 191.194 - 177.668. \\ = &\ 3.527 \end{aligned}$$

Table 6.3: The ANOVA Table

SOURCE	df	SS	MS	F
Regression	1	177.668	177.668	183.921
Error	14	13.526	0.966	
Total	15	191.194		

From the above, the ANOVA table can be constructed as follows.

We can see from this ANOVA table that F is quite large. Infact it leads us to a rejection of the hypothesis of no regression relationship (verify).

We can compute the coefficient of Multiple determination from Table 6.3 above. Thus:

$$\begin{aligned} R^2 &= \frac{SSR}{SST} \\ &= \frac{177.668}{191.195} \\ &= 0.929. \end{aligned}$$

Thus, about 92.9% *of the variation in prices (Y) is explained by the regression model. So, the salary estimates do seem do determine the price of flour. The adjusted* $R^2 = 92.4\%$*.*

The correlation coefficient (r) measures the strength of the linear relationship between the dependent variable and all the independent variables. It is computed from the formula:

$$r = +\sqrt{R^2} \tag{6.38}$$

$$= +\sqrt{\frac{SSR}{SST}}, \tag{6.39}$$

where SSR is the Regression Sum of Squares and SST the Total Sum of Squares. R close to 1 means that there is a good linear relationship between the dependent and independent variables.

Exercise 6.9

A student recorded the 6 test marks she obtained after devoting a particular number of hours of study. The marks are:

Marks (%)	*60*	*50*	*40*	*100*	*10*
Time (hrs)	*2*	*1.5*	*0.5*	*3*	*0*

1. *Estimate* β_0 *and* β_1 *in a simple linear regression model.*

2. *Construct the ANOVA table.*

3. *Show that F = 42.89, and test the null hypothesis that* $\beta_1 = 0$ *at* $\alpha = 0.05$.

6.13 Conclusion

In this Unit, we focused on Simple Linear Regression. We discussed the estimation of the parameters using the Least Squares technique. There are other methods availabe for estimating these parameters. We shall meet these in future modules.

We went on to discuss how to check if any of the assumptions which enable us to use the least squares approach have been violated. This is often ignored by "pseudo-statisticians". Some blame this abuse of Regression Analysis on computers which allow you to use statistical computer packages without looking at the underlying theory behind the techniques.

6.14 Exercises

1. Find the relationship between the correlation coefficient r and β_1.

2. A farmer is carrying out an experiment on the effect of soil acidity levels on yield. The farmer records the yields of 10 fields of equal size but different acidity levels. [A negative acidity level here indicates that the acidity is less than the standard level pegged at 0.]

Field:	1	2	3	4	5	6	7	8	9	10
Level:	4.5	17.7	-16.6	-14	18.6	-10.6	5.8	-8.1	-5.2	7.8
Yield:	75	112	38	120	105	52	116	118	105	110

   (a) Examine graphically the relationship between Level and Yield.

   (b) Is this a statistical relationship? Explain.

   (c) Which is the dependent variable, and which is the independent variable? Explain.

   (d) Fit a simple linear regression model to this data.

   (e) Investigate if any assumptions were violated.

   (f) What do you expect the yield to be for a field with acidity level 0?

   (g) Do you think the field would make a better independent variable?

3. A recent review of salaries at a company in Harare made the recommendation that, depending on the number of years of service X, the minimum salary Y of an employee should be as follows:

x:	5	10	15	20	25	30
y:	39.5	49.0	58.5	68.0	77.5	87.0

   (a) Examine graphically the relationship between number of years of service and minimum salary.

   (b) Do these recommendations represent an exact or statistical relation? Explain using statistics like the sample correlation coefficient.

4. For each of the following pairs of variables, explain whether an exact or statistical relation would most likely hold:

   (a) X=number of beds in a hotel; Y=hotel's annual operating cost.

   (b) X = Volume of a gas; Y = Pressure on the gas.

   (c) X = A departmental store's promotional and advertising expenditure; Y = the company profits.

5. Find the correlation matrix ρ for the data in Question 1 above.

6. The following information was recorded over 10 years, the amount of rainfall, maize production and maize price.

   (a) Which is the dependent variable? Explain.

   (b) What is the regression model linking these variables?

   (c) Which two variables are likely to have a high correlation?

Course outline

Introduction Mathematical Statistics

Lectures 2 hrs a week [24 lectures]
Tutorials 2 hrs a week
Consultations 2 hrs a week
Lecturers: Charles Chimedza and Nothabo Ndebele

Outline

1. Descriptive Statistics [2 lectures]
    - Introduction [Univariate Statistics]
    - Types of data
        - discrete – nominal, ordinal, Interval
        - continuous – ratio
    - Sampling (Sample, Population)
    - Descriptive Statistics
        - Graphical Tables, Pie charts, bar charts, Multiple bar charts histograms (vertical – and – horizontal) (relative frequency), frequency polygons, Cumulative frequency curve. (ogive)
        - Measure statistics (grouped and ungrouped)
            * Location
                · Mean, median, mode, quartiles percentiles – (Bias)
            * Spread
                · Standard deviation, variance (bias), range, interquartile range
            * Box and Whisker plots
            * Skewness and Kurtosis (based in measures of location and spread)
2. Introduction to Probability [1 lecture]
    - Trials, outcomes, Events, sample Space, Venn diagram

  - Set theory – Sets, null set, subset, universal Set,
  - Union, Intersection, complement; mutually exclusive and exhaustive (via set notation), de Morgan?s laws
- Permutations and combinations [3 lectures]
  - Definition
  - Simple problems
  - Scenario leading to hyper geometric distribution
- Define probability (using above) [3 lectures]
- Axioms
- Theorems + proofs
  - $P(A) = 1 - P(A?)$
  - $P(\emptyset) = 0$
  - If event A and B are such that $A \subset B$, then $P(A) \leq P(B)$
  - For each event A, $P(A) \leq 1$
  - $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
  - $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) - P(A \cap B \cap C)$
- Conditional probability
  - Definition, multiplication rule, independent (pairwise, mutual) events;
  - Bayes theorem
  - Tree diagrams

3. Probability Distributions [8 lectures]

- Discrete Probability distributions
  - Definitions (Probability mass function [pmf])
  - Moments [Expectation of form $E(aX^k + b)$], mean, variance, skewness, and Kurtosis
  - Examples of Discrete distributions (uniform, benoulli, binomial, geometric, Poisson (link with Binomial), hyper geometric (no mean and variances) proofs of pmf.
  - Moment Generating Functions [mgf]
- Continuous distributions
  - Definitions– probability density function
  - Moments [Expectation of form $E(aX^k + b)$], mean, variance, skewness, and Kurtosis
  - Examples of continuous distributions ? continuous uniform, normal, exponential (link with Poisson ? tuts), memory less property – tuts.

4. Statistical Inference [4 lectures]

- Mean
  - T tests
  - Z tests
- Unbiasedness of $\bar{x}$ and s
- Types of errors
- Significance, critical values/regions, $p - values$ and power of a test
- Interval estimation and hypotheses

5. Correlation and Regression Analysis [2 lectures]

- Scatter plots
- Correlation analysis
- Simple linear regression model
- Least squares estimation
- (tut ? link ρ and β_1)
- Maybe – inferences about β_1

Notes: 48 hours contact for students via tuts, 1 lecture = 2 hours of work outside class. Give some solutions (odd numbers) No more than 2 hours of consultation per week